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BUBBLEVERSE

Q Journals

Building a Model of the Universe from Observation

Q001-Q043

The chronological research record

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Appendix E — Bubbleverse Q-Journals —

Q001-Q043

Introduction to the Journal Record

The main chapters of this book organize the surviving science logically. This appendix preserves the investigation chronologically.

Each Q-Journal records, where available, the question that was asked, the scientific state entering the question, inherited results, hypotheses, methods, evidence, programmes, workflows, tests, intermediate and final results, failed attempts, negative results, model changes, provenance and the handoff to the next question.

Many early journals are retrospective reconstructions because complete journal records were not originally maintained at the time of those questions. Their retrospective status remains explicitly labelled. Missing historical GitHub information is not filled by invention. Later corrections do not erase earlier journal states. Superseded numbers, failed numerical routes and earlier interpretations remain historical records where they are required to reconstruct the path of the investigation.

The supplied journal package contains two versions of Q009 describing the same Q, date, question and scientific result. The more structured later copy is retained here as the published Q009 record; the earlier editorial duplicate is not reproduced a second time. No standalone Q010 journal was present in the supplied journal package. Its absence is recorded explicitly at its chronological position rather than reconstructed from later knowledge.

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Q001

BUBBLEVERSE — Q-JOURNAL

Q-ID: Q001 Status: CONFIRMED JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION RECONSTRUCTED ON: 2026-09-08

Question:

What observational facts about the universe are robustly established, distinct from model-dependent interpretation?

STARTING STATE

No prior Bubbleverse model. Q001 establishes the empirical baseline.

Core rule: observation $\neq$ measurement $\neq$ reconstruction $\neq$ inference $\neq$ interpretation.

HYPOTHESES

Supported:

⊛ The observable universe is structured, evolving, expanding, and was hotter/denser in the past.

Rejected:

⊛ Λ CDM is directly observed.

⊛ A specific dark-matter particle is established.

⊛ Dark energy is directly detected.

⊛ A hot early universe proves a ($t=0$) singularity.

METHOD & EVIDENCE

Primary-source review and epistemic classification.

Main evidence:

⊛ CMB: Fixsen, Planck, ACT

⊛ Expansion/BAO: DESI

⊛ Supernovae/ H_0 : Riess, Perlmutter, Pantheon+, SHOES

⊛ Lensing/dark matter: Planck lensing, Bullet Cluster

⊛ Early universe: primordial deuterium

⊛ Gravitational waves: GW150914, GW170817

No numerical Bubbleverse program or workflow was required.

KEY RESULT

Robust observations establish a dynamic, expanding universe with a hotter, denser past.

Dark-matter microphysics, dark-energy nature, and several cosmological parameters remain model-dependent or unresolved.

Contradiction retained: Hubble-tension inference chains disagree.

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NEGATIVE RESULTS / MIKAMI

Killed claims:

- ⊗ “ Λ CDM is observed.”
- ⊗ “Dark matter particle detected.”
- ⊗ “Dark energy substance detected.”
- ⊗ “Hot Big Bang proves singularity.”

MODEL CHANGE

Before: No Bubbleverse baseline.

After: Observation-first cosmological framework established.

Change: Foundational model created.

GITHUB / PROVENANCE

Repository / commits / workflows: NOT RECORDED AT TIME OF Q

Programs: None.

Provenance reconstructed from Q001 outputs, manuscript material, handoff context, and cited primary literature.

REPRODUCIBILITY

Recheck the listed primary sources, classify every claim by epistemic level, verify key numerical values, preserve dataset dependencies, and apply the same acceptance criteria.

CONCLUSION

Q001 establishes Bubbleverse’s empirical foundation and permanent rule:

Never silently promote interpretation into observation.

NEXT-Q HANDOFF

Next Q: Q002

Reason:

Cosmic expansion now requires separate analysis of what is directly observed versus model-inferred.

Next question:

What observations establish cosmic expansion, and which parts of the inferred expansion history depend on cosmological modelling?

FINAL ANSWER: Empirical baseline established.

MIKAMI: Four overstrong claims rejected.

MODEL CHANGE: Observation-first framework created.

GITHUB: Historical provenance not recorded.

NEXT: Q002 — cosmic expansion.

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Q002

BUBBLEVERSE — Q-JOURNAL

1. IDENTITY

Q-ID: Q002

Date/time: 2026-08-26 20:31 CEST

Status: CONFIRMED

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

Question:

What do observations show about cosmic expansion, and which conclusions depend on cosmological models?

2. STARTING STATE

Model entering Q: Q001 observational baseline.

Inherited:

Observation must be separated from calibration, reconstruction and interpretation.

Constraint:

Do not equate redshift, $(H(z))$, (H_0) , scale factor or acceleration.

3. HYPOTHESES

Tested:

H1: Multiple probes support expansion.

H2: Redshift alone proves expansion.

H3: Simple tired-light/static models remain viable.

Rejected:

H2 — redshift alone is insufficient.

H3 — simple no-time-dilation models conflict with SN time dilation.

Surviving:

H1 — CONFIRMED.

4. METHOD & EVIDENCE

Methods:

Literature comparison, cross-probe consistency and observation/inference audit.

Evidence:

Hubble 1929; Riess/Perlmutter; Pantheon+; Goldhaber/Blondin SN time dilation;

BAO/eBOSS/DESI; Moresco chronometers; GW170817 standard siren; CMB $(T(z))$.

Programs/workflows:

None verified or required.

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Tests:

Redshift-distance consistency, SN time dilation, BAO, chronometers, standard sirens, simple tired-light falsification.

5. KEY RESULTS

$1+z = a(t_0)$

, $H = \dot{a}$

$a(t_{em})$

a

Redshift is observed; $(a(t))$, $(H(z))$, (H_0) and acceleration are inferred.

Final result:

Cosmic expansion is CONFIRMED by multiple consistent probes. Late acceleration is strongly supported but does not prove (Λ) or dark energy.

Limitations:

Calibration, (rd) , stellar modelling, peculiar velocities and geometric assumptions.

Contradiction:

“We directly observe space expanding” is too strong.

6. NEGATIVE RESULTS / MIKAMI

Killed:

- Redshift alone proves expansion.
- $(H(z))$ is directly observed.
- BAO gives absolute $(H(z))$ without (rd) .
- Acceleration proves (Λ) .
- Simple no-time-dilation tired light.

No computational failures recorded.

7. MODEL CHANGE

Before: Expansion evidence present.

After: Expansion = CONFIRMED; acceleration = strongly supported; mechanism = OPEN.

Change: Epistemic model strengthened.

8. GITHUB / PROVENANCE

Repository/commits/workflows/programs/artifacts:

NOT RECORDED AT TIME OF Q.

Provenance:

Reconstructed from surviving Q material, manuscript state and primary literature. No identifiers invented.

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9. REPRODUCIBILITY

Re-check the cited probes, separate observations from inferred quantities, and test whether the complete evidence set is consistent with expansion and incompatible with simple no-time-dilation alternatives.

10. CONCLUSION

The observable Universe is very strongly consistent with an expanding large-scale geometry. The exact expansion history and cause of acceleration remain more model-dependent.

11. NEXT-Q HANDOFF

Next Q: Q003

Reason:

Determine exactly what (H_0) is and how different observational chains infer it.

FINAL ANSWER: Expansion CONFIRMED.

Q-JOURNAL: This journal.

MIKAMI: Redshift-only, direct- $(H(z))$, assumption-free BAO, acceleration= (Λ) , simple tired light rejected.

MODEL CHANGE: Expansion promoted to CONFIRMED.

GITHUB: NOT RECORDED AT TIME OF Q.

NEXT-Q: Q003 — compare (H_0) inference chains.

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Q003

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q003

Date/time: 2026-08-26 23:55 CEST

Status: CONFIRMED / SUPPORTED

Question: What is H_0 , how is it inferred, and where do calibration and model dependence enter?

2. STARTING STATE

Model entering Q: Early Bubbleverse Hubble-tension framework.

Inherited:

- Q001: observation $\neq$ inference.
- Q002: expansion is robust; redshift $\neq H_0$.

3. HYPOTHESES

Rejected:

- H_0 is directly observed.
- CMB/local/BAO estimates are equivalent or fully independent.

Surviving:

Different methods provide partially independent inference routes.

4. METHOD & EVIDENCE

Method: Map each route from observable calibration/model $\rightarrow H_0$.

Sources: K-002, K-014, K-019, K-024–K-030.

Programs/workflows: NOT RECORDED AT TIME OF Q.

5. KEY RESULTS

a) t_0

$H_0 = 'a$

Key benchmarks:

- **HODN: (73.50 ± 0.81)**

- CMB-SPA, flat Λ CDM: (67.24 ± 0.35)

- DESI DR2+BBN, flat Λ CDM: (68.51 ± 0.58)

HODN vs CMB-SPA: $\sim (7.1\sigma)$.

HODN vs DESI+BBN: $\sim (5\sigma)$.

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Final result: (H_0) is an inferred parameter, not a raw observable. The tension is between different inference chains.

Limitations: No joint likelihood/covariance reconstruction.

6. NEGATIVE RESULTS / MIKAMI

Killed:

- " (H_0) is directly measured."
- "BAO gives model-free absolute (H_0) ."
- " 7.1σ tension = 7.1σ evidence for new physics."

7. MODEL CHANGE

Before: Planck vs SH0ES shorthand.

After: Multi-chain structure using HODN, CMB-SPA, DESI+BBN and alternative probes.

Change: MODIFIED / STRENGTHENED.

8. GITHUB / PROVENANCE

Repository/commits/workflows/results: NOT RECORDED AT TIME OF Q.

Provenance: Reconstructed from Q003 result, source register and handoff material.

9. REPRODUCIBILITY

Verify listed sources, reconstruct each inference chain, map shared dependencies, and compare benchmarks without double counting.

10. CONCLUSION

Q003 established that the Hubble tension is a conflict among distinct inference chains for the same parameter. Its cause remained unresolved; no new physics was established.

11. NEXT-Q HANDOFF

Next Q: Q004

Question: Is the (H_0) tension robust after known systematics, calibration dependencies, model assumptions and independent probes are treated together?

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Q004

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08 19:17 CEST

1. IDENTITY

Q-ID: Q004

Original date/time: NOT RECORDED

Status: SUPPORTED

Question:

Is the difference between the main H_0 inferences robust when known systematics, calibration dependencies, model assumptions, and independent methods are considered together?

2. STARTING STATE

Model entering Q: Bubbleverse H_0 evidence structure after CASE-003.

Inherited:

- HODN: $73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

- CMB-SPA, flat Λ CDM: 67.24 ± 0.35 .

- **DESI DR2 + BBN: 68.51 ± 0.58 .**

Constraint: H_0 is an inferred parameter, not a raw observable; overlapping evidence must not be double-counted.

3. HYPOTHESES

Tested:

H1: Main H_0 tension survives known systematics.

H2: One dominant local systematic explains it.

H3: One CMB/Planck-specific systematic explains it.

Rejected as sufficient single causes: HST Cepheid crowding alone; Planck pipeline alone; SN-Ia rung alone.

Surviving: residual local/systematic combinations, peculiar velocities/selection, stellar calibration, early-/late-Universe model dependence, sound-horizon physics, multi-effect explanations.

4. METHOD & EVIDENCE

Compared covariance/dependence structure and cross-checks across local distance, CMB, BAO+BBN, TRGB/JAGB, megamasers, strong lenses, and standard sirens.

Primary sources: K-024–K-032; supporting K-002, K-014, K-019.

Programs/workflows: NOT RECORDED AT TIME OF Q.

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Numerical tests: simple Gaussian tension calculations and cross-method consistency assessment.

5. KEY RESULTS

- HODN vs CMB-SPA: $\Delta H_0 = 6.26 \pm 0.88$, 7.1σ . $\approx$
- HODN vs DESI DR2 + BBN: 5.0σ . $\approx$
- CCHP JWST-only TRGB/JAGB differences from HODN are only 2σ before shared $\approx$ covariance and are not independent.
- Independent megamaser/lensing/GW routes remain too imprecise to adjudicate alone.

Final result:

The H_0 discrepancy is a robust Type-C parameter tension, Level 3, but its cause is not established.

Limitations: no $\sim 1\%$ fully independent third H_0 route; residual local calibration structure remains.

Contradictions: C-001 persists; C-002 reduced to internal late-Universe methodological tension.

6. NEGATIVE RESULTS / MIKAMI

Killed as full explanations:

- HST Cepheid crowding alone.
- Planck-specific pipeline alone.
- SN-Ia final rung alone.

Reason: independent/cross-pipeline evidence retains substantial high-versus-low H_0 disagreement.

7. MODEL CHANGE

Before: H_0 discrepancy active but robustness not yet fully established.

After: C-001 = ROBUST ANOMALY / TYPE C / LEVEL 3. C-002 = internal methodological tension.

Change: promoted C-001; constrained C-002 and several explanations.

Reason: multi-method robustness and systematic cross-checks.

8. GITHUB / PROVENANCE

Repository: UNKNOWN / NOT RECORDED

Commits/workflows/programs/artifacts: NOT RECORDED AT TIME OF Q.

Provenance: reconstructed from surviving Q004 result, Motor-14 revision, manuscript integration material, and cited source Ids.

9. REPRODUCIBILITY

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To reproduce: recover cited primary sources; reconstruct H_0 benchmark comparisons; preserve stated model assumptions and evidence overlaps; reproduce Gaussian tension estimates; apply the same independence/systematics criteria.

Missing historical software/Git metadata: NOT RECORDED AT TIME OF Q.

10. CONCLUSION

The Hubble tension survives serious treatment of the major known calibration, covariance, model, and cross-method issues. It is strong enough to require explanation, but not to establish new physics.

11. NEXT-Q HANDOFF

Why: the anomaly is established; the remaining problem is discrimination among explanations.

Next Q: Q005

Next question:

Which remaining systematic and cosmological explanation classes can quantitatively reduce the robust H_0 tension while remaining consistent with HODN, CMB-SPA, DESI DR2 + BBN, internal late-Universe calibration structure, and physically independent H_0 routes?

REQUIRED END-OF-Q OUTPUT

1. FINAL ANSWER: Robust H_0 parameter tension confirmed at Level 3; cause unresolved.
2. Q-JOURNAL: This retrospective reconstruction.
3. MIKAMI GRAVEYARD UPDATE: crowding-only, Planck-only, and SN-rung-only explanations killed as sufficient full causes.
4. MODEL CHANGE SUMMARY: C-001 promoted; C-002 constrained.
5. REPRODUCIBILITY / GITHUB REFERENCES: sources preserved; GitHub metadata not recorded.
6. NEXT-Q HANDOFF: Q005 — discriminate surviving explanation classes.

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Q005

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q005

Date/time: 2026-08-29 23:40 CEST

Status: SUPPORTED

Question: Hvilke af de resterende systematiske og kosmologiske forklaringsklasser kan kvantitativt reducere den robuste H0-spænding, samtidig med at de forbliver forenelige med HODN, CMB-SPA, DESI DR2 + BBN, den interne late-Universe calibration structure og de fysisk alternative H0-routes?

2. STARTING STATE

Model entering Q: Robust H0-anomali, Type C / Level 3; årsag ukendt.

Inherited: HODN 73.50 ± 0.81 ; CMB-SPA 67.24 ± 0.35 ($\sim 7.1\sigma$); DESI DR2+BBN 68.51 ± 0.58 .

Constraints: H0 71.5 cosmology-screen; systematik, calibration og alternative H0-ruter $\geq$ skal respekteres; common-backend $\neq$ paper-native reproduction.

3. HYPOTHESES

Tested:

H1: Late-Universe systematik kan forklare spændingen.

H2: EDE/recombination physics kan hæve H0.

H3: Andre udvidelser (N_{eff} , I_{DE} , P_{MF} , α xiodilaton) kan give fuld løsning.

Rejected: Ordinary peculiar velocities/standard cosmic variance som fuld løsning. Generic late-DE, N_{eff} -only m.fl. stærkt svækket som fulde løsninger.

Surviving: $n=3$ EDE = partial; varying- m_e = stærkeste common-backend partial kandidat; IDE/systematik = partial.

4. METHOD & EVIDENCE

Methods: Litteratursyntese + fælles likelihood-backend + multi-start optimering + deterministic basin/globality recovery.

Primary sources: K-024, K-029, K-030, K-033–K-044; især K-036 ($n=3$ EDE) og K-038 ($n=2$ EDE/varying- m_e).

Programs/workflows: "q005_hpc_v14.py"; "q005_globality_recovery_v14.py";

"github/workflows/q005-hpc-v14.yml"; "github/workflows/q005-hpc-v14-globality-recovery.yml".

Tests: H0 71.5 gate; $\Delta\chi^2/AIC$; restart/globality; deterministic basin recovery. $\geq$

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5. KEY RESULTS

- n=3 high-EDE basin: $H_0=70.8533$, $f_{\text{EDE}}=0.0811$, $\chi^2=548.010$; kun $\Delta\chi^2$ 0.112 fra low-fEDE $\approx$ basin, men under H_0 -gaten.

- n=2: tidligere χ^2 1840–1847 var optimizer trapping; recovered low-fEDE basin $\approx$ $H_0=68.4924$, $\chi^2=548.196$.

- varying-m_e: bedste recovered $H_0=70.1572$, $\Delta\chi^2$ 2.59, ΔAIC 0.59; globality stadig $\approx - \approx -$ unresolved.

Final result: Ingen testet forklaringsklasse er etableret som robust fuld H_0 -løsning; flere giver kun partielle forbedringer.

Limitations: Exact global model ranking unresolved; K-036/K-038 native executable provenance mangler.

Contradictions: HODN vs CMB-SPA og DESI-side spænding består.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: Ordinary V14 restarts fandt ustabile minima; n=2 katastrofebasin var numerisk forkert basin, ikke model-falsifikation.

Graveyard: Ordinary peculiar velocities/cosmic variance som fuld løsning; generic late-DE/ N_{eff} -only som førende fuld løsning; fysisk fortolkning af n=2 χ^2 1847. $\approx$

7. MODEL CHANGE

Before: Flere systematiske/kosmologiske klasser åbne; n=2 numerisk patologisk.

After: Ingen fuld løsning; n=3 og varying-m_e beholdes som partial; n=2 reklassificeret low-fEDE/ Λ CDM-like.

Change: Modelrummet indsnævret og numerisk fortolkning korrigeret.

Reason: Q005 litteratur + V14 runs + deterministic recovery.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commits: "b875ea95d9b6390b8f03dac729eb7f5fe84d3118"; recovery

"7293c7f5db4d7ab72b1aaad38fe15f2639e11d5e"

Runs: #24 "33249784822"; #25 "33252632548"; recovery "33271326245"

Result: "q005_v14_globality_recovery_result.json"

Provenance note: Frozen common backend; recovery parent metadata referenced older run21, preserved as limitation.

9. REPRODUCIBILITY

Checkout recorded commits obtain frozen likelihood/data run listed workflows/configs $\rightarrow$ $\rightarrow$ compare JSON/artifacts apply same H_0 , $\Delta\chi^2$ /AIC og globality gates. $\rightarrow$ $\rightarrow$

10. CONCLUSION

H_0 -spændingen består. Ordinary systematics er utilstrækkelige som fuld forklaring. N=3 EDE og varying-m_e kan flytte H_0 opad, men ingen testet kandidat når robust fuld

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reconciliation. Exact common-backend ranking forbliver numerisk usikker uden at ændre hovedkonklusionen.

11. NEXT-Q HANDOFF

Why: Den vigtigste flaskehals er nu, hvilke likelihood/data-komponenter der stopper de overlevende high-H0 mekanismer.

Next Q: Q006

Next question: Hvilke konkrete CMB-, DESI DR2- og

BBN-likelihood-komponenter/parameterkoblinger leverer den dominerende fitstraf, når $n=3$ EDE og varying- m_e forsøger at nå H_0 71.5? $\geq$

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Ingen testet klasse giver robust fuld løsning; $n=3$ EDE og varying- m_e overlever som partial.

Q-JOURNAL: Ovenstående.

MIKAMI GRAVEYARD: Ordinary PV/cosmic variance som fuld løsning; $n=2$ katastrofebasin som fysisk resultat.

MODEL CHANGE: Modelrummet indsnævret; $n=2$ korrigeret; varying- m_e styrket som partial.

REPRODUCIBILITY: Morfindien/Bubbleverse; commits/runs/workflows ovenfor.

NEXT-Q: Q006 — identificér den dominerende likelihood-barriere for high-H0 modeller.

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Q006

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q006 Date/time: 2026-08-30 CEST Status: CONFIRMED

Question:

Hvilke konkrete datasæt, likelihood-komponenter og parameterkoblinger i de aktuelle CMB + DESI DR2 + BBN-inferenskæder leverer den dominerende fitstraf, når $n=3$ EDE og varying electron mass forsøger at flytte H_0 fra cirka 68–70 til området $H_0 \approx 71.5$?

2. STARTING STATE

Model entering Q: Frozen Q005 common backend; $n=3$ EDE and varying- m_e active/constrained candidates.

Inherited results:

Q005: Λ CDM $H_0=68.494422$, $\chi^2=547.482504$.

Q005: $n=3$ EDE $H_0=70.853292$; varying- m_e $H_0=70.157184$, $\chi^2=544.887898$.

Constraints: Fixed backend/data/priors/bounds/nuisance; H_0 profiles at baseline, 71.5, 72.5, 73.5; components Planck low- ℓ , ACT DR6, DESI DR2, BBN.

3. HYPOTHESES

Tested:

H1: Same likelihood component dominates both mechanisms.

H2: $n=3$ EDE is primarily CMB/ACT limited.

H3: varying- m_e transfers penalty toward DESI/BBN.

Rejected: H1 — component attribution differs by mechanism.

Surviving: H2 supported; H3 supported.

4. METHOD & EVIDENCE

Methods: Fixed- H_0 profile optimization with non-overlapping χ^2 decomposition and parameter tracking.

Primary sources: K-029, K-030, K-036, K-037, K-038, K-039.

Programs/workflows:

q006_likelihood_attribution_v1.py

q006_likelihood_attribution_v1_config.yml

.github/workflows/q006-likelihood-attribution-v1.yml

Tests: 8 required profiles; completeness, component-attribution and final-result gates.

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5. KEY RESULTS

n=3 EDE:

H0=71.5: $\Delta\chi^2=+2.032$, ACT +2.133.

H0=72.5: $\Delta\chi^2=+19.877$, ACT +9.343, DESI +9.213.

H0=73.5: $\Delta\chi^2=+10.972$, ACT +8.960.

ACT dominates 3/3 profiles.

varying- m_e :

H0=71.5: $\Delta\chi^2=+6.811$, DESI +3.394.

H0=72.5: $\Delta\chi^2=+3.421$, BBN +2.906.

H0=73.5: $\Delta\chi^2=+4.752$, BBN +2.842.

Final result: High-H0 barriers are mechanism-specific: EDE is primarily ACT-limited;

varying- m_e is primarily DESI/BBN-limited.

Limitations: Common-backend, not exact K-036/K-038 reproduction; global minima not proven. EDE $\Delta\chi^2$ profile is non-monotonic and remains a preserved numerical irregularity.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: RUN #1 and #2 failed before optimization because rendered YAML

filenames became ...pyaml; technical failures, not physics failures.

Mikami Graveyard:

Killed route: "one universal high-H0 likelihood barrier."

Reason: EDE and varying- m_e show different dominant penalties.

7. MODEL CHANGE

Before: Surviving high-H0 candidates treated broadly as facing a generic cosmological barrier.

After: Model-specific constraint topology.

Change: MODIFIED / CONSTRAINED.

Reason: Q006 likelihood decomposition.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commits: RUN #1 15d60435d6199e71662448fce6a1e8506fc1eac7; RUN #2

85ff40dae59be88636d7542de00025fed1f68449; successful-run commit NOT RECORDED.

Runs: 33278320823; 33278861896; successful run ID NOT RECORDED.

Artifact: q006_likelihood_attribution_v1_result.json

Artifact SHA-256: b7ac0f72aad01be7739560c38faff03ee082c4697fa7c6cdd101e209acf7477

9. REPRODUCIBILITY

Checkout recorded version; obtain frozen Q005 inputs; run Q006 workflow at specified H0 grid; verify component χ^2 outputs and PASS gates; compare against recorded result JSON.

Missing successful-run commit/run ID: NOT RECORDED.

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10. CONCLUSION

Q006 established that reducing the sound horizon does not encounter one universal

observational barrier. $n=3$ EDE primarily accumulates ACT DR6 penalty, whereas varying- m_e shifts the dominant cost toward DESI DR2 and BBN. Neither mechanism is globally falsified.

11. NEXT-Q HANDOFF

Why another Q is required: Q006 located the penalties but did not identify which physical ACT observables/degeneracies create the EDE barrier.

Next Q: Q007

Next question:

Hvilke konkrete fysiske observables og parameterdegeneracies i ACT DR6 gør $n=3$ EDE dyr ved H_0 71.5, og hvorfor kan varying electron mass undgå den samme ACT-straf, mens $\geq$ dens fit cost i stedet flyttes mod DESI DR2 og BBN?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: EDE primarily ACT DR6 barrier; varying- m_e DESI/BBN barrier. — —

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Universal common high- H_0 barrier rejected.

MODEL CHANGE SUMMARY: Generic barrier mechanism-specific constraint topology. —

REPRODUCIBILITY / GITHUB REFERENCES: Morfindien/Bubbleverse; Q006

script/config/workflow; failed-run commits above; successful result JSON/hash preserved.

NEXT-Q HANDOFF: Q007 — determine the physical observables and degeneracies producing the different barrier structures.

BUBBLEVERSE Q JOURNALS

0.42

Q007

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q007

Date/time: 2026-08-30 11:42 CEST

Status: SUPPORTED

Question: Which concrete physical observables and parameter degeneracies in ACT DR6 make $n=3$ EDE costly at $H_0 \geq 71.5$, and why can varying electron mass avoid the same ACT penalty while shifting fit cost toward DESI DR2 and BBN?

2. STARTING STATE

Model entering Q: Frozen Q005/Q006 common backend; $n=3$ EDE + varying-me; global minima not established.

Inherited: Q006 found ACT dominant for high- H_0 EDE; varying-me cost shifted mainly to DESI/BBN. Q006 EDE profile was non-monotonic.

Constraints: No reoptimization; preserve exact Q006 points/covariance; no independent TT/TE/EE claim; negative signed contributions allowed.

3. HYPOTHESES

Tested:

H1: One ACT spectrum/region dominates EDE penalty.

H2: Penalty is distributed across correlated spectral structure.

H3: varying-me avoids EDE-like ACT penalty through different recombination degeneracy.

Rejected: Single isolated ACT observable as complete explanation.

Surviving: H2 + H3 — supported; TE/intermediate- ℓ concentration within distributed covariance-coupled penalty.

4. METHOD & EVIDENCE

Methods: Exact covariance attribution $c_i = d_i$ (C

-1d)i; grouped TT/TE/EE and descriptive

multipole bands; EDE vs varying-me control.

Sources: K-036 Poulin et al. arXiv:2505.08051; K-037 ACT DR6 extended models

arXiv:2503.14454; K-038 Toda & Seto, JCAP 02 (2026) 019, arXiv:2508.09025; K-045 Seto &

Toda arXiv:2206.13209.

Programs/workflows: "q007_act_observable_attribution_v2.py"; "q007-act-observable-attribution-v2.yml"; frozen ACT DR6-lite commit "0e0cd2c...".

Tests: 8/8 frozen profiles; covariance closure; completeness; interpretation/final gates.

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5. KEY RESULTS

EDE $\Delta\chi^2_{\text{ACT}}$: 71.5: +2.135; 72.5: +9.348; 73.5: +8.966.

TE attribution: +1.667, +5.914, +6.765 respectively — largest positive spectrum contribution at all three points.

Varying-me $\Delta\chi^2_{\text{ACT}}$: 71.5: +1.042; 72.5: 0.227; 73.5: +0.100. –

Final result: High- H_0 EDE pays a distributed ACT spectral penalty concentrated in TE/intermediate- ℓ ; varying-me largely avoids this ACT cost and transfers constraint pressure toward DESI/BBN.

Limitations: No global-minimum proof; ACT_LOW/MID/HIGH are descriptive; Q006 non-monotonic EDE profile remains unresolved.

Contradiction: None fundamental; numerical profile irregularity preserved.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: Early V1 runs failed because fixed H_0 was incorrectly required from serialized "bestfit"; later ACT-data download timeout. Both technical, not physical failures.

Mikami Graveyard additions:

- "EE alone dominates the ACT EDE penalty" — rejected.
- "Sound-horizon reduction alone is sufficient" — rejected.
- "All early-Universe solutions face the same barrier" — rejected as too crude.

7. MODEL CHANGE

Before: Generic early-Universe/CMB barrier.

After: Model-specific constraint-transfer topology.

Change: MODIFIED / PROMOTED.

Reason: EDE and varying-me reach similar high H_0 /rd r a g through sharply different likelihood-penalty structures.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Commits: Successful Q007 execution commit NOT RECORDED; frozen ACT

"0e0cd2c703c62a0e980470b572602233b27750e1"; CLASS_EDE

"5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Workflow: "q007-act-observable-attribution-v2.yml"

Program: "q007_act_observable_attribution_v2.py"

Output: "q007_act_observable_attribution_v1_result.json"

Provenance: Exact Q006 fixed- H_0 points, no reoptimization; 8/8 complete; closure/completeness/final gates PASS.

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9. REPRODUCIBILITY

Checkout frozen versions obtain Q006 aggregate + ACT DR6-lite run Q007 V2 on eight — —

T C

fixed profiles verify — $\Sigma c_i = d$

—1d and 8/8 completion compare result JSON and apply —
recorded gates.

10. CONCLUSION

Q007 supports a model-specific constraint-transfer picture: $n=3$ EDE distorts CMB spectral

structure enough to incur a strong ACT penalty, especially through TE, whereas varying-me preserves ACT compatibility more effectively but shifts the cost toward DESI/BBN. This does not establish either model as a complete Hubble-tension solution.

11. NEXT-Q HANDOFF

Why: The EDE fixed- H_0 profile remains strongly non-monotonic and globality is undocumented.

Next Q: Q008

Question: Is the Q006 non-monotonic $n=3$ EDE profile an optimizer/basin/global-minimization artefact, or a reproducible property of the frozen likelihood?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: EDE has a distributed ACT spectral penalty dominated positively by TE; varying-me largely avoids it and transfers cost toward DESI/BBN.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Generic universal barrier, sound-horizon-only sufficiency, and EE-dominance rejected.

MODEL CHANGE SUMMARY: Generic barrier model-specific constraint-transfer topology. —

REPRODUCIBILITY / GITHUB: "Morfindien/Bubbleverse"; Q007 V2 workflow/program;
frozen ACT/CLASS commits; successful run commit not recorded.

NEXT-Q HANDOFF: Q008 — determine whether the non-monotonic EDE profile is
numerical/globality artefact or real frozen-likelihood structure.

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Q008

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q008

Date/time: 2026-08-30 / exact completion time NOT RECORDED

Status: SUPPORTED

Question: Is the non-monotonic Q006 (n=3) EDE fixed-(H 0) profile primarily optimizer/basin/incomplete-minimization structure, or a reproducible property of the frozen common-backend likelihood?

2. STARTING STATE

Model entering Q: Q005 V14 frozen common backend; (n=3) EDE active/constrained, not established new physics.

Inherited: Q006 profile ($\Delta \chi$

$2=(+2.03,+19.88,+10.97)$) at (H 0=(71.5,72.5,73.5)); Q007 ACT

attribution applied to those frozen points.

Constraints: Preserve backend, priors, historical points and fixed-(H 0) targets; BEST

OBSERVED $\neq$ GLOBAL MINIMUM.

3. HYPOTHESES

Tested:

H1 optimizer/suboptimal basin; H2 incomplete minimization; H3 reproducible multimodal likelihood; H4 new physical structure.

Rejected: Strong H3 and H4 as explanations of the original (+19.88) peak.

Surviving: H1 strongly supported; H2 supported; residual multi-basin/globality uncertainty remains.

4. METHOD & EVIDENCE

Method: Four start families at (H 0=70.853292,71.5,72.5,73.5); BOBYQA Stage-1 plus independent Stage-2 refinement; basin replication and boundary diagnostics.

Sources/evidence: Q005/Q006/Q007 internal results; K-036 Poulin et al.; ACT DR6 / DESI DR2 context.

Programs/workflows: "q008_globality_validation_v1.py";

"q008_globality_validation_v1_config.yml"; "q008-globality-validation-v1.yml".

Tests: completeness, valid-start quorum, refinement, basin replication, boundaries, material profile change.

5. KEY RESULTS

Best observed (χ

2):

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70.853292: 548.010401 (Q005 lower basin retained)

71.5: 547.154321

72.5: 548.854655

73.5: 549.105584

Relative best-known envelope: 0, 0.856, +0.844, +1.095. –

The old Q006 (H 0=72.5) value changed from +19.877 +0.844, demonstrating a material – basin/minimization artifact.

Limitations: Global minima not established; basin replication unresolved at 70.85, 72.5 and 73.5; boundary warnings at several targets.

Contradiction: Q006's apparent sharp non-monotonic barrier was not numerically robust.

6. NEGATIVE RESULTS / MIKAMI

Failed route: Treating the Q006 (+19.88) peak as robust likelihood structure.

Mikami Graveyard:

Killed: Strong physical interpretation of the original Q006 non-monotonic peak.

Reason: Same frozen backend produced dramatically lower valid solutions.

7. MODEL CHANGE

Before: Q006 profile active; Q007 attribution interpreted on those points.

After: Q006 points superseded as active best-observed profile; preserved historically. Q007 remains valid only for its original points.

Change: MODIFIED / CONSTRAINED.

Reason: Multi-start/refinement recovered materially lower minima.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Commit(s): NOT RECORDED / NOT VERIFIED

Workflow: "q008-globality-validation-v1.yml"

Programs: "q008_globality_validation_v1.py"

Result: "q008_globality_validation_v1_result.json"

Run ID: "Q008-EDE-GLOBALITY-VALIDATION-V1"

Provenance: Frozen Q005 V14 backend; Q006 historical points retained; Q008 generated new best-observed basin searches and refinements.

9. REPRODUCIBILITY

Checkout recorded version; obtain frozen Q005 backend/config; run Q008 workflow at recorded fixed (H 0) targets/start families; compare result JSON; apply the same refinement, replication and material-change gates. Missing commit/run metadata: NOT RECORDED /

NOT VERIFIED.

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10. CONCLUSION

Q008 established that the original Q006 (n=3) EDE non-monotonic profile—especially the (+19.88) penalty at ($H_0=72.5$)—was not robust. Optimizer basin structure/incomplete minimization materially produced the apparent barrier. The corrected values are best observed, not documented global minima.

11. NEXT-Q HANDOFF

Why: Q007's ACT TT/TE/EE attribution was evaluated at superseded Q006 basins.

Next Q: Q009

Question: At the corrected best-observed (n=3) EDE fixed-(H_0) minima, which ACT DR6 TT/TE/EE and multipole regions dominate the residual likelihood cost, and does Q007's TE/intermediate-(ℓ) attribution survive?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Original Q006 non-monotonic profile was materially optimizer/basin-driven; no global-minimum claim.

Q-JOURNAL: This reconstruction.

MIKAMI GRAVEYARD UPDATE: Strong physical interpretation of Q006 (+19.88) peak killed.

MODEL CHANGE SUMMARY: Q006 active profile superseded; Q007 retained only for historical points.

REPRODUCIBILITY / GITHUB: "Morfindien/Bubbleverse"; Q008

program/config/workflow/result listed above; missing commit metadata not invented.

NEXT-Q HANDOFF: Q009 — repeat observable attribution at corrected best-observed minima.

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Q009

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q009

Date/time: 2026-08-30 CEST

Status: CONFIRMED

Question: At the corrected best-observed n=3 EDE fixed-H0 minima, which ACT DR6 TT/TE/EE observables and multipole regions dominate residual likelihood cost, and does

Q007's TE/intermediate- ℓ attribution persist?

2. STARTING STATE

Model entering Q: n=3 EDE; Q005 V14 frozen common backend; Q008 best-observed basins; globality unresolved.

Inherited: Q007 found TE/ACT_MID dominance at old Q006 basins. Q008 superseded those minima with $\chi^2 = 547.154321, 548.854655, 549.105584$ at $H_0 = 71.5, 72.5, 73.5$.

Constraint: Preserve Q005 baseline $H_0=70.853292$, $\chi^2=548.010401$; no reoptimization/new priors/data/bounds.

3. HYPOTHESES

H1: TE/ACT_MID topology persists.

H2: Attribution changes at corrected basins.

H3: Change is target/basin dependent.

Rejected: H1 — failed at all three targets.

Surviving: H2/H3 — supported.

4. METHOD & EVIDENCE

Fixed-point ACT DR6-lite reevaluation using full covariance:

$-1d \)i, \Sigma$

$C_i = di(C$

$ci = \chi$

2.

i

Bands: LOW 600–1499.999; MID 1500–2999.999; HIGH 3000–8500.

Evidence: INTERNAL-Q005, Q007, Q008, Q009; ACT DR6-lite commit "0e0cd2c703c62a0e980470b572602233b27750e1".

Final program: "q009_act_corrected_minima_attribution_v8.py"

Workflow: Bubbleverse Q009 ACT Corrected Minima Attribution V8.

5. KEY RESULTS

$H_0=71.5$: $\Delta\chi^2_{\text{ACT}} = 2.5540$; dominant TT / ACT_LOW. –

$H_0=72.5$: $\Delta\chi^2_{\text{ACT}} = +0.8768$; dominant TT / ACT_LOW.

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$H_0=73.5$: $\Delta\chi^2_{\text{ACT}} = +0.1088$; dominant EE / ACT_MID.

Historical Q007: TE / ACT_MID at all three.

Final result: Q007's TE/intermediate- ℓ topology does not persist at corrected Q008 basins; ACT attribution is basin dependent.

Limitation: Q008 GLOBALITY_GATE remains UNRESOLVED; results apply to best-observed, not proven-global minima.

Contradictions: None fundamental; Q007 remains valid for its original basins.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: V1-V7 failed technically through parent-schema, stale-file, expired-artifact, baseline-provenance and "bestfit"/"fixed_sampled_parameters" integration errors. No scientific failure implied.

Mikami Graveyard:

- Basin-independent TE/ACT_MID high- H_0 EDE barrier — KILLED.
- Q006 high- H_0 minima as active best minima — SUPERSEDED by Q008.

7. MODEL CHANGE

Before: High- H_0 $n=3$ EDE appeared associated with TE/intermediate- ℓ ACT penalty.

After: That topology is historical/basin-conditional; corrected basins show strongly reorganized and much smaller ACT cost.

Change: MODIFIED / CONSTRAINED.

Reason: Q009 V8 fixed-point attribution.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Inspection commit: "a8d1cc9687e5df757f4794224d83ed34df03846"

Program: "q009_act_corrected_minima_attribution_v8.py"

Config: "q009_act_corrected_minima_attribution_v8_config.yml"

Workflow: "q009-act-corrected-minima-attribution-v8.yml"

Run ID: "Q009-ACT-CORRECTED-MINIMA-ATTRIBUTION-V8"

GitHub execution run ID: NOT RECORDED

Result: "q009_act_corrected_minima_attribution_v8_result.json"

9. REPRODUCIBILITY

Checkout recorded versions obtain Q005/Q007/Q008 preserved inputs run V8 → → workflow at $H_0=71.5/72.5/73.5$ verify covariance closure and outputs compare → → dominant spectrum/band against historical Q007.

10. CONCLUSION

The previously inferred TE/intermediate- ℓ ACT barrier was not robust to improved basin selection. At the corrected best-observed EDE points, both the magnitude and location of

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ACT likelihood cost change substantially. This does not establish EDE as a solution because globality remains unresolved.

11. NEXT-Q HANDOFF

Why: With the old ACT barrier removed, the remaining total high-H0 likelihood cost must be localized.

Next Q: Q010

Question: At the corrected best-observed $n=3$ EDE fixed-H0 basins, which non-ACT likelihood components and parameter trade-offs dominate the change in total scientific χ^2 relative to the Q005 baseline?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Q007 TE/intermediate- ℓ topology does not persist; corrected ACT attribution is basin dependent.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Basin-independent TE/ACT_MID barrier killed.

MODEL CHANGE SUMMARY: Q007 interpretation narrowed to historical basins.

REPRODUCIBILITY / GITHUB: V8 files/workflow and commits above.

NEXT-Q HANDOFF: Q010 — locate the remaining likelihood cost.

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Q010

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-09

1. IDENTITY

Q-ID: Q010

Date/time: 2026-08-30 CEST

Status: CONFIRMED

Question: At the corrected Q008 best-observed $n=3$ EDE fixed- H_0 basins, which non-ACT likelihood components and parameter trade-offs dominate the change in total scientific χ^2 relative to Q005 baseline, and where does the remaining high- H_0 cost move when Q007's ACT barrier does not persist?

2. STARTING STATE

Model entering Q: $n=3$ EDE; frozen Q005 V14 backend; Q008 best-observed basins; Q009 corrected ACT attribution.

Inherited:

- Q005 baseline: $H_0=70.853292$, $\chi^2=548.010401$.
- Q008: $\chi^2=547.154321$, 548.854655 , 549.105584 at $H_0=71.5$, 72.5 , 73.5 .
- Q009: ACT cost strongly reorganized; no universal TE/ACT_MID barrier.

Constraint: Fixed-point attribution only; no reoptimization/new data/priors.

GLOBALITY_GATE remains UNRESOLVED.

3. HYPOTHESES

H1: One non-ACT likelihood replaces ACT as universal high- H_0 barrier.

H2: Cost is distributed across likelihoods.

H3: Cost depends on basin and parameter trade-offs.

Rejected: H1 — no single replacement component persists.

Surviving: H2/H3 — confirmed within tested fixed points.

4. METHOD & EVIDENCE

Method: Re-evaluate serialized Q008 vectors on frozen Planck low- ℓ + ACT DR6 + DESI DR2 + BBN likelihood; compare component deltas and parameter shifts.

Evidence: INTERNAL-Q005, Q007, Q008, Q009, Q010-V3.

Programs/workflows:

"q010_nonact_likelihood_attribution_v3.py"

"q010_nonact_likelihood_attribution_v3_config.yml"

"q010-nonact-likelihood-attribution-v3.yml"

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Tests: Q007 ACT-baseline cross-check; Q009 ACT-delta cross-check; component closure; finite-result/final-result gates.

5. KEY RESULTS

Serialized attribution baseline: Planck 390.5709; ACT 159.1372; DESI 11.0431; BBN 11.7761; total 548.9751. Optimizer-native Q005 total remains 548.010401; +0.9647 – roundtrip difference is diagnostic only.

H0=71.5: Planck +1.1417; ACT 2.5540; DESI 0.6380; BBN +0.3475; non-ACT +0.8512. – –

H0=72.5: Planck 0.5307; ACT +0.8768; DESI 0.4580; BBN +0.1231; non-ACT 0.8656. – – –

H0=73.5: Planck +0.1830; ACT +0.1088; DESI 0.5420; BBN +0.5011; non-ACT +0.1421. –

Major parameter reorganization with increasing H0: feDE, n_s, omega_cdm, thetai_scf, log10z_c, logA.

Final result: The old ACT barrier is not replaced by one universal non-ACT barrier; high-H0 constraint geometry is distributed, basin-dependent, and tied to multidimensional parameter trade-offs.

Limitations: Q008 points are best-observed, not proven global minima; external viability not tested.

Contradictions: None fundamental.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts:

- V1: Q008 schema/H0 ingestion error.

- V2: invalid mandatory gate requiring serialized vectors to reproduce optimizer-native χ^2 .

- **V3: PASS.**

Mikami Graveyard:

- “DESI becomes the universal replacement barrier” — KILLED.

- “One invariant likelihood component controls high-H0 EDE” — KILLED for tested basins.

7. MODEL CHANGE

Before: Q007 suggested a dominant ACT TE/intermediate- ℓ bottleneck.

After: Constraint cost is distributed and changes with basin/parameter trajectory.

Change: MODIFIED / CONSTRAINED.

Reason: Q009 + Q010-V3 component attribution.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commit(s): NOT RECORDED IN FINAL HANDOFF

Workflow: “q010-nonact-likelihood-attribution-v3.yml”

Program: “q010_nonact_likelihood_attribution_v3.py”

Config: “q010_nonact_likelihood_attribution_v3_config.yml”

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Run ID: "Q010-NONACT-LIKELIHOOD-ATTRIBUTION-V3"

Result: "q010_nonact_likelihood_attribution_v3_result.json"

Provenance: Q005 frozen backend + Q007 serialized baseline + Q008 corrected vectors +

Q009 ACT cross-checks V3 fixed-vector full-likelihood attribution. —

9. REPRODUCIBILITY

1. Checkout recorded repository version.
2. Obtain Q005/Q007/Q008/Q009 preserved inputs.
3. Run Q010 V3 workflow.
4. Verify ACT cross-checks, closure and final JSON.
5. Compare component deltas and parameter shifts using serialized attribution basis only.

10. CONCLUSION

No single likelihood inherits the high-H0 penalty after the historical ACT barrier disappears. Planck dominates the positive non-ACT cost at 71.5, non-ACT improves the fit at 72.5, and BBN is the largest positive non-ACT term at 73.5, while DESI improves at all three points. The relevant structure is therefore a basin-dependent multidimensional constraint geometry.

11. NEXT-Q HANDOFF

Why another Q is required: The scientific importance of Q008–Q010 depends on whether the corrected basins are robust/global representatives of the frozen likelihood.

Next Q: Q011

Next question: Are the corrected Q008 best-observed $n=3$ EDE fixed-H0 basins at $H_0=71.5$, 72.5 and 73.5 global or sufficiently robust minima under the frozen Q005 V14 likelihood?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: No universal replacement barrier exists; high-H0 cost is distributed and basin/parameter dependent.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Universal DESI/replacement-barrier interpretation killed.

MODEL CHANGE SUMMARY: High-H0 EDE constraint picture changed from single-barrier to multidimensional geometry.

REPRODUCIBILITY / GITHUB REFERENCES: Morfindien/Bubbleverse; Q010 V3 program/config/workflow/result above.

NEXT-Q HANDOFF: Q011 — resolve basin robustness/globality.

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Q011

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q011

Date/time: 2026-09-01 04:41 CEST

Status: CONSTRAINED

Question:

Do the corrected high- H_0 $n=3$ early-dark-energy basins at fixed

-1 Mpc

-1 represent reproducible global or sufficiently certified

$H_0=71.5, 72.5, 73.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$

near-global minima of the frozen Bubbleverse likelihood surface, or is the shallow high- H_0 profile still an optimizer-basin artifact?

2. STARTING STATE

Model entering Q: $n=3$ EDE / ede_{n3} ; Q005 V14 frozen backend; active but conditional.

Inherited:

- Q008 best-observed: χ^2

$2=547.154321, 548.854655, 549.105584$ at $H_0=71.5, 72.5, 73.5$.

- Q008 did not prove globality; optimizer-basin sensitivity remained unresolved.

Constraints: Same datasets, likelihoods, priors, bounds and model; no new physics/configuration changes.

3. HYPOTHESES

Tested:

H1: Q008 minima are reproducible/global or strongly near-global.

H2: Lower competing basins remain.

H3: High- H_0 profile is materially optimizer-basin sensitive.

Rejected: H1 — certification criteria not achieved.

Surviving: H2/H3 — supported; materially lower basins found.

4. METHOD & EVIDENCE

Method: Finite multi-start optimization + refinement at four fixed- H_0 targets on frozen

Q005 V14 likelihood.

Evidence: INTERNAL-Q005; INTERNAL-Q008; R-Q011-EDE-GLOBALITY-CERTIFICATION-004.

Workflow/program:

Bubbleverse Q011 EDE Globality Certification V4; "q011_ede_globality_certification_v4.py".

Tests: Multi-start basin search; refinement; completeness/finite-result/globality-certification gates.

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5. KEY RESULTS

- H 0=71.5:547.154321–546.364260, $\Delta \chi$

2=-0.790061 materially superseded. —

- **H 0=72.5:548.854655–548.624632, $\Delta=-0.230023$.**

- **H 0=73.5:549.105584–548.766646, $\Delta=-0.338938$.**

- H 0=70.853292: search reached χ

2=546.679939.

Final result: High-H 0 minima are BEST-OBSERVED MINIMUM ONLY; globality and strong near-global certification were not established.

Limitations: Finite optimization cannot prove mathematical globality; same underlying frozen likelihood means this is not independent observational evidence.

Contradictions: Q008 H 0=71.5 active minimum contradicted by materially lower Q011 basin.

6. NEGATIVE RESULTS / MIKAMI

Failed route: Promotion of Q008 high-H 0 minima to global/strongly near-global status.

Mikami Graveyard:

Q011-GLOBAL-CERTIFICATION — REJECTED.

Reason: certification gate failed; materially lower competing basin found.

7. MODEL CHANGE

Before: Q008 H 0=71.5, χ

2=547.154321 active best-observed.

After: Q011 H 0=71.5, χ

2=546.364260 active best-observed.

Change: MODIFIED / CONSTRAINED.

Reason: $\Delta \chi$

2=-0.790061, exceeding material threshold 0.50; PAT-001 strengthened.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commit: "e113fb846948f424b865d1303fba8b5988ec9133"

GitHub run: "33363067156"

Workflow: Bubbleverse Q011 EDE Globality Certification V4

Program: "q011_edg_globality_certification_v4.py"

Program SHA256:

"53dc99d00d220671fe16d17136ef626a7db454711120f5861a6677b3879c57e6"

Config: "q011_edg_globality_certification_v4_config.yml"

Config SHA256:

"37baaee48fd4f2bf70807dd2c6baab6e8a138725f8b70987773d4ddad5b8467b"

Parent: Q008 run "33307023434".

9. REPRODUCIBILITY

Checkout recorded commit restore Q005 V14 backend/data run Q011 V4 workflow at -- --

recorded fixed-H 0 targets compare generated minima/gates with R-Q011-EDE- —

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GLOBALITY-CERTIFICATION-004 apply material-improvement threshold $\rightarrow$ 0.50 and certification criteria.

10. CONCLUSION

Q011 did not establish global minima. It demonstrated that the high- H_0 EDE likelihood

profile remains optimizer-basin sensitive, with the $H_0=71.5$ Q008 minimum materially superseded. EDE remains active but conditional; physical viability is unchanged.

11. NEXT-Q HANDOFF

Next Q: Q012

Reason: Q009/Q010 attribution at $H_0=71.5$ used the now-superseded Q008 point.

$2=546.364260$ best-observed basin, how is the

Next question: At the new $H_0=71.5$, χ

likelihood change distributed between ACT DR6 and Planck low- ℓ , DESI DR2 and BBN, and do the previous attribution conclusions survive?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: No; global/strong near-global certification was not achieved.

Q-JOURNAL: This journal.

MIKAMI GRAVEYARD UPDATE: Q011 global-certification claim rejected.

MODEL CHANGE SUMMARY: Q008 $H_0=71.5$ minimum superseded; PAT-001 strengthened.

REPRODUCIBILITY / GITHUB: Commit "e113fb...9133"; run "33363067156"; Q011 V4 program/config.

NEXT-Q HANDOFF: Q012 — recompute likelihood attribution at the new $H_0=71.5$ basin.

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Q012

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q012

Date/time: 2026-09-01 CEST

Status: CONFIRMED

Question: At the new Q011 best-observed ($n=3$) EDE minimum at fixed -1 M pc

($H_0=71.5$ km s

$^{-1}$), how is the total (χ

2) change divided between ACT DR6 and non-

ACT likelihoods, and does the previous Q009/Q010 attribution survive?

2. STARTING STATE

Model: MOD-EDE-N3 / frozen Q005 V14 / ACTIVE-CONSTRAINED.

Inherited: Q011: ($H_0=71.5$, χ

$^2=546.364260$), BEST-OBSERVED ONLY. Q009/Q010 contained

attribution for the superseded Q008 vector.

Constraint: Fixed-vector evaluation only; no reoptimization, new priors, bounds, model, or datasets.

3. HYPOTHESES

H1: Previous attribution survives qualitatively.

H2: New basin materially changes attribution.

H3: A broad likelihood sign reverses.

Rejected: H3 — no broad sign reversal.

Surviving: H1 + H2 — qualitative structure survives, quantitative attribution changes.

4. METHOD & EVIDENCE

Method: Exact Q011 vector reevaluated on common serialized attribution basis; ACT full-covariance decomposition plus Planck low- (ℓ) , DESI DR2 and BBN components.

Sources: INTERNAL-Q005; INTERNAL-Q009; INTERNAL-Q010-V4; R-Q011-EDE-GLOBALITY-

CERTIFICATION-004.

Program: "q012_likelihood_attribution_v1.py"

Workflow: Q012 EDE Likelihood Attribution V1

GitHub run: 33465056836.

Tests: frozen-backend, exact-vector, component closure, ACT quadratic closure/cross-check, no-reoptimization, final-result gate — PASS.

5. KEY RESULTS

Serialized attribution:

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- ($\Delta\chi^2_{\text{total}}$

$2 = -2.612490$)

- **ACT DR6: 3.372038** –

- Planck low- (ℓ) : +0.976557

- **DESI DR2: 0.632194** –

- **BBN: +0.415184**

- non-ACT total: +0.759548

ACT: EE 1.888310; TE 1.980919; TT +0.497192. Strongest improving region: ACT_MID; – –

strongest cell: EE $\times$ ACT_MID = 2.788494. –

Final result: Q009/Q010 attribution survives qualitatively, but their active ($H_0=71.5$)

numbers are superseded. The new basin is even more ACT-driven.

Limitations: Q011 remains BEST-OBSERVED ONLY; external EDE viability not tested.

Optimizer-native and serialized (χ

2) tracks must not be mixed.

Contradictions: None.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: None scientifically relevant in Q012.

Graveyard: Old Q009 ACT value 2.554002 and Q010 non-ACT +0.851228 killed as active –

71.5 attribution; preserved historically for Q008 vector.

7. MODEL CHANGE

Before: Active 71.5 attribution unresolved after Q011.

After: Active Q012 attribution established.

Change: MOD-EDE-N3 remains ACTIVE / CONSTRAINED; attribution updated, no physical promotion.

Reason: Fixed-vector Q012 decomposition.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commit: "02afd47ae12daae89e9c894ae7bee7576724e85b"

Program SHA256:

"771439e47c9440818821fbbd9e1c18f276d949e69a4697d65d54818184657a2c"

Config: "q012_likelihood_attribution_v1_config.yml"

Config SHA256:

"c1368581202d8756ca5bd6e6569449b5ba6fb8cf25d59df404274949fcdaac36"

Artifact: "q012-likelihood-attribution-v1-final", ID 9784591538, SHA256

"ad492704891475c5acecb240859bef13636597e54d354b513fdc33a0c8bde651".

9. REPRODUCIBILITY

Checkout recorded commit restore Q005 V14 inputs run Q012 workflow at exact Q011 – –

vector compare component/result outputs require recorded PASS gates and closures. – –

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10. CONCLUSION

The improved ($H_0=71.5$) EDE basin is primarily favored by ACT DR6, while Planck low- (ℓ) and BBN oppose it and DESI DR2 mildly favors it. Q012 resolves the internal attribution question but does not establish globality or physical EDE viability.

11. NEXT-Q HANDOFF

Next Q: Q013

Question: When model definitions, parameterizations, priors, dataset overlap and evidence independence are treated explicitly, is the active Q011/Q012 ($n=3$) EDE region at fixed ($H_0=71.5$) compatible with relevant external EDE constraints and independent observational chains, or is it already excluded or substantially weakened?

Reason: Internal attribution is closed; external viability is now the main bottleneck.

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Q009/Q010 qualitative attribution survives; active numerical 71.5 attribution is superseded by Q012 and is more strongly ACT-driven.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Historical Q009/Q010 values removed from active use; preserved as provenance.

MODEL CHANGE SUMMARY: Attribution updated; MOD-EDE-N3 remains constrained, not established.

REPRODUCIBILITY / GITHUB REFERENCES: Commit "02afd47...", run 33465056836, Q012 V1 program/config/artifact above.

NEXT-Q HANDOFF: Q013 — external applicability and viability.

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Q013

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q013

Date/time: 2026-09-01 05:42 CEST

Status: CONSTRAINED

Question: When model definition, parameterization, priors, dataset overlap and evidence independence are treated explicitly, is the active Q011/Q012 (n=3) EDE region at fixed ($H_0=71.5$) compatible with relevant external constraints, or already excluded/materially weakened?

2. STARTING STATE

Model entering Q: MOD-EDE-N3 — ACTIVE BUT CONDITIONAL.

Inherited: Q011: ($H_0=71.5, f_{EDE}=0.096582859, \chi^2=546.364260$), BEST-OBSERVED ONLY.

Q012: ACT-driven improvement, ($\Delta\chi^2_{ACT}=-3.372038$).

Constraint: Same-model comparisons and dataset independence must be explicit; no cross-model exclusion.

3. HYPOTHESES

H1: External evidence permits the Q011 region.

H2: External evidence excludes it.

H3: Evidence is dataset/model dependent and materially weakens but does not universally exclude it.

Rejected: H2 as universal claim; K-038 uses (n=2), not Q011's (n=3).

Surviving: H3 — SUPPORTED.

4. METHOD & EVIDENCE

Method: Same-model literature comparison; model-definition, prior/statistical-method and dataset-overlap audit.

Primary sources:

K-036 — Poulin et al., PRD 113, 063519 (2026), arXiv:2505.08051.

K-038 — Toda & Seto, JCAP 02 (2026) 019, arXiv:2508.09025.

K-039 — Efstathiou, Rosenberg & Poulin, PRL 132, 221002 (2024), arXiv:2311.00524.

K-040 — Khalife et al./SPT-3G, PRD 113, 103546 (2026), arXiv:2507.23355.

Internal: Q005, R-Q011, R-Q012.

Programs/workflows: No new numerical Q013 run; research/evidence synthesis only.

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5. KEY RESULTS

- K-036 uses ($n=3$) and permits roughly ($f E D E=0.09\pm0.03$, $H_0=71.0\pm1.1$), but shares ACT/DESI/Planck data.

- K-038's ($f E D E<0.014$) is ($n=2$) and cannot directly exclude Q011.

- K-039 is same-model ($n=3$); profile limits approximately ($f E D E<0.094$, $H_0<70.2$) at (σ^2_{mod}), materially pressuring Q011.

- K-040 SPT-only allows ($f E D E<0.12$) but does not independently prefer ($H_0\approx71.5$).

Final result: Q011's ($n=3$) high- (H_0) region is not universally excluded, but materially externally weakened and not independently certified.

Limitations: No direct common-likelihood ($\Delta\chi^2$)

2) evaluation of the exact Q011 region; K-039

limits are separate 1D profiles.

Contradiction: Previous K-036/K-038 same-model conflict dissolved.

6. NEGATIVE RESULTS / MIKAMI

Killed: "K-038 directly excludes Q011 ($n=3$). — wrong model mapping.

Killed: "K-036 independently confirms Q011." — substantial dataset overlap.

Killed: "External evidence has already ruled out all high- (H_0) ($n=3$) EDE." — too strong.

7. MODEL CHANGE

Before: MOD-EDE-N3 — ACTIVE BUT CONDITIONAL.

After: ACTIVE BUT MATERIALLY EXTERNALLY WEAKENED — NOT EXTERNALLY

CERTIFIED.

Change: CONSTRAINED.

Reason: Same- $(n=3)$ Planck NPIPE resistance plus lack of independent SPT high- (H_0) replication.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse.

Q013 commit/workflow/program: NOT RECORDED AT TIME OF Q.

Inherited Q011: commit "e113fb846948f424b865d1303fba8b5988ec9133", run "33363067156".

Inherited Q012: commit "02afd47ae12daae89e9c894ae7bee7576724e85b", run "33465056836".

Provenance: External primary literature compared against frozen Q005/Q011/Q012 model state.

9. REPRODUCIBILITY

Reproduce by retrieving K-036/K-038/K-039/K-040, confirming (n), datasets and statistical treatment; retrieve Q011/Q012 recorded states; compare model compatibility and overlap

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using the same acceptance rule: same-model evidence may constrain Q011, overlapping evidence is not independent replication.

10. CONCLUSION

The active (n=3) EDE basin survives as a candidate, but its physical interpretation is substantially weaker than its internal ACT-driven fit alone suggests. Planck NPIPE provides genuine same-model pressure, while SPT-only does not independently reproduce the high- (H_0) preference.

11. NEXT-Q HANDOFF

Next Q: Q014.

Why: Direct external likelihood penalty remains unmeasured.

Next question: For the same (n=3) EDE model, what likelihood penalty does the $(H_0=71.5)$ region receive in Planck NPIPE and SPT-3G D1 when each chain is correctly reprofiled and overlapping data are not double counted?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Not universally excluded; materially externally weakened and not independently certified.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: K-038 direct-exclusion and K-036 independent-confirmation interpretations killed.

MODEL CHANGE SUMMARY: MOD-EDE-N3 materially externally weakened. —

REPRODUCIBILITY / GITHUB: Morfindien/Bubbleverse; Q011/Q012 provenance above; Q013 execution metadata not recorded.

NEXT-Q HANDOFF: Q014 — direct Planck NPIPE/SPT-3G same-model likelihood test.

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Q014

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

Q-ID: Q014 | Completed: 2026-09-02 | Status: SUPPORTED

1. IDENTITY

Question: Når den samme $n=3$ EDE-model anvendes, hvor stor ekstern likelihood-penalty får high- H_0 -regionen omkring det aktive Q011-punkt ved $H_0=71.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ i Planck NPIPE- og SPT-3G D1-kæderne, når hver kædes egne nuisance-parametre og relevante kosmologiske parametre profileres korrekt og datasetoverlap ikke dobbeltregnes?

2. STARTING STATE

Model entering Q: MOD-EDE-N3 — ACTIVE / externally weakened / not externally certified.

Inherited: Q011 high- H_0 best-observed basin at $H_0=71.5$; Q012 ACT-dominated

improvement; Q013 external mapping K-039/K-040.

Frozen constraints: $n=3$ only; Q011 remains BEST-OBSERVED MINIMUM ONLY; chain-native nuisance profiling; no Planck+SPT χ^2 sum; no HODN/SHOES prior; SPT+DESI secondary due overlap.

3. HYPOTHESES

H1: $H_0=71.5$ receives large external penalty across CMB chains.

H2: External viability is dataset/likelihood dependent.

H3: Exact Q011/ACT parameter vector is externally portable.

Rejected: H1 as universal claim; H3.

Surviving: H2 — supported.

4. METHOD & EVIDENCE

Profile-likelihood comparison of each chain's free- H_0 minimum, fixed $H_0=71.5$ reprofile, and exact Q011 shared-physics vector. Multistart stability, monotonicity, finite-result, provenance and fixed free embedding gates required. —

Sources: K-039 Planck NPIPE $n=3$; K-040 SPT-3G D1 $n=3$; K-029 DESI DR2 secondary.

Workflow: Bubbleverse Q014 $n=3$ EDE External Viability V12.

Program: "q014_external_viability_v12.py".

5. KEY RESULTS

Fixed $H_0=71.5$ profile penalties:

Planck NPIPE approximation $\Delta\chi^2=+12.873928$; SPT D1-only $+0.342067$; SPT D1+DESI $+0.157387$.

Exact Q011-vector penalties:

Planck $+101.858574$; SPT-only $+22.112326$; SPT+DESI $+21.511108$.

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Final result: High- H_0 $n=3$ EDE viability is strongly dataset/likelihood dependent; the exact ACT/Q011 vector is not externally portable.

Limits: Planck/SPT are common-backend, not exact paper-native K-039/K-040 reproductions; no proof of mathematical globality.

Contradiction: Planck materially penalizes $H_0=71.5$ while SPT does not.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: V1–V11:

extraction/setup/package/import/nuisance/objective/multistart/wrapper/nesting/embedding failures; none accepted as Q014 physics. V12 supersedes them.

Graveyard:

- Universal large high- H_0 external penalty — REJECTED by SPT.
- Exact Q011-vector as generally externally viable — REJECTED by all tested chains.
- K-038 $n=2$ as direct $n=3$ test — INAPPLICABLE.

7. MODEL CHANGE

Before: MOD-EDE-N3 active but materially externally weakened, not certified.

After: ACTIVE BUT STRONGLY DATASET/LIKELIHOOD DEPENDENT; $H_0=71.5$ survives SPT profiling but is penalized by Planck; exact Q011/ACT vector not externally portable.

Change: CONSTRAINED / REFINED.

8. GITHUB / PROVENANCE

Repo: Morfindien/Bubbleverse

Head: "ab42b1937ca935f498a006ead2c034954ce1d81f"

GitHub run: "33597153303"

Result: "R-Q014-EDE-EXTERNAL-VIABILITY-009"

Artifact: "q014-external-viability-v12-final", ID "9837116589", digest

"sha256:7ac4b9a2ef6f9d64bebaa55cec50fa8cf84cfaba7bc00fe8cd055ee183e65856"

Jobs: 39/39; all mandatory gates PASS.

9. REPRODUCIBILITY

Checkout recorded head; recover frozen Q011 vector and Q014 inputs; run V12 workflow with frozen $n=3$ configuration; verify 39/39 outputs and artifact; apply identical stability, monotonicity, provenance and embedding gates.

10. CONCLUSION

Q014 closes: Planck and SPT do not impose the same penalty on reprofiled high- H_0 $n=3$

EDE. Planck gives substantial resistance, SPT almost none, while all tested external chains strongly reject the exact ACT-optimized Q011 parameter vector.

11. NEXT-Q HANDOFF

Next Q: Q015

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Why: Determine what physically/statistically drives the validated Planck-SPT difference.

Question: Which CMB likelihood components, observables, multipole ranges and chain-native nuisance/calibration directions drive Planck's $\Delta\chi^2 +12.87$ versus SPT's $\Delta\chi^2 +0.34$ at $\approx$

$H_0=71.5?$

REQUIRED END-OF-Q OUTPUT:

FINAL ANSWER: Dataset-/likelihood-dependent viability; exact Q011 vector externally rejected.

Q-JOURNAL: This entry.

MIKAMI: Universal penalty + exact-vector portability killed.

MODEL CHANGE: MOD-EDE-N3 constrained/refined.

REPRODUCIBILITY: Repo/run/artifact above.

NEXT-Q: Q015.

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Q015

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q015

Date/time: 2026-09-02 19:00 CEST

Status: CONFIRMED

Question: At fixed $H_0=71.5$ in the same $n=3$ EDE common backend, which CMB components, observables, multipoles and chain-native calibration directions explain the Planck NPIPE-approximation $\Delta\chi^2=+12.873928$ versus SPT-3G D1-only $+0.342067$?

2. STARTING STATE

Model entering Q: MOD-EDE-N3; frozen `class_edc` backend, commit "5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited: Q011 best-observed $H_0=71.5$ basin; Q012 ACT attribution; Q014 Planck/SPT external profiles and rejection of exact-Q011-vector portability.

Constraints: No cross-chain χ^2

2 summing/comparison; Q011 remains BEST-OBSERVED

MINIMUM ONLY; Planck/SPT endpoints independently reprofiled.

3. HYPOTHESES

Tested:

H1: Planck's $+12.87$ is mainly CMB.

H2: Planck/SPT penalties have similar TT/TE/EE structure.

H3: SPT calibration hides a large penalty.

Rejected: H1, H2, H3.

Surviving: Dataset-/likelihood-specific spectral response; causal mechanism remains OPEN.

4. METHOD & EVIDENCE

Methods: Within-chain decomposition; covariance-aware TT/TE/EE and ℓ -bin attribution; nuisance/calibration endpoint comparison.

Sources: K-036 ACT+DESI; K-039 Planck NPIPE; K-040 SPT-3G D1; SPT D1 spectra arXiv:2506.20707; Q011–Q014 internal results.

Programs/workflows: Q015-CMB-ATTRIBUTION-V3; V2 endpoint jobs + V3 certification/merge.

5. KEY RESULTS

Planck: CMB $+4.541019$, low- z $+5.319582$, constraints $+3.013327$ total $\rightarrow +12.873928$. High- ℓ : TT $+4.837082$, TE $+0.021571$, EE -0.582330 ; strongest band $\ell=500-999$, $+3.452059$. A pl an c k moves $\sim 1.918\sigma$.

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SPT: primary +0.743683, lensing -0.351681, τ -0.044647, Tcal -0.005287 total - +0.342067.

Primary: TE +1.138190, TT -0.557279, EE +0.168448. Tcal moves only $\sim 0.018\sigma$.

Final result: Planck/SPT difference is not one universal EDE CMB penalty; it arises from different chain components and covariance-coupled spectral responses.

Limitations: Attribution $\neq$ causation; branches are common-backend approximations, not exact K-039/K-040 reproductions.

Contradictions: None fundamental.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: V1 failed "KeyError: use_fg_residual_model" — software transport failure.

V2 final merge failed an invalid raw-covariance/serialized-likelihood identity gate;

endpoint science remained valid and was certified unchanged by V3.

Graveyard:

- "Planck CMB = +12.87" — killed.
- Uniform Planck/SPT CMB penalty — killed.
- SPT Tcal as main driver — killed under tested conditions.

7. MODEL CHANGE

Before: C-EDE-002 = broad Planck/SPT high- H_0 discrepancy.

After: Discrepancy localized to different component/TT-TE-EE/ ℓ /calibration structures.

Change: FC-002 strengthened; C-EDE-002 refined; FC-001 remains active.

Reason: Q015 covariance-aware attribution.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

V2 run: "33629190153"

Head: "78a65aae897fca18132a5c0758bccdbbc6a09402"

Result: "R-Q015-EDE-CMB-ATTRIBUTION-003"

Workflow: "Q015-CMB-ATTRIBUTION-V3"

V3 final GitHub run ID: NOT RECORDED

Final archive SHA-256:

"80b774f66c9f0650e2c36f19633a2daa6fbee69d1dbde2a9df91e286520ca8d4"

9. REPRODUCIBILITY

Checkout recorded versions obtain Q014 endpoints/likelihood data reproduce V2 — —

endpoint attribution preserve covariance allocations and bridge terms run V3 — —

certification verify final PASS and recorded outputs. —

10. CONCLUSION

Q015 resolves where the Planck/SPT difference appears, not why it exists. Planck's large

total is partly non-CMB and its high- ℓ response is TT-dominated; SPT's primary response is

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TE-dominated and substantially compensated by TT and lensing. High- H_0 $n=3$ EDE viability is therefore strongly dataset/likelihood dependent.

11. NEXT-Q HANDOFF

Next Q: Q016

Why: The remaining uncertainty is mechanistic.

Question: On matched ACT/Planck/SPT observables and overlapping multipoles, does the differing high- H_0 response reduce to experiment-/likelihood-specific structure, or does a common physical CMB residual survive?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Q015 resolved the Planck/SPT attribution; no universal CMB penalty exists.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Three simplified explanations killed; V1/V2 failures retained as technical provenance.

MODEL CHANGE SUMMARY: C-EDE-002 refined; FC-002 strengthened; FC-001 remains open.

REPRODUCIBILITY / GITHUB: Morfindien/Bubbleverse; V2 run 33629190153; Q015-CMB-ATTRIBUTION-V3; final archive SHA above.

NEXT-Q HANDOFF: Q016 — matched ACT/Planck/SPT spectral-response test.

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Q016

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q016

Date/time: 2026-09-04 06:25 CEST

Status: SUPPORTED

Question: While Planck NPIPE, SPT-3G D1 and ACT DR6 are examined on a deliberately matched $n=3$ EDE surface with comparable CMB observables and overlapping multipole ranges, can the documented difference in high- H_0 response be explained by experiment-/likelihood-specific frequency, foreground, calibration, sky or covariance structure, or does a common residual pattern survive across the experiments?

2. STARTING STATE

Model entering Q: MOD-EDE-N3 — ACTIVE / CONSTRAINED; "mwt5345/class_ed", commit "5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited: Q014 showed strong Planck vs weak SPT high- H_0 response; Q015 found different covariance-aware Planck/SPT attribution patterns; exact Q011 vector already rejected as universal endpoint.

Constraints: TT+TE+EE, $\ell=600-2000$; fixed $H_0=71.5$ vs free profile; no cross-chain χ^2 sum; chain-native nuisance reprofiling.

3. HYPOTHESES

Tested:

H1: response difference is experiment-/likelihood-specific.

H2: one common residual pattern survives across Planck/SPT/ACT.

H3: apparent Planck difference is optimizer/globality artefact.

Rejected: H2 — 0/9 matched residual cells shared the same non-zero sign. H3 — Planck low- H_0 basin reproduced by multistart certification.

Surviving: H1 — supported; exact instrumental sub-cause remains unresolved.

4. METHOD & EVIDENCE

Methods: matched free/fixed profiling; covariance-aware residual decomposition by TT/TE/EE and multipole band; multistart/globality checks.

Sources: K-036, K-039–K-046; Planck NPIPE/CamSpec, SPT-3G D1, ACT DR6.

Programs/workflows: Q016 V16 matched-surface certification; "q016-cmb-mechanism-v4.yml".

Key run: GitHub run "33836015653", SHA "91bc0736a00e2454d4e37324e5d3a1a7fe7813d6".

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5. KEY RESULTS

Planck matched primary penalty: $\Delta\chi^2=+4.67204$, mainly $\ell=600-999$ (+3.22423).

SPT: +0.01465; ACT: +0.01512, both with strong internal cancellations.

Common residual sign: 0/9 cells.

Final result: Q016-A — EXPERIMENT-SPECIFIC LOCALIZATION.

Limitations: No unique frequency/calibration/foreground/covariance cause identified; no combined inter-experiment significance; SPT/ACT full-MF secondary diagnostics technically incomplete.

Contradictions: No fundamental cosmological contradiction identified.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: V1 dependency/bootstrap failure; V2 path/API failure; V3 Planck/ACT API mismatches; SPT full-MF shape mismatch; ACT full-MF missing "mflike". Technical failures only.

Mikami Graveyard additions:

- Common Planck/SPT/ACT residual-direction hypothesis — killed by 0/9 sign concordance.
- Planck optimizer-artifact explanation — killed by V16 multistart reproduction.
- Uniform CMB high- H_0 penalty — rejected.

7. MODEL CHANGE

Before: MOD-EDE-N3 active with unresolved optimizer/dataset-response ambiguity.

After: MOD-EDE-N3 remains active but explicitly strongly dataset-/likelihood-dependent.

Change: constrained / refined.

Reason: certified Planck-SPT-ACT response split without common residual direction.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Commit: "91bc0736a00e2454d4e37324e5d3a1a7fe7813d6"

Workflow: ".github/workflows/q016-cmb-mechanism-v4.yml"

Results: "R-Q016-EDE-MATCHED-CMB-SURFACE-016"; "R-Q016-EDE-CMB-MECHANISM-004"

Artifacts: merged "9923364927"; final gates "9923368121".

Provenance: V16 certified endpoints reused; V4 decomposed matched residual/covariance response without reoptimizing parent endpoints.

9. REPRODUCIBILITY

Checkout recorded commits obtain frozen Planck/SPT/ACT likelihood inputs reproduce — —

V16 free/fixed endpoints run V4 mechanism workflow verify covariance closure and — —

0/9 sign test apply Q016 A/B/C decision rule. —

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10. CONCLUSION

Q016 establishes that the tested high- H_0 $n=3$ EDE response is not one common CMB

residual expressed with different strength. Planck shows a substantial penalty, while SPT and ACT remain nearly indifferent through different cancellation structures. The result localizes the discrepancy to experiment-/likelihood-response structure, not to a proven instrumental systematic or new physics.

11. NEXT-Q HANDOFF

Next Q: Q017

Why: Planck is now the principal unresolved mechanism bottleneck.

Next question: Is Planck's $H_0=71.5$ penalty, especially at $\ell=600-999$, localized to specific full-multifrequency frequency/calibration/foreground/covariance directions, or broadly distributed?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Q016-A — EXPERIMENT-SPECIFIC LOCALIZATION.

Q-JOURNAL: This journal.

MIKAMI GRAVEYARD UPDATE: Common residual, uniform penalty and optimizer-artifact routes rejected.

MODEL CHANGE SUMMARY: MOD-EDE-N3 retained but strengthened as dataset-/likelihood-dependent.

REPRODUCIBILITY / GITHUB: "Morfindien/Bubbleverse"; run "33836015653"; SHA "91bc0736..."; V16 + V4 results above.

NEXT-Q HANDOFF: Q017 — isolate the Planck-specific high- H_0 penalty mechanism.

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Q017

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

Q-ID: Q017 | Date: 2026-09-04 | Status: INCONCLUSIVE

QUESTION: On the certified Q016 $n=3$ EDE Planck NPIPE/CamSpec surface, is the high- H_0 penalty at $H_0=71.5$ —especially the $\ell=600\text{--}999$ contribution—localized to specific Planck TT frequency, calibration, foreground or covariance directions in a full multifrequency likelihood, or does the penalty remain broadly distributed after nuisance-aware reprofiling?

STARTING STATE

Model: MOD-EDE-N3; $n_{\text{scf}}=3$; mwt5345/class_edc commit
5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Inherited: Q016 certified Planck endpoints: FREE $H_0=67.988328967159$; FIXED $H_0=71.5$.

Q016 localized most matched Planck penalty to $\ell=600\text{--}999$; Q015 showed strong

covariance-coupled 143×217 structure.

Constraint: freeze cosmology/endpoints; full-MF Planck nuisance-only reprofiling; no causal-systematic claim.

HYPOTHESES

H1: penalty localizes to one specific Planck direction.

H2: penalty remains broadly distributed.

H3: neither classification meets preregistered thresholds.

Rejected: H1, H2 under Q017 thresholds.

Surviving: H3 — INDETERMINATE / multidirectional response.

METHOD & EVIDENCE

Likelihood: planck_NPIPE_highl_CamSpec.TTTEEE. Nine calibration/foreground nuisances reprofiled; 3 baseline starts/endpoint; 2 starts/lock group; covariance-aware frequency/block decomposition.

Sources: K-039 (arXiv:2311.00524), K-044 (arXiv:2205.10869), K-045 (arXiv:2510.09430), K-046 Cobaya docs; internal Q015/Q016.

Workflow: Bubbleverse Q017 Planck Direction Localization V3; GitHub run 33842355384.

Mandatory stability, closure, nesting and FINAL_RESULT_GATE: PASS.

KEY RESULTS

Full-MF fixed free objective: +17.3979348723; raw high- ℓ χ^2 : +17.2433166263. –

$\ell=600\text{--}999$ TT abs shares: $143\times 217=0.54791$; $143\times 143=0.38458$; $217\times 217=0.06751$. Effective frequency number=2.20913.

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Nuisance lock shares: calibration=0.45469; fg143×217=0.18883; fg217=0.17999; fg143=0.17648. Strongest covariance pair share=0.17533.

Final: penalty survives nuisance reprofiling, but neither specific-localized nor broadly-distributed criteria pass. Classification: INDETERMINATE.

Limit: attribution ≠ causal systematic; Q016 and Q017 objectives are different likelihood constructions.

NEGATIVE RESULTS / MIKAMI

V1: static-gate implementation failure; no scientific result.

V2: workflow path typo

("q005h pcv24r e quir e me nt s.t x t "v s"q005h pcv14r e quir e me nt s.t x t "); scientific execution never started.

Mikami: killed strong claims "143×217 is the cause", "calibration is the cause", and "penalty is demonstrably broadly distributed".

MODEL CHANGE

Before: FC-CMB-002 allowed possible specific Planck frequency/calibration/foreground/covariance localization.

After: FC-CMB-002 remains active only as a multidirectional Planck-specific likelihood/covariance candidate.

Change: CONSTRAINED / MODIFIED.

Reason: no single tested direction crossed localization thresholds.

GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Execution commit: 277bb274e5783831303b542681e8cf87a70d31c2

Backend commit: 5a131c91d657dd9a7c6364cc45b038710f8d0d97

Result: R-Q017-EDE-PLANCK-DIRECTION-LOCALIZATION-003

Artifact: q017-final-v3, ID 9925585139, SHA-256

a45ba197f20595013d390136e00f8fd83f40a70fb9e9e1d83f2048f2d70320db.

REPRODUCIBILITY

Checkout recorded commit obtain frozen Q016 endpoint lock/Planck likelihood run V3 -- workflow with frozen config compare baseline/lock/decomposition/final JSON outputs -- apply unchanged preregistered classification thresholds.

CONCLUSION

The Planck high- H_0 penalty is robust to the tested full-MF nuisance reprofiling, but its $\ell=600-999$ structure is neither sufficiently concentrated in one frequency/nuisance/covariance direction nor sufficiently broad to satisfy the opposite criterion. Q017 therefore closes as INDETERMINATE, not technically unresolved.

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NEXT-Q HANDOFF

Next Q: Q018.

Question: Which explicit likelihood ingredients account for the difference between the

Q016 matched Planck penalty (+5.823856774789) and Q017 full-MF penalty

(+17.397934872301) at the same certified cosmological endpoints?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: INDETERMINATE; strong Planck penalty survives, no unique or broad direction certified.

Q-JOURNAL: this entry.

MIKAMI GRAVEYARD UPDATE: single-direction causal claims rejected.

MODEL CHANGE SUMMARY: FC-CMB-002 constrained to multidirectional candidate.

REPRODUCIBILITY / GITHUB: Morfindien/Bubbleverse; run 33842355384; commit 277bb274...; result R-Q017-EDE-PLANCK-DIRECTION-LOCALIZATION-003.

NEXT-Q HANDOFF: Q018 likelihood-bridge analysis.

BUBBLEVERSE Q JOURNALS

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Q018

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q018

Date/time: 2026-09-04 09:15 CEST

Status: CONFIRMED

Question: When the same certified Q016 n=3 EDE Planck FREE and H 0=71.5 FIXED

endpoints are evaluated through a controlled bridge between Q016 CamSpec-lite and Q017

full-multifrequency CamSpec, what explains the penalty increase from +5.823856774789 to +17.397934872301, and is it one component or several coupled components?

2. STARTING STATE

Model entering Q: MOD-EDE-N3, ACTIVE / CONSTRAINED; nsc f=3; frozen backend

mw5345/class_edecommit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Inherited results:

- Q016: FREE H 0=67.988328967159, FIXED H 0=71.5, matched penalty +5.823856774789.

- Q017: full-MF penalty +17.397934872301, localization INDETERMINATE.

Constraints: same endpoints/model/backend; no cross-chain χ^2 addition; architecture gap not physical χ^2 ; Q017 not reopened.

3. HYPOTHESES

Tested:

- H1: objective/prior semantics causes most of the gap.
- H2: shared Aplanck,calTE,calEEnuisance freedom causes most of the gap.
- H3: one clean additive likelihood component explains the gap.

Rejected:

- H1: native vs likelihood-only differs by only 0.235. $\approx$
- H2/H3: large penalty survives shared3 locking; no valid additive isolation.

Surviving:

- Coupled full-MF likelihood-architecture dependence: SUPPORTED.

4. METHOD & EVIDENCE

Methods: four controlled full-MF bridge surfaces, two endpoints, three multistarts each.

Primary evidence/sources:

- K-039 arXiv:2311.00524
- K-044 arXiv:2205.10869
- K-045 arXiv:2510.09430

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- K-046 Cobaya 3.5.6 documentation/source
- Q016/Q017 internal certified results.

Programs/workflows:

- Q018-PLANCK-LIKELIHOOD-BRIDGE-V3

- Bubbleverse Q018 Planck likelihood bridge V3
- 24 profile jobs.

Tests performed: identity, endpoint, completeness, multistart, finite-result, Q017-native reproduction, non-additivity gates.

5. KEY RESULTS

Important results:

- full_native = 17.395372421241
- full_likelihood_only = 17.160171713960
- foreground6_only = 15.617680433608
- shared3_only = 11.764134763087

Final result: The extra full-MF Planck penalty is produced by coupled likelihood-architecture effects, not one independently separable component.

Uncertainties / limitations:

- shared3_only unstable: spreads 5.28 fixed, 6.03 free. $\approx \approx$
- frequency/covariance/foreground contributions remain inseparable as additive physical terms.

Contradictions discovered: None fundamental.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts:

- V1 failed endpoint-lock schema gate; no scientific compute.
- V2 failed endpoint API/config compatibility; no scientific minimizer.

Mikami Graveyard additions:

- **OBJECTIVE-SEMANTICS-ONLY.**
- **SINGLE CLEAN ADDITIVE COMPONENT.**
- SHARED3-ONLY as sufficient explanation.
- Architecture gap 11.574078... as physical χ^2 component.

7. MODEL CHANGE

Before: FC-CMB-002 = possible multidirectional Planck-specific likelihood/covariance response.

After: FC-CMB-002 = MULTIDIRECTIONAL, LIKELIHOOD-CONSTRUCTION-DEPENDENT

PLANCK RESPONSE.

Change: modified / constrained.

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Reason: Q018 shows the magnitude of the Planck high- H_0 penalty depends materially on likelihood construction.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commit: 1883370cab6bc1bf94c48300015f8306ef1dfe47

Workflow: Bubbleverse Q018 Planck likelihood bridge V3

Program/result ID: R-Q018-EDE-PLANCK-LIKELIHOOD-BRIDGE-003

GitHub Actions run: 33846317737

Result files:

- q018_merged_v3.json

- q018_tests_v3.json

Provenance note: reused certified Q016 endpoints and Q017 full-MF architecture on frozen MOD-EDE-N3 backend.

9. REPRODUCIBILITY

1. Checkout recorded commit.
2. Use frozen backend and Q016 endpoints.
3. Run Q018 V3 workflow with 4 surfaces $\times$ 2 endpoints $\times$ 3 restarts.
4. Compare merged/test JSON outputs.
5. Require mandatory gates PASS and preserve non-additivity rule.

10. CONCLUSION

Q018 confirms that Q016 and Q017 are valid but distinct likelihood objectives. Their

penalty difference must not be interpreted physically as an additive χ^2 term. Native prior/objective semantics is subdominant; the dominant change belongs to the coupled full-multifrequency Planck likelihood architecture.

11. NEXT-Q HANDOFF

Why another Q is required: Q018 tested fixed certified endpoints but did not determine whether likelihood choice shifts the preferred cosmological EDE solution itself.

Next Q: Q019

Next question: Does separately reprofiling MOD-EDE-N3 under CamSpec-lite and full-MF CamSpec materially shift the preferred H_0 and EDE parameters, or mainly alter the high- H_0 penalty?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Coupled likelihood-architecture dependence; no single additive component identified.

Q-JOURNAL: This journal.

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MIKAMI GRAVEYARD UPDATE: objective-semantics-only, shared3-only, single-component, and physical-gap interpretations killed.

MODEL CHANGE SUMMARY: FC-CMB-002 refined to likelihood-construction-dependent multidirectional Planck response.

REPRODUCIBILITY / GITHUB REFERENCES: Morfindien/Bubbleverse; commit 1883370...; run 33846317737; R-Q018-EDE-PLANCK-LIKELIHOOD-BRIDGE-003; q018_merged_v3.json; q018_tests_v3.json.

NEXT-Q HANDOFF: Q019 — test whether likelihood dependence changes the preferred MOD-EDE-N3 cosmology itself.

BUBBLEVERSE Q JOURNALS

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Q019

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q019

Date/time: 2026-09-04 14:35 CEST

Status: SUPPORTED

Question: When MOD-EDE-N3 is independently reprofiled under Q016 matched/CamSpec-lite and Q017 full-multifrequency CamSpec, does likelihood choice materially shift the preferred EDE/ H_0 region, or mainly change the penalty at frozen high- H_0 endpoints?

2. STARTING STATE

Model entering Q: MOD-EDE-N3, n_scf=3; ACTIVE TEST MODEL. Backend

"mwt5345/class_ede", commit "5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited: Q016 matched Planck penalty +5.8238568 at $H_0=71.5$; Q017 full-MF penalty +17.3979349, INDETERMINATE; Q018 likelihood-architecture dependence.

Constraints: Same model/backend; harmonized objective semantics; no cross-chain χ^2 summation; Q018 architecture gap nonphysical; threshold 1.0 operational, not significance.

3. HYPOTHESES

H1: Likelihood choice mainly changes high- H_0 penalty.

H2: Likelihood choice materially shifts the multidimensional preferred region.

H3: Difference is primarily optimizer/globality artifact.

Rejected: H1 as sufficient description; H3 as principal explanation after V3 stability.

Surviving: H2 — SUPPORTED under frozen Q019 criterion.

4. METHOD & EVIDENCE

Methods: Independent lite/full-MF reprofiling; fixed- $H_0=71.5$ profiles; reciprocal cross-profile tests; targeted multistart/globality repair; Q016 endpoint reproduction.

Sources: K-039, K-044, K-045, K-046; Q016–Q018 internal validated chain.

Programs/workflows: "q019_planck_cosmology_reprofile_v1.py"; "q019_globality_v3.py"; ".github/workflows/q019-planck-globality-repair-v3.yml".

Tests: 12/12 V3 continuations; stability, reference reproduction, frozen-classification and interpretation-safety gates.

5. KEY RESULTS

Lite free: $H_0=67.9726651$, fEDE=0.0462741.

Full-MF free: $H_0=68.1061207$, fEDE=0.0439411.

Cross-costs: full-MF optimum lite = +0.8120; lite optimum full-MF = +13.5564. — —

BUBBLEVERSE Q JOURNALS

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Reprofiled $H_0=71.5$ penalties: lite +5.0620; full-MF +15.3993.

V3 best-two spread = 0.05387; 12/12 complete; FINAL_RESULT_GATE=PASS.

Final result: Similar H_0 /fEDE coordinates do not imply equivalent multidimensional preferred regions; likelihood choice materially shifts MOD-EDE-N3 likelihood geometry.

Limitations: No unique physical/systematic cause identified; internal Planck reanalysis, not independent observation.

Contradiction: No fundamental contradiction; earlier “only penalty changes” interpretation rejected.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: V1 remained INDETERMINATE from stability/reference gates; V2 suffered endpoint-vector and PyYAML technical failures.

Graveyard additions:

PLANCK-OPTIMIZER-ARTIFACT — rejected as principal explanation.

“ONLY HIGH- H_0 PENALTY CHANGES” — rejected as sufficient description.

7. MODEL CHANGE

Before: Planck response known to depend on likelihood architecture mainly through penalty differences.

After: Response classified as multidirectional and likelihood-construction-dependent in full parameter geometry.

Change: PRÆCISERET / STRENGTHENED.

Reason: Asymmetric cross-profile costs with validated globality.

8. GITHUB / PROVENANCE

Repository: “Morfindien/Bubbleverse”

Q019 V3 commit: “71238a5dd27bba4c9f399dfaac638814bed7361e”

Run: “33868899612”

Result: “R-Q019-EDE-PLANCK-COSMOLOGY-REPROFILE-003”

Workflow: “github/workflows/q019-planck-globality-repair-v3.yml”

Programs: “q019_globality_v3.py”, “q019_tests_v3.py”

Provenance: Q019 V1 profiles/cross-tests + V3 targeted globality and Q016 reference validation.

9. REPRODUCIBILITY

Checkout recorded commits; obtain frozen Planck/CamSpec inputs; run Q019 workflow with frozen model/objective; compare generated JSON/results; apply unchanged 1.0 cross-cost criterion and mandatory gates.

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10. CONCLUSION

Q019 supports a material likelihood-construction dependence of the preferred

multidimensional MOD-EDE-N3 region. The projected H_0 and $f\text{EDE}$ minima remain close, but full-MF strongly penalizes the lite preferred vector while the reverse cost is small. No Planck systematic or new physics is established.

11. NEXT-Q HANDOFF

Why: The parameter/nuisance direction causing the asymmetry remains unknown.

Next Q: Q020

Question: Which coupled cosmological and full-MF nuisance directions drive the asymmetric Q019 cross-profile geometry?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Likelihood choice materially changes the multidimensional preferred MOD-EDE-N3 region despite similar $H_0/f\text{EDE}$ minima.

Q-JOURNAL: This journal.

MIKAMI GRAVEYARD UPDATE: Optimizer-artifact and “penalty-only” explanations rejected as principal/sufficient explanations.

MODEL CHANGE SUMMARY: Planck response refined to multidirectional, likelihood-construction-dependent parameter geometry.

REPRODUCIBILITY / GITHUB: “Morfindien/Bubbleverse”; commit “71238a5d...”; run “33868899612”; Q019 V3 workflow/result.

NEXT-Q HANDOFF: Q020 — identify the coupled parameter direction producing the asymmetry.

BUBBLEVERSE Q JOURNALS

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Q020

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08 20:13 CEST

1. IDENTITY

Q-ID: Q020

Original completion: 2026-09-04

Status: CONFIRMED

Question: Which multidimensional cosmological/nuisance direction drives the asymmetric

Q019 cross-profile response—lite cost +0.812 versus full-MF cost +13.556—and can it be $\approx$

localized without treating correlated likelihood contributions as independent physical χ^2 components?

2. STARTING STATE

Model: MOD-EDE-N3; nsc f=3; frozen "mwt5345/class_e" commit

"5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited: Q018: coupled likelihood-architecture dependence; Q019: likelihood choice shifts preferred MOD-EDE-N3 region, with preserved asymmetric cross-profile costs.

Constraint: No cross-chain χ^2 summation; Shapley/interactions diagnostic only, not causal/independent physical components.

3. HYPOTHESES

Tested: response dominated by EDE timing, primordial parameters, matter parameters, H0, or distributed coupling.

Rejected: H0-only driver; approximately equal four-block distribution; independent- χ^2 interpretation.

Surviving: Limited multidimensional direction dominated by the primordial block; physical/systematic cause remains unresolved.

4. METHOD & EVIDENCE

Method: 16 hybrid cosmological vertices $\times$ 2 likelihood architectures $\times$ 2 nuisance restarts =

64 profiles. Blocks: EDE timing (f E D E, log10 z_c , θ_i), primordial (ns, log A, τ), matter (ω_b, ω_c, d_m),

H0. Native nuisance parameters reprofiled; Shapley and pair interactions calculated.

Sources: K-039 arXiv:2311.00524; K-044 arXiv:2205.10869; K-045 arXiv:2510.09430; K-046

Cobaya 3.5.6 documentation.

5. KEY RESULTS

Full-MF absolute Shapley fractions: primordial 71.58%, EDE timing 17.27%, matter 9.15%,

H0 2.00%. Primordial signed change: +10.5344. Largest pair interaction: matter $\times$ H0

5.7189. –

BUBBLEVERSE Q JOURNALS

0.42

Full-MF geometry span: 14.1286; lite: 0.8632. Historical Q019 costs remain +13.5564 and +0.8120 and are not replaced.

Final result: LIMITED MULTIDIMENSIONAL DIRECTION LOCALIZED.

Limitation: Block-level only; does not establish ns alone, an instrumental systematic, or new physics.

6. NEGATIVE RESULTS / MIKAMI

Failed attempt: Q020 V1 final gate failed because it incorrectly compared a reprofiled geometry span with Q019's historical cross cost using different baselines.

Graveyard: H0-only explanation; additive-independent χ^2 interpretation; V1 final validation semantics.

Important: V1's 64 raw profiles remain valid.

7. MODEL CHANGE

Before: Q019 established likelihood-dependent preferred regions, but the responsible multidimensional direction was unresolved.

After: Full-MF asymmetry is localized primarily to the primordial block; likelihood response is explicitly architecture-dependent and non-additive.

Change: PRÆCISERET / CONSTRAINED.

8. GITHUB / PROVENANCE

Repo: "Morfindien/Bubbleverse"

Q020 V1 run: "33876066510", commit "7ca5fa06418eb8b6df4c1e6c4077e679fbfdc989"

Q020 V2 run: "33878880397", commit "b0c0a8fd89408bac91502e15d42c69820e913ab6"

Result: "R-Q020-EDE-PLANCK-CROSS-PROFILE-DIRECTION-002"

Programs: "q020_direction_localization_v1.py", "q020_finalize_repair_v2.py",
"q020_tests_v2.py"

Workflows: "q020-planck-cross-profile-direction-v1.yml", "q020-cross-profile-gate-repair-v2.yml"

Artifact: "q020-final-v2"; FINAL_RESULT_GATE = PASS.

9. REPRODUCIBILITY

Checkout recorded commits; obtain frozen Q019 endpoints/source lock; run V1 64-profile grid; apply V2 aggregation/tests; verify absolute Q019 cross-objective reproduction, endpoint non-regression, restart stability, history preservation and no gate relaxation.

10. CONCLUSION

Q020 establishes that the large Q019 full-MF cross-profile asymmetry is not primarily an H0 effect. Under the tested four-block decomposition it is dominated by the primordial (ns,log A , τ) direction, while CamSpec-lite shows a much smaller, more distributed response. The attribution is functional likelihood geometry, not causal physics.

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11. NEXT-Q HANDOFF

Next Q: Q021

Why: The parameter direction is localized, but the spectra/frequency/nuisance structure producing it is not.

Question: Which full-MF spectra/frequency combinations and nuisance couplings carry the

Q020 primordial direction, and can they be connected quantitatively to the earlier

ell/frequency response without treating correlated contributions as independent χ^2 ?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Primordial block dominates full-MF response ($\sim 71.58\%$); H0 alone $\sim 2\%$; localization confirmed.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: H0-only driver and additive-independent χ^2 interpretation rejected; V1 gate semantics superseded.

MODEL CHANGE SUMMARY: Q019 likelihood-choice effect refined to a primordial-dominated, architecture-dependent multidimensional geometry.

REPRODUCIBILITY / GITHUB: Runs "33876066510", "33878880397"; commits and files above.

NEXT-Q HANDOFF: Q021 — localize the corresponding spectrum/frequency/nuisance mechanism.

BUBBLEVERSE Q JOURNALS

0.42

Q021

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08 20:20 CEST

1. IDENTITY

Q-ID: Q021

Original completion: 2026-09-04, time not fully recorded

Status: CONFIRMED

Question: Within the frozen $n=3$ early-dark-energy model, what internal structure of the primordial block ($\text{ns}, \log A, \tau, r, \text{eio}$) produces the dominant full-multifrequency Planck cross-profile response identified in Q020, and does that structure remain materially different on matched CamSpec-lite?

2. STARTING STATE

Model: MOD-EDE-N3; nsc f=3; mwt5345/class_edc commit

"5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited: Q019 established asymmetric matched/full-MF likelihood geometry; Q020

localized 71.58% of absolute full-MF Shapley response to primordial block.

Constraint: Same frozen model/likelihood chain; attribution is functional, not independent physical χ

2.

3. HYPOTHESES

Tested: ns-only; $\log A$ -only; τ -only; pair interaction; three-way/distributed coupling; likelihood-construction dependence.

Rejected/constrained: No single parameter or pair robustly dominates. Robust matched-vs-full-MF internal difference not established.

Surviving: PRIMORDIAL THREE-WAY / DISTRIBUTED COUPLING.

4. METHOD & EVIDENCE

Method: 2 likelihood constructions $\times$ 8 primordial vertices $\times$ 3 coherent optimizer restarts = 48 profiles; Möbius/Harsanyi and exact Shapley decomposition.

Sources: K-039 Efstathiou, Rosenberg & Poulin (2024), arXiv:2311.00524; K-044 Rosenberg, Gratton & Efstathiou (2022), arXiv:2205.10869; K-046 Cobaya 3.5.6.

Runs: "Q021-PRIMORDIAL-INTERNAL-STRUCTURE-V1"; repaired by "Q021-PRIMORDIAL-

INTERNAL-STRUCTURE-REPAIR-V2".

5. KEY RESULTS

Full-MF: DISTRIBUTED 3/3; ns largest individual term 3/3, absolute fraction 0.2885–0.4891, never 0.50; $\geq$ ns Shapley 7.03–8.78.

Matched: DISTRIBUTED 3/3; leader changes between $\log A \times \tau$, $\text{ns} \times \log A$, and ns.

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0.42

Full-vs-matched TV range: 0.24068–0.49801 around threshold 0.25 —

SENSITIVE_TO_OPTIMIZER_BASIN.

Final result: Q020 primordial dominance is internally a distributed coupled structure with persistent ns prominence, not an ns-only effect.

Limitation: Full-MF maximum vertex objective spread 7.7236; matched 0.9149. Detailed attribution remains basin-sensitive.

Contradiction: None fundamental.

6. NEGATIVE RESULTS / MIKAMI

Failed V1 interpretation: Process-dependent function hashes falsely broke likelihood identity; best-per-vertex selection produced a non-coherent “Frankenstein” decomposition.

Graveyard: Single-parameter localization; robust single-pair localization; authoritative V1 aggregation.

Preserved: All 48 V1 numerical profiles remain valid.

7. MODEL CHANGE

Before: Primordial block dominant, internal driver unresolved.

After: Primordial response = distributed three-way coupling, ns most persistent individual direction.

Change: MODIFIED / STRENGTHENED.

Reason: Stable distributed classification across all coherent restarts.

8. GITHUB / PROVENANCE

Repo: “Morfindien/Bubbleverse”

Parent V1 Actions run: “33882434170”

V1 head: “03b20081220e9a584458c203fc798cfd39dfd692”

Program: “scripts/q021_finalize_repair_v2.py”

Workflow: “.github/workflows/q021_primordial_internal_structure_repair_v2.yml”

Authoritative result: “R-Q021-EDE-PRIMORDIAL-INTERNAL-STRUCTURE-002”

V2 Actions run/commit: NOT RECORDED IN AVAILABLE JOURNAL MATERIAL.

Provenance: V2 reused all 48 V1 profiles; 0 new likelihood evaluations.

9. REPRODUCIBILITY

Checkout recorded backend/Q chain obtain Q021 V1 48 profiles run V2 coherent- — —

restart finalizer verify mandatory gates PASS reproduce distributed classifications, — —

Möbius/Shapley ranges and TV test.

10. CONCLUSION

Within the frozen MOD-EDE-N3 construction, the full-MF primordial response is robustly distributed rather than localized to one parameter. ns is consistently the strongest

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individual direction but never reaches the predefined dominance threshold. Detailed matched/full-MF decomposition remains optimizer-basin sensitive.

11. NEXT-Q HANDOFF

Next Q: Q022

Question: Within the frozen MOD-EDE-N3 full-multifrequency Planck likelihood, what distinguishes the coherent optimizer basins observed in Q021, and are they primarily cosmological, nuisance/foreground, or numerical?

FINAL ANSWER: Distributed primordial coupling; persistent but non-dominant ns.

MIKAMI: Single-driver interpretations killed; V1 aggregation superseded.

MODEL CHANGE: Q020 strengthened and internally refined.

REPRO/GITHUB: "Morfindien/Bubbleverse"; V1 run "33882434170"; V2 result "R-Q021-EDE-

PRIMORDIAL-INTERNAL-STRUCTURE-002".

NEXT: Q022 — identify the origin of optimizer-basin structure.

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Q022

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

Q-ID: Q022

Date/time: 2026-09-04 ~21:23 CEST

Status: CONFIRMED

1. IDENTITY

Question: Within the frozen MOD-EDE-N3 full-multifrequency Planck likelihood, what distinguishes the coherent optimizer basins observed in Q021, and are they primarily associated with different cosmological compromises, nuisance/foreground compromises, or numerical optimizer structure?

2. STARTING STATE

Model: MOD-EDE-N3; nsc f=3; mwt5345/class_edc commit

"5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited: Q020 — primordial-block dominance; Q021 — distributed primordial coupling with substantial full-MF multistart spread.

Constraint: Same frozen objective/model/priors; preserve coherent restarts; no cross-likelihood χ

2 summation.

3. HYPOTHESES

Tested:

H1 pure cosmology basin; H2 pure nuisance/foreground basin; H3 start/incomplete-minimization artifact; H4 stable mixed multibasin structure.

Rejected: H1/H2: both parameter sectors move materially. H3 for tested masks: exact-endpoint continuation and tighter refinement did not collapse solutions.

Surviving: H4 — STABLE_MIXED_COSMOLOGY_NUISANCE_MULTIBASIN_STRUCTURE.

4. METHOD & EVIDENCE

Method: Read existing Q021 full-MF endpoints; quantify cosmology/nuisance/foreground separation; select masks 3, 6, 7; reseed exact Q021 endpoints on identical objective; apply symmetric tighter refinement.

Sources: K-039 Efstathiou, Rosenberg & Poulin (2024), arXiv:2311.00524; K-044 Rosenberg, Gratton & Efstathiou (2022), arXiv:2205.10869; K-046 Cobaya 3.5.6; K-047 Cobaya minimizer documentation.

Programs/workflow: "q022_globality_continuation_v2.py"; ".github/workflows/q022-optimizer-basin-continuation-v2.yml".

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5. KEY RESULTS

V1 mean normalized RMS: cosmology 1.6065, nuisance 1.8976, foreground 1.8337;
cosmology/nuisance ratio 0.8466.

V2 refined:

- mask 3: spread 3.5728, joint RMS 2.0246
- mask 6: spread 0.4725, joint RMS 3.0266
- mask 7: spread 3.0928, joint RMS 2.3535

Final result: Stable full-MF basins involve both cosmology and nuisance/foreground structure and do not collapse under the tested continuation/refinement procedure.

Limitations: Numerical geometry only; not proof of exact mathematical minima, physical modes, Planck systematics, or new physics.

Contradictions: None with Q021; Q022 resolves its basin-origin uncertainty.

6. NEGATIVE RESULTS / MIKAMI

Killed: PURE-COSMOLOGY-BASIN; PURE-NUISANCE-BASIN; SIMPLE-CROSS-START-COLLAPSE under tested conditions.

Failed attempts: No scientifically relevant technical failure recorded.

7. MODEL CHANGE

Before: Mixed basin structure observed but origin unresolved.

After: Stable mixed cosmology+nuisance multibasin structure established within frozen machinery.

Change: CONSTRAINED / REFINED.

Reason: Exact endpoint reseeding + symmetric refinement.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Q021 parent run: "33882434170"

Q022 V2 commit: "16a7902c3be7d1967adaae9ee380baf0d17e903a"

Results: "R-Q022-EDE-FULLMF-OPTIMIZER-BASIN-001"; "R-Q022-EDE-FULLMF-OPTIMIZER-BASIN-002"

Program: "q022_globality_continuation_v2.py"

Workflow: ".github/workflows/q022-optimizer-basin-continuation-v2.yml"

Numeric Q022 Actions run ID: NOT RECORDED.

9. REPRODUCIBILITY

Checkout recorded commit; obtain Q021 raw endpoints from run "33882434170"; run Q022

V2 workflow with masks 3/6/7 and frozen backend; compare endpoint/objective separations; apply collapse criterion requiring both objective spread ≤ 0.50 and joint RMS ≤ 0.10 .

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10. CONCLUSION

The decisive full-MF optimizer families are stable numerical basins containing material cosmological and nuisance/foreground differences. Near-equal objective values do not imply the same solution, as mask 6 demonstrates. No physical cause is established.

11. NEXT-Q HANDOFF

Next Q: Q023

Question: Which specific cosmological and nuisance/foreground parameter combinations distinguish the stable full-multifrequency basin families, and are those directions reproducibly shared across masks 3, 6, and 7?

FINAL ANSWER: Stable mixed cosmology+nuisance multibasin structure confirmed within frozen numerical machinery.

MIKAMI: Pure-cosmology, pure-nuisance and simple-collapse explanations rejected.

MODEL CHANGE: Basin origin refined; no new physical model promoted.

REPRODUCIBILITY: Repo/run/commit/workflow above.

NEXT-Q: Q023 — identify the parameter directions defining the stable basin families.

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Q023

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q023

Date/time: 2026-09-04 21:50 CEST

Status: CONSTRAINED

Question: Which specific cosmological and nuisance/foreground parameter combinations distinguish the stable full-multifrequency basin families identified in Q022, and are the distinguishing directions reproducibly shared across masks 3, 6, and 7?

2. STARTING STATE

Model entering Q: MOD-EDE-N3; (nsc f=3); frozen class_edc commit

"5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited results:

Q021: primordial structure is distributed, not (ns)-only.

Q022: stable mixed cosmology+nuisance multibasin structure exists for masks 3, 6, 7.

Constraints: Reuse validated endpoints; no new likelihood evaluations unless necessary; no physical causation from parameter motion; Q021/Q022 remain closed.

3. HYPOTHESES

Tested:

H1: Basin families share one reproducible normalized parameter direction across masks.

H2: Directions are mask-dependent/non-universal.

H3: Recurrent individual parameters alone define the common direction.

Rejected: H1 and H3 under fixed-lineage criterion; no seed-pair direction reproduces across all masks.

Surviving: H2 — mask-dependent/non-universal directions. A weaker recurrence of some individual parameters remains.

4. METHOD & EVIDENCE

Methods: Read-only normalized endpoint-vector comparison; cosine similarity across masks; parameter ranking and group-energy decomposition.

Primary evidence/sources: Q021/Q022 validated Bubbleverse endpoints; K-039 Efstathiou, Rosenberg & Poulin (2024); K-044 Rosenberg, Gratton & Efstathiou (2022); Cobaya technical sources K-046/K-047.

Programs/workflows:

"q023_basin_directions_v2.py"

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"q023_basin_directions_v2_config.yml"

"q023_tests_v2.py"

".github/workflows/q023-read-only-basin-directions-v2.yml"

Tests: Endpoint completeness; three-pair direction; finite outputs; zero-new-likelihood; parent identity; no-overclaim; Q021/Q022 non-reopening. All mandatory gates PASS.

5. KEY RESULTS

Median cross-mask ($|\cos\theta|$):

0-1: 0.190; 0-2: 0.316; 1-2: 0.220.

Only mask 3 7 for pair 1-2 reached $\rightarrow (|\cos\theta| \approx 0.719)$; mask 6 did not reproduce it.

Recurring large components included "amp_143x217", (H 0), (ω_b/ω_c d m), (A pl anc k), "amp_217", "amp_143", and ($\log_{10} zc$).

Final result: "MASK_DEPENDENT_OR_NONUNIVERSAL_BASIN_DIRECTIONS".

Limitations: Fixed seed-lineage may not equal permutation-invariant basin identity; no posterior-volume or physical-cause inference.

Contradictions: None with Q022; stable basins can exist without one universal separating direction.

6. NEGATIVE RESULTS / MIKAMI

Failed attempt: Q023 V1 failed "Q022_FINAL_UNIQUENESS_GATE", hits=3, due to an authoritative-file selector bug; technical failure only.

Mikami Graveyard:

Strong universal basin-direction hypothesis — killed under tested lineage definition.

Reason: 0/3 seed-pair directions reproduced across all masks.

7. MODEL CHANGE

Before: Stable mixed multibasins; concrete common direction unknown.

After: Stable mixed multibasins retained; separating directions classified mask-dependent under tested definition.

Change: CONSTRAINED / PRECISED.

Reason: Validated Q023 cross-mask endpoint geometry.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Execution commit: "ea5af9a0b25090eb8472adb3c9574dcb4e1efe5c"

GitHub Actions run: "33913077353"

Result: "R-Q023-EDE-FULLMF-BASIN-DIRECTIONS-002"

Artifact: "q023-final-v2"; supplied ZIP SHA-256

"0aae77875aeeccf3b5d7245f6b03a83a50a249188d8eb5f25a3066088f4e567ad"

Outputs: "q023_result_v2.json", "q023_tests_v2.json", "q023_final_v2.json".

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Provenance note: Existing validated Q021/Q022 endpoints were analyzed read-only; 0 new likelihood evaluations.

9. REPRODUCIBILITY

Checkout recorded commits; obtain preserved Q021/Q022 endpoints; run Q023 V2 workflow/config; compare generated JSON with recorded final artifact; apply identical cross-mask cosine/reproducibility criterion.

10. CONCLUSION

Q023 shows that Q022's stable full-MF basins are separated by mixed cosmological, calibration/nuisance, and foreground directions, but no tested direction is reproducibly shared across masks 3, 6, and 7. The result concerns internal frozen likelihood geometry, not physical causation or independent observational evidence.

11. NEXT-Q HANDOFF

Why another Q: Fixed seed labels may not identify equivalent basin families across masks.

Next Q: Q024

Next question: After permutation- and sign-invariant matching of the stable Q022 full-MF basin endpoints across masks 3, 6, and 7, do reproducible basin-family directions emerge, or does Q023's mask-dependent structure persist?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Stable basins persist, but their fixed-lineage separating directions are mask-dependent/non-universal.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Universal shared basin-direction hypothesis rejected under **Q023 criterion.**

MODEL CHANGE SUMMARY: Q022 retained; interpretation narrowed from possible universal direction to mask-dependent multidimensional geometry.

REPRODUCIBILITY / GITHUB: "Morfindien/Bubbleverse"; run "33913077353"; commit "ea5af9a0b25090eb8472adb3c9574dcb4e1efe5c"; Q023 V2 program/workflow/artifact above.

NEXT-Q HANDOFF: Q024 — test permutation-/sign-invariant basin-family matching.

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Q024

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q024

Date/time: 2026-09-04 22:23 CEST

Status: SUPPORTED

Question: Can Q022 stable full-multifrequency basin families be matched reproducibly across masks 3, 6, and 7 using permutation- and sign-invariant endpoint geometry, and does this alter Q023's non-universal-direction conclusion?

2. STARTING STATE

Model entering Q: MOD-EDE-N3, n_scf=3, frozen class_edc commit

5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Inherited:

- Q022: stable multibasin structure across masks 3/6/7.
- Q023: basin directions non-universal under fixed seed lineage.

Constraint: Read-only endpoint analysis; no new likelihood evaluations.

3. HYPOTHESES

Tested:

H1: Q023 non-alignment is only a seed-label permutation artefact.

H2: A permutation/sign-invariant common basin structure exists.

Rejected:

H1/H2 — exhaustive matching produced no shared edge.

Surviving:

Mask-dependent/non-universal basin directions under tested geometry.

4. METHOD & EVIDENCE

Methods: Enumerated all 36 joint mask-6/mask-7 family permutations relative to mask 3; compared three basin edges using absolute cosine similarity.

Sources/evidence:

K-039; K-044; K-046; K-047; Q021–Q023 endpoint chain.

Programs/workflows:

Q024_basin_matching_v1.py

Q024_basin_matching_v1_config.yml

Q024_tests_v1.py

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.github/workflows/q024-read-only-basin-matching-v1.yml

Tests: permutation completeness, sign invariance, mask/seed scope, zero-new-likelihood gate, physical-overclaim gates, FINAL_RESULT_GATE.

5. KEY RESULTS

36/36 permutations tested.

Best candidate:

- Shared edges: 0/3
- Global median $|\cos| = 0.374489$
- Minimum edge median = 0.316164
- Maximum shared-edge count across all candidates = 0
- Best-minus-permutation-median = 0.0
- New likelihood evaluations = 0

- FINAL_RESULT_GATE = PASS

Final result: Exhaustive permutation/sign-invariant matching does not recover a reproducibly shared cross-mask basin direction.

Limitations: Endpoint-coordinate geometry only; no proof about nonlinear/covariance/spectrum-space representations or physical cause.

Contradictions: None.

6. NEGATIVE RESULTS / MIKAMI

Failed route:

“Universal basin direction hidden only by seed-family permutation.”

Mikami Graveyard:

CONN-Q024-SEED-PERMUTATION-RESCUE — REJECTED; all 36 mappings failed.

7. MODEL CHANGE

Before: Q023 non-universality constrained by possible seed-label mismatch.

After: Non-universality strengthened under exhaustive permutation/sign-invariant matching.

Change: CONSTRAINED / strengthened.

Reason: No permutation produced a passing shared edge.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Known related commit:

B8e16629bbc6592ee11f0d90f78c887a913ac4c1 — workflow relocation only.

Parent provenance:

Q021 raw run 33882434170

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Q022 commit 16a7902c3be7d1967adaae9ee380baf0d17e903a

Q023 run 33913077353 / commit ea5af9a0b25090eb8472adb3c9574dcb4e1efe5c

Q024 exact Actions run / execution commit: NOT RECORDED.

Result:

R-Q024-EDE-FULLMF-BASIN-MATCHING-001

9. REPRODUCIBILITY

Checkout recorded code/backend; obtain Q021/Q022 endpoint artifacts; run Q024 workflow with masks 3/6/7 and seeds 0/1/2; verify 36 permutations, zero new likelihood calls, recorded cosine metrics and PASS gates.

10. CONCLUSION

Q024 strengthens Q023: stable optimizer basins exist, but their separation directions are not reproducibly universal across masks 3, 6, and 7 under either fixed-lineage or exhaustive permutation/sign-invariant matching. This remains internal computational geometry, not evidence of a physical systematic or new physics.

11. NEXT-Q HANDOFF

Next Q: Q025

Question: Which likelihood components, spectra, frequency combinations, and parameter-group interactions generate the strengthened mask dependence of the Q022 basin geometry?

Reason: Simple label-permutation rescue is exhausted; the next informative target is the likelihood structure producing the mask dependence.

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: No robust permutation/sign-invariant common basin direction was found.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Seed-permutation rescue rejected.

MODEL CHANGE SUMMARY: Q023 conclusion strengthened; Q022 remains intact.

REPRODUCIBILITY / GITHUB REFERENCES: Morfindien/Bubbleverse; files and provenance above.

NEXT-Q HANDOFF: Q025 — likelihood/spectrum/frequency attribution.

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Q025

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q025

Date/time: 2026-09-04 23:08 CEST

Status: SUPPORTED

Question: Which likelihood components, spectra, frequency combinations, and parameter-group interactions are responsible for the strengthened mask dependence of the stable

Q022 full-multifrequency basin geometries across masks 3, 6, and 7?

2. STARTING STATE

Model: MOD-EDE-N3; n_scf=3; frozen "mwt5345/class_ede" commit
"5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Inherited: Q022 stable full-MF multibasins; Q023 non-universal basin directions; Q024 exhaustive relabelling failed to restore common directions.

Constraints: Masks 3/6/7 fixed; Q022 endpoints preserved; no optimization; no Q024 rerun; primordial coordinates frozen within each mask; functional attribution $\neq$ physical causality.

3. HYPOTHESES

Tested: universal parameter-group driver; universal spectrum/frequency driver; foreground-only explanation; distributed covariance/interaction structure.

Rejected: universal single-driver and pure-foreground explanations.

Surviving: distributed, mask-dependent cosmology–nuisance compensation with covariance-coupled multifrequency structure.

4. METHOD & EVIDENCE

Method: 8 fixed-vector parameter-block vertices per basin edge $\times$ 9 edges = 72 likelihood evaluations, 0 optimizations; covariance/spectrum/multipole and Shapley/interaction attribution.

Sources: Planck NPIPE/CamSpec context; Efstathiou et al. 2024 (arXiv:2311.00524); Rosenberg et al. 2022 (arXiv:2205.10869); Efstathiou & Gratton (arXiv:1910.00483).

Program/workflow: "q025_fullmf_component_attribution_v2.py";
"q025_fullmf_component_attribution_v2_config.yml"; "q025_tests_v2.py";
".github/workflows/q025-fullmf-component-attribution-v2.yml".

5. KEY RESULTS

- Masks 3 and 7: cosmology dominates all three edge decompositions.

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- Mask 6: shared nuisance dominates all three; large cosmology×shared-nuisance interactions (13.79, 10.65). ≈ – –
- "143×217" largest absolute spectrum term on 7/9 edges; "143×143" on 2/9.
- TT generally largest observable contribution; strong signed covariance cancellation, often concentrated at $\ell < 1000$.

Final result: Stable full-MF basin geometry is supported by distributed mask-dependent compensation and covariance-coupled TT multifrequency structure, not one universal component.

Limitations: Internal dependent Planck-likelihood reanalysis; no physical/systematic causality established.

Contradictions: None.

6. NEGATIVE RESULTS / MIKAMI

Failed: Q025 V1, run "33917809217", SHA "470c11eafd7249d520c401c4a91ae7fd8fa6ede9"; endpoint loader failed ("Q022_ENDPOINT_GATE=FAIL"), 0/72 evaluations, no scientific result.

Graveyard: universal single parameter-group driver; universal single-spectrum driver; pure-foreground explanation; independent-additive-component interpretation.

7. MODEL CHANGE

Before: full-MF basin directions known to be mask-dependent/non-universal.

After: their internal support is localized to mask-dependent cosmology–nuisance compensation plus covariance-coupled multifrequency structure.

Change: CONSTRAINED / PRECISED.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Successful run: "33918779905"

Head SHA: "b2db6827b1caf5405e6a34bff221b6ae4813c1d0"

Artifact: "q025-final-v2", ID "9954333941", SHA-256

"104b584c9212a65e7f06cae64a732dac6f58b801cb8feec34c7f49d1bfddac0f".

Provenance: Q021/Q022 endpoints Q023/Q024 geometry Q025 fixed-vector attribution; – – same frozen Planck/CamSpec evidence chain.

9. REPRODUCIBILITY

Checkout recorded commits; obtain preserved Q021/Q022 parent outputs; run Q025 V2 workflow with frozen masks/model; verify 72/72 evaluations, 0 optimizer calls and all gates PASS; compare "q025-final-v2".

10. CONCLUSION

Q025 explains the previously established mask dependence at the likelihood level: masks

3/7 are cosmology-dominated, while mask 6 is shared-nuisance-dominated, with strong

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covariance cancellation and recurring $143 \times 217 / 143 \times 143$ TT structure. This is a numerical-likelihood result, not evidence of calibration failure, bad data, or distinct physical cosmologies.

11. NEXT-Q HANDOFF

Next Q: Q026

Why: Mask 6 is qualitatively different and its shared-nuisance dominance remains unexplained.

Question: Can mask 6's shared-nuisance-dominated geometry be localized to specific nuisance parameters and covariance/frequency structures, or is it irreducibly distributed?

FINAL ANSWER: Distributed mask-dependent full-MF attribution with strong covariance-coupled TT frequency structure.

MIKAMI UPDATE: Single-driver explanations killed.

MODEL CHANGE: Precised/constrained; no fundamental model reversal.

REPRODUCIBILITY: Run "33918779905", SHA "b2db6827...", artifact "q025-final-v2".

NEXT-Q: Q026 — localize mask-6 compensation structure.

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Q026

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q026

Date/time: 2026-09-04 ~23:49 CEST

Status: CONFIRMED

Question: Why does mask 6 become shared-nuisance dominated while masks 3 and 7 remain cosmology dominated, and can this be localized to specific nuisance coordinates, cosmology–nuisance couplings, or CamSpec covariance/data-model structure without assuming a systematic?

2. STARTING STATE

Model: MOD-EDE-N3; (nsc f=3); frozen "class_edu" commit

"5a131c91d657dd9a7c6364cc45b038710f8d0d97"; "planck_NPIPE_highl_CamSpec.TTTEEE".

Inherited: Q022 stable mixed basins; Q025: masks 3/7 cosmology dominated, mask 6 shared-nuisance dominated with strong TT multifrequency/covariance structure.

Constraint: Fixed validated endpoints; no optimizer; no Q024 permutation rerun; functional diagnostics ≠ physical causality.

3. HYPOTHESES

Tested:

H1 generic shared-nuisance response.

H2 (Aplanc k)-centered response.

H3 specific cosmology×nuisance coupling associated with mask-sensitive covariance structure.

Rejected: Generic/equal nuisance explanation; calTE- or calEE-dominated explanation.

Surviving: (Aplanc k)-centered mask-6 geometry, dominated by (ωcd m× Aplanc k).

4. METHOD & EVIDENCE

Method: 252 fixed-vector likelihood evaluations across 9 basin edges; individual nuisance switches plus all 18 cosmology×shared-nuisance pairs.

Sources: Planck/NPIPE/CamSpec; Q022/Q025 validated parents; Rosenberg et al. 2022 (arXiv:2205.10869); Efstathiou & Gratton (arXiv:1910.00483).

Program/workflow: "q026_mask6_nuisance_localization_v1.py"; ".github/workflows/q026-mask6-nuisance-localization-v1.yml"; run "33921255497".

Tests: All mandatory gates PASS; "FINAL_RESULT_GATE=PASS"; 252/252 finite evaluations; 0 optimizer evaluations; 0 Q024 reruns.

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5. KEY RESULTS

Mask-6 median absolute nuisance switches:

($\Delta \text{pl anc k}=8.6626$), ($\text{calTE}=0.0239$), ($\text{calEE}=0.0374$).

Strongest mask-6 pair interactions:

($\omega \text{cd m} \times \Delta \text{pl anc k}=25.1681$);

($\text{H O} \times \Delta \text{pl anc k}=7.7736$);

($\text{f E D E} \times \Delta \text{pl anc k}=4.7882$).

Strongest edges 0–2 and 1–2 coincide with large cancelling TT

($143 \times 143 / 143 \times 217 / 217 \times 217$) covariance terms around ($\ell=600!-!999$).

Final result: Mask 6 is not generically shared-nuisance dominated; its distinctive likelihood

geometry is predominantly ($\Delta \text{pl anc k}$)-centered, especially through ($\omega \text{cd m} \times \Delta \text{pl anc k}$).

Limitations: Local functional geometry only; no calibration/systematic/physical-causality claim.

Contradictions: None.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: None scientifically relevant in Q026.

Graveyard: Uniform shared-nuisance explanation; calTE-dominated; calEE-dominated; equal cosmology $\times$ nuisance structure. Calibration-failure claim remains NOT ESTABLISHED.

7. MODEL CHANGE

Before: Mask 6 = shared-nuisance dominated.

After: Mask 6 = predominantly ($\Delta \text{pl anc k}$)-centered, strongest pair ($\omega \text{cd m} \times \Delta \text{pl anc k}$).

Change: MODIFIED / CONSTRAINED interpretation.

Reason: Q026 coordinate and pair localization.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Commit: "7cb4d98c01196c22b407b6f4fd4118179afea3de"

Run: "33921255497"

Result: "R-Q026-EDE-FULLMF-MASK6-COUPPING-001"

Artifact: "q026-final-v1"; ID "9955684272"; SHA-256

"fc1970c0e531a5a9157e2054aea91d54cbca9297a0599634710f2ceaa854cc87".

9. REPRODUCIBILITY

Checkout recorded commit obtain validated Q022/Q025 parents run Q026 workflow --
with frozen backend/likelihood verify 252 evaluations and mandatory gates compare --
final artifact/result ID.

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10. CONCLUSION

Q026 resolves the parameter-side origin of mask 6's unusual basin attribution: the effect is

overwhelmingly (Aplanc k)-centered, with (ω d m $\times$ Aplanc k) the dominant local functional interaction. The same strongest edges also contain extreme cancelling TT multifrequency covariance structure near ($\ell=600!-!999$), but causal direction remains unresolved.

11. NEXT-Q HANDOFF

Why: The bridge between parameter compensation and residual/covariance structure remains unknown.

Next Q: Q027

Question: Can the mask-6 (Aplanc k)-(ω d m) compensation be traced primarily to residual-vector changes, inverse-covariance weighting, their interaction, or broader mask-sensitive CamSpec geometry?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Mask-6 shared-nuisance dominance localizes predominantly to (Aplanc k), especially (ω d m $\times$ Aplanc k); no physical systematic established.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Generic/equal shared-nuisance, calTE-dominated and calEE-dominated terminal explanations rejected.

MODEL CHANGE SUMMARY: Q025 block-level result retained but refined to (Aplanc k)-centered geometry.

REPRODUCIBILITY / GITHUB: Run "33921255497"; commit "7cb4d98..."; result "R-Q026-EDE-

FULLMF-MASK6-COUPLING-001".

NEXT-Q HANDOFF: Q027 — identify residual/covariance mechanism behind the localized mask-6 geometry.

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Q027

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q027

Date/time: UNKNOWN / NOT RECORDED

Status: CONFIRMED

Question: Can the mask-6 (A pl anc k)–(wc d m) compensation identified in Q026 be traced to the TT (143×143), (143×217), and (217×217), ($\ell=600 - 999$) blocks, or is it distributed through broader mask-sensitive geometry?

2. STARTING STATE

Model entering Q: MOD-EDE-N3, (nsc f=3), frozen mwt5345/class_e de commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97; CamSpec TTTEEE masks 3/6/7.

Inherited results: Q025 distributed covariance-coupled full-MF structure; Q026 strong mask-6 (A pl anc k)-centred coupling, strongest pair (wc d m× A pl anc k).

Constraints: reuse Q022 endpoints; no reoptimization; preserve signed covariance accounting.

3. HYPOTHESES

Tested:

H1: TT ($\ell=600 - 999$) triad is the dominant/local mask-6 explanation.

H2: Interaction is distributed through broader residual/covariance geometry.

H3: Within-mask effect arises from residual changes under fixed precision.

Rejected: H1 — localization is similar across masks and tiny after precision weighting.

Surviving: H2/H3 — supported.

4. METHOD & EVIDENCE

Methods: 36 fixed-vector likelihood evaluations; residual vector $\otimes$, precision (P), (Pr), signed pointwise and block decomposition.

Sources: K-039, K-044, K-048, K-049; CamSpec/NPIPE computational chain.

Programs/workflows: q027_residual_covariance_bridge_v1.py;

q027_residual_covariance_bridge_v1_config.yml; q027_tests_v1.py; q027-residual-covariance-bridge-v1.yml.

Tests: Q026 pair reproduction; precision invariance; decomposition completeness; finite metrics; cross-mask comparison; no-causal-overclaim gate.

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5. KEY RESULTS

Mask-6 median TT-triad shares: absolute pointwise 0.4445; residual (L^2) 0.3018; $\approx \approx$ precision-weighted residual (L^2) 0.00232. $\approx$

Cross-mask absolute pointwise medians: mask3 0.444585; mask6 0.444548; mask7 0.444749.

Q026 pair reproduction max difference: 0.0.

Final result: DISTRIBUTED_BEYOND_TT600_999_TRIAD.

Limitations: no valid cross-mask precision swap; residual-vs-precision cross-mask causation remains non-identified.

Contradictions: Earlier possibility that TT600–999 alone explained mask 6 was rejected.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: None scientifically required; no optimizer/sampler reruns.

Mikami Graveyard addition: “TT 143×143/143×217/217×217 at ($\ell=600 - 999$) is a sufficient local explanation of mask-6 coupling.” Killed by cross-mask localization and precision-weighted result.

7. MODEL CHANGE

Before: Strong mask-6 (Aplanc k)-centred geometry with possible TT600–999 localization.

After: Broader distributed mask-sensitive full-MF residual/covariance geometry.

Change: CONSTRAINED / MODIFIED.

Reason: Q027 localization decomposition.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commit: Q027 execution commit UNKNOWN / NOT RECORDED

Workflow: q027-residual-covariance-bridge-v1.yml

Programs: q027_residual_covariance_bridge_v1.py; q027_tests_v1.py

Outputs: q027_merged_v1.json; q027_tests_v1.json; q027_final_v1.json

Result ID: R-Q027-EDE-FULLMF-RESIDUAL-COVARIANCE-BRIDGE-001

Run: Q027-FULLMF-RESIDUAL-COVARIANCE-BRIDGE-V1

FINAL_RESULT_GATE: PASS

9. REPRODUCIBILITY

Checkout frozen backend and recorded Q022/Q025/Q026 state; run Q027 workflow with masks 3/6/7 and authoritative endpoints; reproduce 36 fixed evaluations; verify Q026 pair difference 0.0 and all gates PASS.

10. CONCLUSION

The strong mask-6 (Aplanc k $\times$ wd m) interaction is not sufficiently localized to the selected TT ($\ell=600 - 999$) triad. Within each mask, the identifiable mechanism is residual change under

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a fixed inverse covariance. The broader cross-mask split between residual/data structure and precision/covariance structure remains unresolved.

11. NEXT-Q HANDOFF

Why another Q is required: Cross-mask residual-vs-precision attribution is still non-identified.

Next Q: Q028

Next question: Can cross-mask non-universality be separated into residual/data-vector versus mask-specific precision effects using a mathematically valid common-support counterfactual construction?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: DISTRIBUTED_BEYOND_TT600_999_TRIAD.

Q-JOURNAL: This reconstruction.

MIKAMI GRAVEYARD UPDATE: TT600–999 triad as sufficient local explanation rejected.

MODEL CHANGE SUMMARY: Interpretation narrowed to broader distributed residual/covariance geometry.

REPRODUCIBILITY / GITHUB REFERENCES: R-Q027-EDE-FULLMF-RESIDUAL-COVARIANCE-

BRIDGE-001; listed program/workflow/result files.

NEXT-Q HANDOFF: Q028 — valid common-support residual-vs-precision decomposition.

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Q028

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q028

Date/time: 2026-09-05 11:34 CEST

Status: CONSTRAINED

Question: Can the established cross-mask non-universality of the frozen MOD-EDE-N3 full-MF CamSpec likelihood be separated into residual/data-vector effects versus mask-specific inverse-covariance/precision effects using a mathematically valid common-support counterfactual construction, without changing the physical model or reoptimizing the authoritative Q022 endpoints?

2. STARTING STATE

Model entering Q: MOD-EDE-N3, n_scf=3; frozen backend mwt5345/class_eide @ 5a131c91d657dd9a7c6364cc45b038710f8d0d97; CamSpec TTTEEE.

Inherited results: Q022 stable multibasin endpoints; Q026 strong mask-6 A_planck-centred coupling; Q027 residual change under fixed inverse covariance, but cross-mask residual-vs-precision attribution unresolved.

Constraints: preserve Q022 endpoints/model/likelihood; no optimization, sampling or Q024 rerun; frozen Criterion B+ tolerance 2×10^{-4} .

3. HYPOTHESES

Tested:

H1: A valid common-support residual/precision separation exists.

H2: Edge-dependent precision prevents unique target-functional attribution.

H3: Precision variation materially drives the $\omega_{\text{cdm}} \times A_{\text{planck}}$ cross-lineage interaction.

Rejected: H2/H3 — admissible precision choices are functionally equivalent far below tolerance.

Surviving: H1 — supported only for the frozen Q028 target functional/construction.

4. METHOD & EVIDENCE

Methods: exact labelled common support; Schur-valid quadratic action; factorial functional

$F = q_{11} q_{10} q_{01} + q_{00}$; cross-donor Criterion B+; signed two-factor Shapley. — —

Sources: K-039, K-044, K-048, K-049, K-046; same broad Planck/NPIPE/CamSpec evidence family.

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Program/workflow: q028_common_support_counterfactual_v8.py;

q028_common_support_counterfactual_v8_config.yml; q028_tests_v8.py;

.github/workflows/q028-common-support-counterfactual-v8.yml.

Tests: 36 fixed-vector evaluations; 9 family + 9 precision jobs; Criterion B+ completeness; common-support validity; Shapley closure; FINAL_RESULT_GATE.

5. KEY RESULTS

Common support: $9915/9915 = 1.0$.

Global max precision edge-choice spread: $1.3457679415 \times 10^{-10}$ vs 2×10^{-4} threshold.

Criterion B+: PASS for masks 3, 6, 7.

Precision-side Shapley: 0.0 in all 9 comparisons; residual side carries all non-negligible target-functional differences.

Final result: VALID_COMMON_SUPPORT_SEPARATION.

Limitations: residual $r=d$ m is not raw-data-only; result is non-causal, – model/likelihood/functional conditional and not independent observational evidence.

Contradictions: None with Q019–Q027; Q028 V1/V2 interpretations superseded.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: V1 bitwise criterion too strong; V2 state-level invariance mathematically over-restrictive; V3 symmetry gate; V4 runner shutdown; V5 ell-band schema; V6 checkpoint naming; V7 missing classifier tolerance.

Mikami Graveyard additions: “materially different precision operators drive the Q028 target” — rejected by Criterion B+; state-level q-invariance as required criterion — rejected.

7. MODEL CHANGE

Before: cross-mask residual-versus-precision attribution non-identified.

After: target-functional attribution identified as residual-side under functionally equivalent admissible precision choices.

Change: MODIFIED / CONSTRAINED.

Reason: Q028 V8 Criterion B+ + Shapley PASS.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Execution commit: 7f5d35efeb54b4a62d6dfb8f7074014b4fabc190

Run: 33957937309

Result: R-Q028-EDE-FULLMF-COMMON-SUPPORT-COUNTERFACTUAL-008

Artifact: q028-final-v8, ID 9967068363, SHA256

7dfbd1f27251ff311dd1de6118539dd691ab03d9afdb08539b3b13586c50d5af

Validation: FINAL_RESULT_GATE PASS.

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9. REPRODUCIBILITY

Checkout recorded commit; obtain frozen Q021/Q022/Q025/Q026/Q027 parents; run V8 workflow with frozen model/likelihood/endpoints; compare q028_merged_v8.json, q028_tests_v8.json and q028_final_v8.json; apply Criterion B+ 2×10^{-4} and Shapley $\leq$ closure 10^{-8} .

10. CONCLUSION

Within the frozen MOD-EDE-N3 full-MF CamSpec construction, the cross-lineage $\omega_{\text{cdm}} \times A_{\text{planck}}$ target-functional difference can be validly separated: admissible precision choices are functionally indistinguishable, while the difference resides on the residual side. This does not identify whether the residual difference originates in data or model predictions.

11. NEXT-Q HANDOFF

Why required: r=d m remains internally unresolved. –

Next Q: Q029

Next question: Can the Q028 residual-side target-functional differences be validly separated into observed data-vector versus model-prediction contributions, including cosmological, nuisance, foreground and calibration components?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Valid common-support separation; residual-side dominates, precision-side = 0.0 for all 9 final comparisons.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Precision-driven Q028 target and required state-level invariance killed.

MODEL CHANGE SUMMARY: Cross-mask attribution changed from non-identified to residual-side identified for the frozen target functional.

REPRODUCIBILITY / GITHUB REFERENCES: Morfindien/Bubbleverse; commit 7f5d35ef...; run 33957937309; V8 workflow/program/tests; artifact q028-final-v8.

NEXT-Q HANDOFF: Q029 — decompose the residual itself without reopening Q028 precision identifiability.

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Q029

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q029

Date/time: 2026-09-05 11:59 CEST

Status: CONFIRMED

Question: Within Q028's exact 9915-row common support, can the residual-side $\omega_{cd} \times A_{pl} \text{ anc } k$ differences be uniquely separated into data-vector versus model-prediction contributions, including cosmology, nuisance, foreground and calibration?

2. STARTING STATE

Model: MOD-EDE-N3; nsc f=3; frozen "mwt5345/class_ede" commit "5a131c91d657dd9a7c6364cc45b038710f8d0d97"; "planck_NPIPE_highl_CamSpec.TTTEEE".

Inherited:

- Q026: strongest mask-6 pair interaction = $\omega_{cd} \times A_{pl} \text{ anc } k$.
 - Q028 V8: exact 9915/9915 support; residual-side dominates target-functional variation; precision-side 0 for the defined target. $\approx$
- Constraints: Q022 endpoints frozen; no reoptimization/sampling; preserve $r=d-m$; no causal attribution.

3. HYPOTHESES

H1: Unique data/model decomposition exists.
H2: Exact interaction-preserving decomposition exists.
H3: Model term uniquely splits into cosmology/foreground/nuisance/calibration.
Rejected: H1, H3 — irreducible cross-terms prevent unique attribution.
Surviving: H2 — CONFIRMED.

4. METHOD & EVIDENCE

Method: Exact algebraic expansion of the Q028 factorial functional plus CamSpec likelihood-semantics check.
T P (d -ms)
 $F=q_{11}-q_{10}-q_{01}+q_{00}, q_s=(d -ms)$
Sources: K-039, K-044, K-046, K-048, K-049. K-047 unresolved.
Programs/workflows: No new Q029 computation. Q028 V8 and Q027 validated outputs inherited.
Tests: algebraic cancellation; cross-lineage interaction/path-independence test; model-component additivity check.

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5. KEY RESULTS

T P Δ

T P m)

$2m+\Delta$

$2(m$

$F=-2d$

And exactly:

T P d)=0.

$2(d$

Δ

Thus no standalone data-only term exists. Data enter through the data-model factorial

T P Δ

coupling $-2d$

$2m$.

Cross-lineage interaction:

T P (Δ

I D M= $-2(d A-dB)$

$2mA-\Delta$

$2mB$).

If I D M $\neq 0$, data/model attribution is path-dependent. Calibration is multiplicative, so cosmology/calibration are likewise not uniquely additive.

Final result: Exact interaction-preserving decomposition exists; unique data-vs-model or unique multicomponent attribution does not.

Limitations: functional, likelihood-construction-dependent, non-causal; no new observational evidence.

Contradictions: None with Q028; Q028 interpretation is narrowed.

6. NEGATIVE RESULTS / MIKAMI

Killed:

- "RESIDUAL = RAW DATA"

- unique data/model attribution
- unique cosmology/calibration attribution
- Shapley allocation as physical identification

Failed runs: None in Q029.

7. MODEL CHANGE

Before: Q028 established residual-side rather than precision-side dominance, but lower-level residual attribution remained unresolved.

After: residual functional contains irreducible data-model and model-model interactions.

Change: PRÆCISERET / CONSTRAINED.

Reason: exact Q029 algebra.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Q027: run "33924573651", commit "df03e30712690e61ff799558598405a22c0bacbb".

Q028 V8: run "33957937309", commit "7f5d35c0725530039162adfa4d71be288e8b4462";

artifact "q028-final-v8", ID "9967068363", SHA-256

"7dfbd1f27251ff311dd1de6118539dd691ab03d9afdb08539b3b13586c50d5af".

Q029: no new run/program; mathematical derivation from validated inherited state.

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9. REPRODUCIBILITY

1. Checkout frozen model/Q028 versions.

2. Preserve Q028 9915-row support and Q022 endpoints.

T P (d -ms).

3. Reconstruct $q_s=(d -ms)$

4. Expand the four-state factorial combination algebraically.

T P d)=0 and the stated cross-interaction identity.

5. Verify Δ

2(d

10. CONCLUSION

Q029 establishes that Q028's residual-side result cannot be promoted to a unique data-

driven, model-driven, cosmology-driven or calibration-driven explanation. The

mathematically identifiable structure is an exact data-model coupling plus a model-

quadratic response. Any finer two-way allocation requires convention and is not causal

identification.

11. NEXT-Q HANDOFF

Why: The exact two terms are identified mathematically but not yet quantified across the
nine mask $\times$ edge diagnostics.

Next Q: Q030

Question: Quantify the signed data-model coupling and model-quadratic terms on the
frozen Q028 common support, reusing existing checkpoints where possible.

FINAL ANSWER: Unique attribution rejected; exact interaction-preserving decomposition
confirmed.

MIKAMI GRAVEYARD: raw-data-side, unique data/model split, unique
cosmology/calibration split killed.

MODEL CHANGE: interpretation narrowed; physical model unchanged.

REPRODUCIBILITY: inherited Q027/Q028 validated GitHub chain; no Q029 run.

NEXT-Q: Q030 — numerical decomposition of the two exact surviving terms.

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Q030

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q030

Date/time: 2026-09-05 13:25 CEST

Status: CONFIRMED

Question: Using the exact 9915-row common support and frozen Q022 endpoints, can the

Q029 identity

T P Δ

T Pm)

2m+Δ

2(m

F=-2d

Be reconstructed for all nine mask×edge diagnostics, including signed terms and spectrum/multipole structure, without imposing a causal allocation?

2. STARTING STATE

Model entering Q: MOD-EDE-N3, nsc f=3; frozen mwt5345/class_eде commit "5a131c9..."; CamSpec TTTEEE.

Inherited:

- Q028 V8: exact 9915-row common support; SHA "60881658..."; validated fixed precision structure.

- Q029: exact decomposition exists, but unique data/model or cosmology/calibration attribution is mathematically non-identifiable.

Constraints: Frozen Q022 endpoints; no reoptimization, sampling, Q024 rerun, causal allocation, or tolerance rescue.

3. HYPOTHESES

Tested:

H1: Q029 identity is numerically reconstructible for all 9 diagnostics.

H2: Original 10

-8 validation gate is attainable with stable arithmetic.

H3: Interaction remains covariance-coupled and distributed in spectrum/ℓ space.

Rejected: Numerical precision floor above 10

-8; narrow TT600-999-only explanation;

unique causal allocation.

Surviving: Exact functional decomposition + Q029 non-identifiability; CamSpec-conditional distributed geometry.

4. METHOD & EVIDENCE

Methods: Reused frozen Q022/Q028 objects; evaluated T D M, T M M, and F; spectrum/ℓ pair decomposition; final stable factorial-bilinear precision audit.

Sources: K-039, K-044, K-046, K-048, K-049; Morfindien/Bubbleverse.

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Programs/workflows:

"q030_interaction_preserving_decomposition_v2.py";

"q030_original_gate_precision_audit_v1.py";

".github/workflows/q030-original-gate-audit-finalization-r1.yml".

Tests: 13/13 final mandatory tests PASS; original 10

-8 gates retained.

5. KEY RESULTS

- All 9 mask×edge diagnostics reconstructed.
- $|T M M \rangle > |T D M \rangle$ in 9/9, functional only.
- Largest totals: mask6 E02 F=-29.5462638258; E12 F=-25.1680892024.
- Dominant absolute spectrum pair: 143×217 217×217, 9/9. →
- Dominant absolute ℓ pair: 1500-2000 self-pair, 9/9.
- Strong signed covariance cancellation throughout.
- Final max identity error: 2.09×10^{-13} .
- Max group closure error: 5.48×10^{-14} .
- Max $|\Delta_{2r}| = 4.55 \times 10^{-13}$.
- $2m + \Delta$

Final result: Q029's exact decomposition is numerically realized and passes the original

Q030 validation contract.

Limitations: Frozen MOD-EDE-N3/CamSpec construction; no unique physical cause identified.

Contradictions: None physical; Q028 provenance SHA misrecord corrected.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts:

- Q030 V1: wrong Q028 support SHA technical failure. →
- V2: scientific values produced, but original 10
- 8 closure failed from cancellation-sensitive arithmetic.
- V3: passed only after new 10
- 7 validation config; retained as historical, not final.

Graveyard additions:

- Unique data/model causal split.
- Unique cosmology/calibration split.
- TT600-999-only explanation.
- Claim that V2 failure falsified Q029 identity.

7. MODEL CHANGE

Before: Exact Q029 identity mathematical but not yet fully validated numerically.

After: Exact identity numerically confirmed on all 9 frozen diagnostics; distributed covariance structure quantified.

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Change: STRENGTHENED / PRECISED.

Reason: Stable original-gate audit PASS.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Key commits: "c0d01b8..." (V2), "4cbe1585..." (finalization).

Runs: "33961469874", "33963221940".

Final result: "R-Q030-ORIGINAL-GATE-PRECISION-AUDIT-001"

Artifact: "q030-original-gate-precision-audit-final-v1", ID "9968592047", SHA256 "7685d769..."

9. REPRODUCIBILITY

Checkout recorded commits obtain frozen Q022/Q028 inputs run recorded Q030 → →

workflows compare generated JSON/artifacts apply original → → 10

→ 8, 10

→ 11, and 2×10

→ 4

gates unchanged.

10. CONCLUSION

Q030 confirms that the Q029 interaction-preserving decomposition is numerically valid on

the frozen CamSpec system. The large mask6 structure consists of strongly cancelling covariance-coupled terms; it cannot be uniquely assigned to data, model, cosmology, calibration, or a physical systematic.

NEXT-Q HANDOFF

Why: Main remaining uncertainty is whether this detailed geometry is CamSpec-specific.

Next Q: Q031

Question: Does the Q030 spectrum/multipole interaction geometry persist under an independently validated alternative Planck high- ℓ likelihood construction?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: CONFIRMED under frozen Q030 configuration.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Unique causal allocations and TT600–999-only explanation remain dead.

MODEL CHANGE SUMMARY: Internal likelihood geometry strengthened/precised; no new physics claim.

REPRODUCIBILITY / GITHUB: Morfindien/Bubbleverse; runs "33961469874", "33963221940"; final artifact above.

NEXT-Q HANDOFF: Q031 — cross-likelihood robustness.

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Q031

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q031

Date/time: 2026-09-05, successful run completed 18:54 UTC

Status: REJECTED

Question: Does the distributed, multibasin, mask-dependent, covariance-coupled $n=3$ EDE geometry found in frozen Planck NPIPE/CamSpec survive materially under an independent Planck PR4/NPIPE likelihood implementation?

2. STARTING STATE

Model entering Q: MOD-EDE-N3; $n_{\text{scf}}=3$; active/constrained;
class_edex@5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Inherited results: Q021–Q030 established distributed primordial response, stable mixed cosmology–nuisance CamSpec basins, no universal basin direction, distributed attribution, $\omega_{\text{cdm}} \times A_{\text{planck}}$ coupling, common-support separation, and exact validated factorial geometry.

Constraint: test portability without changing frozen physics, retrospectively redefining basin criteria, summing cross-likelihood objectives, or forcing nuisance equivalence.

3. HYPOTHESES

Tested:

H1: Q022-type stable multibasin geometry survives in independent HiLLiPoP.

H2: Geometry remains separated but fails reproducible stable-basin support.

H3: Implementation fails, leaving portability unresolved.

Rejected: H1 — stable_basin_count=0. H3 — implementation and validation passed.

Surviving: H2 — implementation-specific likelihood geometry; residual separated endpoints remain.

4. METHOD & EVIDENCE

Method: HiLLiPoP PR4/NPIPE TTTEEE v4.3, same frozen $n=3$ backend; 9 mapped multistarts + mandatory tighter refinement; frozen Q022 stability criterion $\Delta_{\text{objective}} 0.50 \leq$ and normalized RMS 0.10 with repeated support. $\leq$

Sources: HiLLiPoP/Planck PR4 literature; Tristram et al. 2024; Efstathiou, Rosenberg & Poulin 2024; McDonough et al. 2024.

Program/workflow: q031_planck_portability_v1.py; .github/workflows/q031-planck-portability-v4.yml.

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Tests: independent-implementation, completeness, multibasin, final-result/claim-boundary validation.

5. KEY RESULTS

9/9 primary starts completed; 9/9 refinements completed.

Stable_basin_count=0.

MATERIAL_STABLE_MULTIBASIN_GATE=FAIL.

Direction gate=NOT_TESTABLE; downstream primordial/compensation gates=NOT_RUN by preregistered stop logic.

Near-degenerate separated examples:

M3-S0 M7-S2: Δ objective=0.2335; RMS=3.4915. $\leftarrow$

M6-S2 M7-S1: Δ objective=0.3803; RMS=7.8245. $\leftarrow$

Final result: Q021-Q030 stable CamSpec geometry does not port to HiLLiPoP under the frozen criterion.

Limitation: one independent likelihood tested; mechanism causing non-portability remains unidentified.

Contradiction: none fundamental; CamSpec vs HiLLiPoP creates implementation-structure tension.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: V1 environment/setup; V2 Python/cache configuration; V3 Q022 stage/refinement artifact collision. All technical, no scientific result.

Mikami Graveyard: "Cross-implementation Planck n=3 EDE likelihood geometry" — rejected under Q031. Universal interpretation of Q021-Q030 deleted; internal CamSpec results preserved.

7. MODEL CHANGE

Before: Q021-Q030 geometry potentially generalizable beyond CamSpec.

After: IMPLEMENTATION-SPECIFIC INTERNAL LIKELIHOOD GEOMETRY.

Change: CONSTRAINED / GENERALIZATION REJECTED.

Reason: validated independent HiLLiPoP portability failure.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Successful commit: ff621d6e0a425a976bb7a55f9194a0de8a87f18f

Workflow: q031-planck-portability-v4.yml

Run: 33975028436

Result: R-CASE031-EDE-PLANCK-PORTABILITY-001

Artifact: q031-final-v1, ID 9975019737

Validated file: q031_final_validated_v1.json

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HiLLiPoP commit: a09ddde3e7ce11df99f74685feb1f1764cafb251.

9. REPRODUCIBILITY

Checkout recorded commit; obtain frozen Q021/Q022 parents and HiLLiPoP PR4 data; run V4 workflow with recorded backend/criteria; compare q031_final_validated_v1.json; require the same frozen portability gates.

10. CONCLUSION

Q031 rejected cross-implementation portability of the stable Q022-type CamSpec basin

geometry. The internal Q021–Q030 calculations remain valid, but their scope is narrowed to the frozen CamSpec implementation. No Planck systematic, calibration failure, EDE falsification, or new physics was established.

11. NEXT-Q HANDOFF

Next Q: Q032

Reason: portability failure is established; its implementation-level cause is not.

Next question: Can the CamSpec–HiLLiPoP portability failure be localized to foreground modelling, calibration/nuisance structure, covariance, frequency/support construction, or a distributed interaction among them?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Portability rejected under the frozen Q031 test.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Universal cross-implementation geometry killed; internal CamSpec geometry retained.

MODEL CHANGE SUMMARY: General Planck interpretation implementation-specific → CamSpec geometry.

REPRODUCIBILITY / GITHUB: Morfindien/Bubbleverse; run 33975028436; commit ff621d6e...; V4 workflow; validated result above.

NEXT-Q HANDOFF: Q032 — identify the cause of CamSpec–HiLLiPoP non-portability.

BUBBLEVERSE Q JOURNALS

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Q032

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q032

Date/time: 2026-09-06, final result; exact completion time not independently recorded here

Status: CONFIRMED

Question: Can the CASE-031 portability failure be localized to one or a small preregistered set of differences between the frozen Planck NPIPE/CamSpec full-multifrequency likelihood and HiLLiPoP PR4/NPIPE, or does the loss of stable multibasin structure remain distributed across implementation changes?

2. STARTING STATE

Model entering Q: MOD-EDE-N3; n_scf=3; mwt5345/class_edc

5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Inherited: Q022 historical stable CamSpec multibasin classification; CASE-031 HiLLiPoP portability failure (st abl eba sinc ount=0).

Constraints: same model/priors/basin thresholds; 9 starts M3/M6/M7; no new seeds; no

Q024/Q030 reruns; no causal A_planck interpretation.

3. HYPOTHESES

Tested:

H1: removed TT SUPPORT causes loss.

H2: TE or EE causes loss.

H3: a small combination of SUPPORT/TE/EE restores stable geometry.

Rejected: H1–H3; no single, pairwise, or full three-sector restoration recovered stable multibasin structure.

Surviving: broader likelihood/execution-semantics dependence.

4. METHOD & EVIDENCE

Method: common-support bridge followed by corrected exact-scalar-start CamSpec add-back hierarchy: SUPPORT, TE, EE all pairs FULL_NATIVE. → →

Sources: K-044 CamSpec PR4; K-046 Cobaya 3.5.6 technical implementation; HiLLiPoP v4.3 technical source.

Programs/workflows: q032_exact_start_addback_v4.py; .github/workflows/q032-exact-start-addback-v4.yml.

Run: GitHub Actions 34015845246.

Tests: exact scalar reference/start provenance; covariance selection; finite likelihood; completeness; globality; interpretation; FINAL_RESULT_GATE.

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5. KEY RESULTS

Exact-start TT3 baseline: stable_basin_count=0.

SUPPORT, TE, EE: all 0.

SUPPORT+TE, SUPPORT+EE, TE+EE: all 0.

FULL_NATIVE SUPPORT+TE+EE: 0.

Final result:

NO_CLEAN_LOCALIZATION_WITHIN_TESTED_CAMSPEC_INFORMATION_SECTORS.

Limitations: methodological likelihood geometry only; does not identify

physical/systematic cause or confirm/falsify EDE.

Contradiction: historical Q022 stable classification is not reproduced under genuine scalar exact-start semantics.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: Q032 V1 nuisance leakage technical failure; Q032 V3 optimizer failure — plus incorrect non-scalar start semantics no scientific result. —

Graveyard: clean SUPPORT-only, TE-only, EE-only, pairwise, and three-sector restoration explanations killed. V2 claim that removed information was materially required is superseded.

7. MODEL CHANGE

Before: historical CamSpec geometry treated as stable multibasin, with CASE-031 showing non-portability.

After: historical endpoint geometry retained, but stable-basin interpretation is execution-semantics dependent; no small CamSpec sector explains portability loss.

Change: MODIFIED / CONSTRAINED.

Reason: Q032 V4 exact-start full-native result.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

V4 execution HEAD: fbde93aee01ec3bed183207518ff315c96a45c3c

Workflow: .github/workflows/q032-exact-start-addback-v4.yml

Program: q032_exact_start_addback_v4.py

Result: R-Q032-EDE-CAMSPEC-ADDBACK-004

Run: 34015845246

Provenance: Q022 endpoint vectors + frozen CamSpec/model configuration; V4 corrected

Cobaya refs to literal scalar starts.

9. REPRODUCIBILITY

Checkout recorded commit obtain frozen inputs/Q022 vectors run V4 workflow — —

unchanged compare baseline/tier/control outputs apply basin criteria Δ objective 0.50, — — $\leq$

RMS 0.10, support 2 and mandatory gates. $\leq \geq$

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10. CONCLUSION

Q032 found no clean localization of the CASE-031 portability failure to SUPPORT, TE, EE, or

their combinations. Even full-native CamSpec remains non-stable under corrected exact-start semantics. The historical geometry is therefore best treated as likelihood/execution-dependent endpoint geometry, not an execution-invariant stable-basin property.

11. NEXT-Q HANDOFF

Why: determine whether launch/reference semantics alone explain the Q022 Q032 ↔ difference.

Next Q: Q033

Question: Under frozen full-MF CamSpec MOD-EDE-N3, can the historical Q022 stable classification versus Q032 V4 non-stability be explained by launch-point/reference-distribution semantics alone with all other components matched?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: No clean small-sector localization; execution semantics materially matter.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: SUPPORT/TE/EE localization routes killed.

MODEL CHANGE SUMMARY: Q022 stable-basin interpretation constrained to historical execution semantics.

REPRODUCIBILITY / GITHUB: R-Q032-EDE-CAMSPEC-ADDBACK-004, run 34015845246,

HEAD fbde93a....

NEXT-Q HANDOFF: Q033 — isolate launch/reference semantics.

BUBBLEVERSE Q JOURNALS

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Q033

BUBBLEVERSE — Q-JOURNAL

1. IDENTITY

Q-ID: Q033

Date/time: 2026-09-06 21:45 CEST

Status: CONFIRMED

Journal type: RETROSPECTIVE RECONSTRUCTION

Reconstructed on: 2026-09-08

Question: Can Q022's stable-multibasin versus Q032 V4's non-stable FULL_NATIVE result be explained by launch/reference-distribution semantics alone?

2. STARTING STATE

Model: MOD-EDE-N3; (nsc f=3); frozen "mwt5345/class_edc" commit "5a131c91...".

Inherited: Q022 "STABLE_MIXED_COSMOLOGY_NUISANCE_MULTIBASIN_STRUCTURE";

Q032 V4 "stable_basin_count=0", "NO_CLEAN_LOCALIZATION...".

Constraint: Preserve model, likelihood, numerical settings and historical execution; test only launch/reference semantics.

3. HYPOTHESES

H1 / HYP-Q033-A: Q022 used non-scalar refs while Q032 used scalar refs.

H2 / HYP-Q033-B: Launch semantics are insufficient; another implementation difference exists.

Rejected: H1 — historical runtime reconstruction showed Q022 refs were already scalar.

Surviving: H2 — strengthened.

4. METHOD & EVIDENCE

Method: Source/provenance audit of Q019 Q021 Q022 versus Q032 V4; no new likelihood → → optimization.

Programs: "q033_launch_semantics_protocol_audit_v2.py"; config/source-lock/tests;

workflow ".github/workflows/q033-launch-semantics-protocol-audit-v2.yml".

Evidence: Q022 run "33902262660"; Q032 V4 run "34015845246"; Cobaya 3.5.6 semantics; frozen historical source blobs.

5. KEY RESULTS

- Q022 sampled non-primordial refs: scalar before/after recentering.

- Q032 V4 sampled refs: scalar.

- Remaining sampled-set difference: Q032 additionally samples "n_s", "logA", "tau_reio";

Q022 freezes them.

- Final result: "LAUNCH_REFERENCE_SEMANTICS_ALREADY_IDENTICAL_NOT_CAUSAL".

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Limit: Cause of Q022/Q032 discrepancy remains unresolved.

Contradiction: Stable Q022 versus non-stable Q032 remains a technical parameterization/implementation tension, not a physical anomaly.

6. NEGATIVE RESULTS / MIKAMI

Failed attempt: Q033 V1 preflight: "NONSCALAR_HISTORICAL_REFERENCE_GATE=FAIL"; no scientific likelihood result.

Graveyard: HYP-Q033-A and planned 18-job distribution-vs-scalar campaign — killed because the proposed historical distribution arm never existed.

7. MODEL CHANGE

Before: Q022/Q032 difference partly attributed to launch/reference semantics.

After: That explanation removed; "D-Q033-REMAINING-001" registered: frozen vs sampled primordial block.

Change: Technical interpretation corrected; physical EDE/ H_0 conclusions unchanged.

8. GITHUB / PROVENANCE

Repo: "Morfindien/Bubbleverse"

Q022 commit: "16a7902c..."

Q032 commit: "fbde93aee..."

Q033 files added: "b03a1ac5..."; workflow placement "614b227c..."

Result: "R-Q033-EDE-LAUNCH-SEMANTICS-AUDIT-002"

Q033 GitHub run ID: NOT RECORDED IN SUPPLIED FINAL ARTIFACT.

9. REPRODUCIBILITY

Checkout recorded versions obtain Q022/Q032 artifacts run Q033 V2 audit with frozen → →
source lock verify scalar-ref identity and sampled-set delta require → →

"FINAL_RESULT_GATE=PASS".

10. CONCLUSION

NO. Launch/reference representation cannot explain the Q022/Q032 classification difference because both executed paths used scalar pointlike refs. The smallest identified remaining difference is whether "n_s", "logA", "tau_reio" are frozen or sampled.

11. NEXT-Q HANDOFF

Next Q: Q034

Question: Under otherwise matched FULL_NATIVE CamSpec conditions, does freezing versus sampling "n_s", "logA", "tau_reio" reproduce the stable-versus-non-stable classification difference?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: NO — launch/reference semantics were already identical.

Q-JOURNAL: This entry.

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MIKAMI GRAVEYARD UPDATE: HYP-Q033-A + unnecessary 18-job V1 route killed.

MODEL CHANGE SUMMARY: Launch-semantics explanation removed; frozen-vs-sampled primordial block becomes next candidate.

REPRODUCIBILITY / GITHUB: Repo, commits, workflow and result Ids above.

NEXT-Q HANDOFF: Q034 tests "n_s/logA/tau_reio" freeze versus sampling.

BUBBLEVERSE Q JOURNALS

0.42

Q034

BUBBLEVERSE — Q-JOURNAL

1. IDENTITY

Q-ID: Q034

Date/time: 2026-09-06-07

Status: CONSTRAINED

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08 20:40 CEST

Question: Does fixing $\text{ns}, \log A, \tau, r, \text{eio}$ in the later full-native CamSpec construction restore the stable multibasin structure reported historically in Q022?

2. STARTING STATE

Model entering Q: MOD-EDE-N3 / ACTIVE, CONSTRAINED / not established new physics.

Inherited:

- Q032 V4 FULL_NATIVE: stable_basin_count = 0.
- Q033: scalar-vs-distribution launch semantics rejected; fixed-vs-sampled primordial block remained candidate difference.

Constraint: Change only primordial fixed/sampled status; preserve backend, likelihood construction, starts, optimizer and classifier.

3. HYPOTHESES

H1: Primordial freezing restores stable basins.

H2: It does not.

Rejected: H1.

Surviving: H2 under the later common-geometry classifier.

4. METHOD & EVIDENCE

Method: Artifact-locked matched isolation. Q032 FULL_NATIVE sampled control reused;

Q034 fixed $\text{ns}, \log A, \tau, r, \text{eio}$ and repeated required refinement.

Primary evidence: Morfindien/Bubbleverse; Q032 V4 artifacts; frozen mwt5345/class_edc commit "5a131c91d657dd9a7c6364cc45b038710f8d0d97".

Programs/workflows:

"q034_primordial_freeze_isolation_v1.py"

"q034_primordial_freeze_isolation_v1_config.yml"

".github/workflows/q034-primordial-freeze-isolation-v1.yml"

Run: GitHub Actions "34059649759", head "8b8035795a9e6064583379afb8663cdc28961764".

5. KEY RESULTS

- Sampled Q032 control: stable_basin_count = 0.

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- Frozen Q034 arm: `stable_basin_count = 0`; `stable_multibasin = false`.
- After refinement: 9 singleton clusters, not repeated stable basins.

Final result: Freezing the primordial block is insufficient to restore stable multibasin structure under the later classifier.

Limitation: Historical Q022 used a different basin classifier; global Q022 Q034 → comparison therefore remains unresolved.

Contradiction: Historical Q022 “stable” and later “non-stable” labels are not classifier-equivalent.

6. NEGATIVE RESULTS / MIKAMI

Failed hypothesis: Fixed-vs-sampled primordial parameters alone explain the discrepancy.

Graveyard: “D-Q033-REMAINING-001” — rejected as sufficient under the Q031/Q032/Q034 common-geometry classifier.

7. MODEL CHANGE

Before: Primordial parameterization remained leading unresolved implementation difference.

After: Insufficient under later classifier; classifier semantics become conclusion-critical.

Change: CONSTRAINED.

Reason: Frozen arm still produced zero stable basins.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commit: “8b8035795a9e6064583379afb8663cdc28961764”

Workflow: “q034-primordial-freeze-isolation-v1.yml”

Artifact: “q034-final-v1”, ID “9998302123”

SHA-256: “0df519645906dfaaf3b270d9791a74c0f33f25e9fdcfa8ca344ad2add70f8a87”

9. REPRODUCIBILITY

Checkout recorded commits obtain Q032 decisive artifacts run Q034 workflow with — — frozen primordial coordinates verify nine completed/refined endpoints apply identical — — later graph-classifier thresholds and compare final artifact.

10. CONCLUSION

Q034 rejects primordial freezing as a sufficient explanation within the later classifier. It

does not invalidate Q022, Q032, or Q034 raw results. The remaining discrepancy is methodological because Q022 and Q031–Q034 used different basin-classification definitions.

11. NEXT-Q HANDOFF

Next Q: Q035

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Why: Determine whether the stable/non-stable discrepancy survives use of identical classifiers.

Next question: When the decisive Q022, Q032 V4 FULL_NATIVE and Q034 frozen endpoint sets are evaluated under both classifier definitions, does the reported discrepancy survive classifier harmonization?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: Primordial freezing does not restore stable basins under the later classifier.

Q-JOURNAL: This record.

MIKAMI GRAVEYARD UPDATE: Fixed-vs-sampled primordial block killed as sufficient explanation under later classifier.

MODEL CHANGE SUMMARY: MOD-EDE-N3 unchanged physically; methodological interpretation constrained.

REPRODUCIBILITY / GITHUB: Run "34059649759"; head "8b803579..."; artifact "q034-final-v1".

NEXT-Q HANDOFF: Q035 — classifier harmonization.

BUBBLEVERSE Q JOURNALS

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Q035

BUBBLEVERSE — Q-JOURNAL

Q-ID: Q035

Date/time: 2026-09-08 20:42 CEST

Status: CONFIRMED

Question: When the same decisive Q022, Q032 V4 FULL_NATIVE and Q034 frozen-primordial endpoints are evaluated under both the historical Q022 classifier and later

Q031/Q032 classifier, does the reported stable/non-stable discrepancy survive harmonization?

1. STARTING STATE

Model: MOD-EDE-N3; n_scf=3; class_edc "5a131c91d657dd9a7c6364cc45b038710f8d0d97"; active/constrained, not established new physics.

Inherited: Q022 = historically STABLE_MULTIBASIN; Q032/Q034 = non-stable under later common-geometry classifier. Q033 rejected launch/reference semantics as cause.

Constraint: Existing endpoints only; no new likelihood, optimization, sampling, seeds, thresholds or physics changes.

2. HYPOTHESES

H1: Discrepancy survives same-classifier comparison. REJECTED.

H2: Classifier semantics materially explain discrepancy. CONFIRMED.

H3: Primordial freeze/sample difference is required explanation. REJECTED as necessary/sufficient.

3. METHOD & EVIDENCE

Deterministic artifact-only cross-classification of the same Q022/Q032/Q034 endpoint sets with both source-locked classifier definitions.

Evidence: Q022 run "33902262660"; Q032 run "34015845246"; Q034 run "34059649759".

Programs: "q022_globality_continuation_v2.py"; "q031_planck_portability_v1.py"; "q032_planck_tt3pair_bridge_v2.py"; "q035_classifier_harmonization_v1.py".

Tests: endpoint completeness; reference replay; historical-all-stable; later-all-nonstable; all-sets-flip; same-classifier-discrepancy-removed; physical-safety; FINAL_RESULT_GATE.

4. KEY RESULTS

Historical Q022 classifier: Q022/Q032/Q034 = STABLE.

Later common-geometry classifier: Q022/Q032/Q034 = NON-STABLE.

Thus both same-classifier cross-case discrepancies = FALSE and all endpoint sets flip together with classifier choice.

Final result:

"CLASSIFIER_SEMANTICS MATERIALLY EXPLAINS REPORTED DISCREPANCY".

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Limitations: Does not determine which classifier is physically preferable; does not establish identical raw geometry, Planck systematics, EDE evidence/falsification or new physics.

Contradiction discovered: CASE-031 negative portability interpretation is weakened because Q022 also fails its stable-basin gate under the same later classifier.

5. NEGATIVE RESULTS / MIKAMI

Original Q035 GitHub workflow failed with "FileNotFoundException" from Q022 artifact path layout; technical failure only, no scientific evidence.

Graveyard: same-classifier assumption; primordial freeze/sample as necessary/sufficient cause; interpretation that Q022 stable Q032/Q034 non-stable demonstrated — disappearance of physical basin structure.

6. MODEL CHANGE

Before: Q022 stable vs later non-stable treated as potentially genuine cross-case geometry change.

After: Stable/non-stable label explicitly classifier-dependent; raw endpoints retained.

CASE-031 portability conclusion reopened.

Change: Methodological interpretation modified; physical MOD-EDE-N3/H₀ status unchanged.

7. GITHUB / PROVENANCE

Repo: "Morfindien/Bubbleverse"

Commits: Q022 "16a7902..."; Q031 "ff621d6..."; Q032 "fbde93a..."; Q034 "8b80357..."; Q035 program "b1166e8..."; Q035 V2 workflow "e12bab3...".

Workflow: ".github/workflows/q035-classifier-harmonization-v2.yml"

Outputs: "q035_classifier_crosswalk_v1.json/.csv", "q035_tests_v1.json", source-lock/config files.

Note: Successful Q035 V2 Actions run ID NOT RECORDED / NOT AVAILABLE.

8. REPRODUCIBILITY

Checkout recorded versions obtain authoritative Q022/Q032/Q034 artifacts run Q035 — — crosswalk unchanged reproduce both classifier outputs require recorded gates and — — final classification. No endpoint rounding or threshold tuning.

9. CONCLUSION

The former Q022-versus-Q032/Q034 stable/non-stable discrepancy disappears when classifier choice is controlled. The endpoints remain valid; the basin label is not classifier-invariant.

10. NEXT-Q HANDOFF

Next: Q036 — HIGH

BUBBLEVERSE Q JOURNALS

0.42

Question: Under identical Q031 common coordinates/scales and without binary stable labels, do Q022 CamSpec and CASE-031 HiLLiPoP endpoints show a material reproducible geometric difference supporting implementation-specific non-portability?

Start motor: NUMERICAL / HPC / MOTOR-BUILDER; preferably artifact-only.

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: No; classifier semantics materially explain the reported discrepancy.

Q-JOURNAL: This entry.

MIKAMI: Same-classifier assumption and freeze/sample global explanation retired.

MODEL CHANGE: Basin ontology classifier-dependent; physical model unchanged. →

REPRODUCIBILITY: "Morfindien/Bubbleverse", sources/runs/commits above.

NEXT-Q: Q036 — harmonized CamSpec-HiLLiPoP raw-geometry portability test.

BUBBLEVERSE Q JOURNALS

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Q036

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08 20:43 CEST

1. IDENTITY

Q-ID: Q036

Date/time: 2026-09-07 09:44 CEST

Status: CONFIRMED

Question: Do Q022 CamSpec and CASE-031 HiLLiPoP endpoints show a material, reproducible difference under the same Q031 common coordinates/scales when stable/non-stable labels are not the sole discriminator?

2. STARTING STATE

Model entering Q: MOD-EDE-N3; ACTIVE / CONSTRAINED / NOT ESTABLISHED NEW

PHYSICS.

Inherited:

Q035: classifier semantics explain the former stable/non-stable discrepancy.

Q022 CamSpec + CASE-031 HiLLiPoP validated endpoint sets.

Constraints: Seven Q031 common coordinates; locked scales; threshold 0.10; no cross-likelihood χ^2 /objective subtraction or summation; no new likelihood/optimizer runs.

3. HYPOTHESES

Tested:

H1: Material continuous geometry difference survives.

H2: Difference disappears under common geometry.

H3: Classifier semantics explain the entire portability signal.

Rejected: H2/H3 — continuous location and shape differences remain large.

Surviving: H1 — CONFIRMED within tested geometry.

4. METHOD & EVIDENCE

Method: Deterministic artifact-only comparison of 9 matched endpoints plus 36 internal pairwise distances.

Evidence: Q022 run 33902262660; CASE-031 run 33975028436; Q035 classifier harmonization.

Program: q036_endpoint_geometry_portability_v1.py

Workflow: [.github/workflows/q036-endpoint-geometry-portability-v5.yml](#)

Authoritative run: 34105942065

Tests: identity/provenance; common-coordinate/scale lock; finite results; permutation invariance; Q031 replay; decision replay; no-cross-likelihood-objective gate.

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5. KEY RESULTS

Matched median RMS = 2.3623; 9/9 > 0.10.

Pairwise median drift = 2.2590; 36/36 > 0.10.

Centroid shift = 1.5371.

Q031 replay error = 8.88×10^{-16} .

FINAL_RESULT_GATE = PASS.

Final result: MATERIAL_REPRODUCIBLE_COMMON_GEOMETRY_DIFFERENCE.

Limitations: Does not identify the cause, a physical Planck systematic, preferred likelihood, or evidence for/against EDE. CamSpec/HiLLiPoP share Planck information.

Contradictions: None with Q035; Q035 concerns classifier labels, Q036 continuous geometry.

6. NEGATIVE RESULTS / MIKAMI

Failed attempt: Q036 V4 failed provenance gate from one incorrect artifact digest; technical only, superseded by V5.

Graveyard additions:

“No material CamSpec–HiLLiPoP geometry difference” — rejected.

Stable/non-stable label as sole portability evidence remains dead from Q035.

7. MODEL CHANGE

Before: CASE-031 portability interpretation weakened after Q035.

After: NARROWED_IMPLEMENTATION_SPECIFIC_NON_PORTABILITY_REHABILITATED.

Change: MODIFIED / REFINED.

Reason: Continuous common-geometry difference survives classifier harmonization.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Q036 execution commit: 095f90565cf694a7c3383d1fbe736707ab187751

Parent commits: Q022 16a790...; Q031 ff621d...

Workflow: q036-endpoint-geometry-portability-v5.yml

Outputs: q036-final-v1, q036-tests-v1, q036-return-handoff-v1, q036-run-manifest-v1.

Provenance: Existing validated Q022/Q031 artifacts only; no new likelihood evaluations.

9. REPRODUCIBILITY

Checkout recorded commits obtain locked Q022/Q031 artifacts run Q036 V5 workflow — —
compare outputs with q036-final-v1 apply locked 0.10 location/shape criteria. — —

10. CONCLUSION

Q036 confirms a large reproducible CamSpec–HiLLiPoP endpoint-geometry difference

under identical normalized coordinates/scales. Q035 remains correct: the old binary classifier argument was invalid, but continuous non-portability survives.

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11. NEXT-Q HANDOFF

Why: Cause of the implementation-dependent geometry remains unresolved.

Next Q: Q037

Question: Does the same material continuous geometry difference survive when Q032's exact TT common-support CamSpec/HiLLiPoP endpoints are compared using the Q036 metric?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: YES — material continuous common-geometry difference confirmed.

Q-JOURNAL: This entry.

MIKAMI: No-difference route rejected; classifier-only portability route remains dead.

MODEL CHANGE: CASE-031 narrowed non-portability rehabilitated.

REPRODUCIBILITY: Q036 V5, run 34105942065, listed commits/artifacts.

NEXT-Q: Q037 — common-support replay.

BUBBLEVERSE Q JOURNALS

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Q037

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q037

Date/time: 2026-09-07, CEST

Status: SUPPORTED

Question: When Q032's validated CamSpec/HiLLiPoP exact-TT common-support endpoints are tested with Q036's continuous seven-coordinate geometry, Q031 locked scales and 0.10 threshold, does the material location/shape difference persist?

2. STARTING STATE

Model: MOD-EDE-N3; ACTIVE / CONSTRAINED / NOT ESTABLISHED NEW PHYSICS.

Inherited: Q032 exact-TT matched endpoints; Q035 classifier discrepancy resolved; Q036 continuous geometry difference established.

Constraint: Artifact-only deterministic replay; same 7 coordinates/scales/threshold; no new likelihood/optimizer execution.

3. HYPOTHESES

H1: Exact matched TT support removes the geometry difference.

H2: Difference persists under matched TT support.

H3: Evidence is technically inconclusive.

Rejected: H1, H3.

Surviving: H2 — supported.

4. METHOD & EVIDENCE

Method: Compare 9 paired CamSpec/HiLLiPoP endpoints in normalized 7-D geometry; test matched displacement, centroid and pairwise-shape drift.

Evidence: Q032 authoritative endpoints, run 33994305721, commit "4dc873a5e880d40858d831a3b421456728f0c032".

Program: "q037_common_support_endpoint_geometry_v1.py" ("Q037-TTGEOM-V1")

Workflow: ".github/workflows/q037-common-support-endpoint-geometry-v1.yml"

Tests: finite/replay, permutation invariance, common-support identity, locked-scale identity, threshold and preservation gates.

5. KEY RESULTS

Matched displacement: median 2.93179, max 5.22954, 9/9 >0.10.

Centroid displacement: 0.83115.

Pairwise drift: median 1.63679, max 4.75557, 36/36 >0.10.

Replay max absolute difference: 0.0; FINAL_RESULT_GATE: PASS.

BUBBLEVERSE Q JOURNALS

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Final result: The material CamSpec–HiLLiPoP geometry difference persists on exact matched TT support.

Limitation: No causal implementation component identified; shared Planck PR4/NPIPE data are not independent observations.

Contradiction: None; Q037 narrows Q036.

6. NEGATIVE RESULTS / MIKAMI

Failed scientific attempts: None recorded. Earlier repository-registration 403 was technical and later superseded.

Graveyard: “Unequal TT support is sufficient/primary explanation” — rejected for the tested bridge because the difference survives exact common support.

7. MODEL CHANGE

Before: C-036-IMPL active; TT-support mismatch remained plausible.

After: C-036-IMPL strengthened; tested TT-support mismatch weakened as primary cause.

Change: CONSTRAINED.

Reason: 9/9 matched and 36/36 pairwise differences remain material.

8. GITHUB / PROVENANCE

Repository: “Morfindien/Bubbleverse”

Commits: “9ff5f29d6ae1f5f5f702ccb98e034fb1075326ba”;
“c49d9885a5bfl5c59d228c7c036cf”

Result: “R-Q037-EDE-TT-COMMON-SUPPORT-GEOMETRY-001”; “q037_final_v1.json”

GitHub scientific run ID: NOT RECORDED.

Provenance: Deterministic replay of Q032 validated endpoints; no new physics execution.

9. REPRODUCIBILITY

Checkout recorded commits obtain Q032 artifacts run Q037 workflow/program — — —
compare with “q037_final_v1.json” require same locked geometry and 0.10 criterion plus —

FINAL_RESULT_GATE PASS.

10. CONCLUSION

Exact matching of TT information support does not remove the CamSpec–HiLLiPoP endpoint-geometry difference. This strengthens an unresolved implementation-level discrepancy but establishes no physical Planck systematic, EDE detection/falsification, H_0 change or new physics.

11. NEXT-Q HANDOFF

Next Q: Q038

Question: Is the surviving common-support geometry difference concentrated in a reproducible minimal subset of the seven coordinates, or distributed across several coordinates?

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REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: YES — the material geometry difference persists under exact matched TT support.

Q-JOURNAL: This record.

MIKAMI: TT-support mismatch rejected as sufficient/primary explanation for the tested discrepancy.

MODEL CHANGE: C-036-IMPL strengthened/constrained; cause unresolved.

REPRODUCIBILITY: Q032 run 33994305721 + Q037 program/workflow/commits above.

NEXT-Q: Q038 — coordinate localization.

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Q038

BUBBLEVERSE — Q-JOURNAL

JOURNAL TYPE: RETROSPECTIVE RECONSTRUCTION

RECONSTRUCTED ON: 2026-09-08

1. IDENTITY

Q-ID: Q038

Date/time: 2026-09-07 13:48 CEST

Status: CONFIRMED

Question: When Q037's paired CamSpec/HiLLiPoP exact-common-support endpoints are decomposed coordinate-by-coordinate under the seven Q031 locked scales, is the surviving implementation-geometry difference concentrated in a reproducible minimal subset or distributed across several coordinates?

2. STARTING STATE

Model: MOD-EDE-N3; nsc f=3; ACTIVE / CONSTRAINED / NOT ESTABLISHED NEW PHYSICS.

Frozen "mwt5345/class_ede" commit "5a131c91...".

Inherited: Q035 removed classifier semantics as a valid geometry discriminator; Q036 established a material continuous implementation difference; Q037 showed exact TT common support does not remove it.

Constraints: Same seven Q031 coordinates/scales; no new likelihood/CLASS/optimizer/sampler runs; no cross-likelihood objective arithmetic; causal attribution forbidden.

3. HYPOTHESES

Tested:

H1: discrepancy concentrated in 2 coordinates. $\leq$

H2: discrepancy distributed across 3 coordinates. $\geq$

H3: apparent localization fails robustness tests.

Rejected: H1 — no 1-, 2-, or 3-coordinate subset met the preregistered joint criterion.

Surviving: H2 — distributed multivariate geometry.

4. METHOD & EVIDENCE

Deterministic artifact-only decomposition of validated Q037 endpoints. All 127 non-empty subsets tested in matched-location, centroid and pairwise-shape channels; full capture 0.80; leave-one-label-out 0.70 with 8/9 passes. $\geq \geq \geq$

Primary evidence: Q032/Q037 endpoint artifacts; CamSpec/HiLLiPoP Planck PR4/NPIPE methodology (K-044, K-049; comparison K-045/K-052).

Program/workflow: "q038_coordinate_localization_v1.py"; ".github/workflows/q038-coordinate-localization-v1.yml".

GitHub run: "34121000822"; head "02ddda7c13a53cd7ad20880bf527a86e17ac4d79".

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5. KEY RESULTS

Minimal qualifying subset size: 4

Unique subset: $\omega_b, \omega_c, d, m, f, E, D, E, \theta_i, sc, f$

Capture: 90.89% matched / 88.48% centroid / 87.20% pairwise-shape.

LOO robustness: 9/9 in all three channels.

FINAL_RESULT_GATE: PASS.

Final result: The surviving CamSpec/HiLLiPoP implementation-geometry difference is DISTRIBUTED, not concentrated in one or two common coordinates.

Limitations: Parameter-space localization is not causal attribution; CamSpec/HiLLiPoP are not independent observations; no H_0 benchmark or EDE viability conclusion changes.

Contradictions: None.

6. NEGATIVE RESULTS / MIKAMI

Failed routes: Single-coordinate, H_0 -only, A Pl anc k-only and two-coordinate concentration explanations are insufficient under the frozen criterion.

Mikami Graveyard:

- 2-coordinate concentration — rejected. $\leq$
- Simple H_0 -only / A Pl anc k-only explanation — weakened.

Reason: neither appears in the unique minimal robust explanation of the complete geometry.

7. MODEL CHANGE

Before: C-036-IMPL = material implementation-dependent geometry difference; cause unresolved.

After: C-036-IMPL = material, exact-common-support-surviving, distributed multivariate geometry difference; cause unresolved.

Change: PRÆCISERET / CONSTRAINED.

MOD-EDE-N3 status: unchanged.

8. GITHUB / PROVENANCE

Repository: "Morfindien/Bubbleverse"

Parent Q037 run: "34115280043", head "c49d9885..."

Q038 run: "34121000822", head "02ddda7c..."

Program: "q038_coordinate_localization_v1.py"

Workflow: ".github/workflows/q038-coordinate-localization-v1.yml"

Result: "R-Q038-EDE-COORDINATE-LOCALIZATION-001"

Provenance: deterministic reanalysis of validated Q037/Q032 endpoint lineage; no new observational data or likelihood execution.

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9. REPRODUCIBILITY

Checkout recorded commits obtain Q037 endpoint artifact run Q038 → →
program/workflow with locked Q031 scales and preregistered thresholds verify subset →
captures/LOO tests require FINAL_RESULT_GATE PASS and classification replay. →

10. CONCLUSION

Q038 establishes that the surviving CamSpec/HiLLiPoP implementation difference is

genuinely multivariate under the tested geometry. At least four common coordinates are required for a robust compact representation. The calculation narrows the methodological problem but does not identify which likelihood implementation component causes it.

11. NEXT-Q HANDOFF

Why: Causal implementation source remains unresolved.

Next Q: Q039

Question: Can a controlled matched intervention isolate one implementation block as sufficient to explain/reduce C-036-IMPL, or are coupled implementation changes required?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: DISTRIBUTED; unique minimal robust subset = $\omega_b + \omega_c d_{m+f} E D E + \theta_{i,sc} f$.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: 2-coordinate concentration rejected; simple $H_0 / \leq A Pl$ and k-only narratives weakened.

MODEL CHANGE SUMMARY: C-036-IMPL refined to distributed multivariate geometry;

MOD-EDE-N3 unchanged.

REPRODUCIBILITY / GITHUB: "Morfindien/Bubbleverse"; runs "34115280043",
"34121000822"; Q038 workflow/program above.

NEXT-Q HANDOFF: Q039 — causal implementation-block attribution.

BUBBLEVERSE Q JOURNALS

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Q039

BUBBLEVERSE — Q-JOURNAL

1. IDENTITY

Q-ID: Q039

Date/time: 2026-09-08 20:46 CEST

Status: INCONCLUSIVE

Question: When CamSpec and HiLLiPoP on Q032 exact TT common support are investigated with preregistered, native-parameter-respecting matched interventions in their implementation blocks, can one single implementation block materially explain/reduce the Q037/Q038 distributed common-geometry difference, or are coupled changes across multiple blocks required?

2. STARTING STATE

Model: MOD-EDE-N3, nsc f=3, ACTIVE / CONSTRAINED / NOT ESTABLISHED NEW PHYSICS.

Inherited: Q037 common-support discrepancy: matched 2.93179, centroid 0.83115, pairwise

1.63679. Q038: discrepancy DISTRIBUTED; minimal robust subset $\omega_b, \omega_c, d, m, f, E, D, E, \theta_i, sc, f$.

Constraints: Frozen Q032 TT support/backend/scales; threshold 0.10; no cross-likelihood $\leq$ objective comparison; native parameter semantics only.

3. HYPOTHESES

Tested:

H1 relative calibration explains discrepancy.

H2 off-diagonal precision coupling explains discrepancy.

H3 calibration+precision explains discrepancy.

H4 native foreground profiling freedom explains discrepancy.

Rejected as sufficient: H1–H4; all remained far above threshold, 0/9 LOO sufficient.

Surviving: deeper coupled likelihood/data/foreground/covariance construction remains possible but unproven.

4. METHOD & EVIDENCE

Methods: Matched implementation interventions; BOBYQA reoptimization; full-sample geometry + 9 leave-one-label-out tests.

Sources: Rosenberg et al. 2022, arXiv:2205.10869; Efstathiou & Gratton 2021,

arXiv:1910.00483; Tristram et al. 2024, DOI 10.1051/0004-6361/202348015; Jense et al. 2026,

arXiv:2510.09430; HiLLiPoP commit a09ddde3; Cobaya 3.5.6.

5. KEY RESULTS

Calibration: 2.84864 / 0.87392 / 1.48583.

Precision: 2.60701 / 0.84119 / ~1.62217.

Calibration+precision: 3.07861 / 1.19445 / 1.87403.

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Foreground-profile: 2.98293 / 1.05547 / 1.63532, LOO sufficient 0/9.

Final result: No tested scientifically matchable single block explains C-036-IMPL. Deeper coupled construction remains unresolved.

Limitations: Full foreground/core-likelihood harmonization cannot be isolated as a clean native single-block intervention. No physical systematic or new physics established.

6. NEGATIVE RESULTS / MIKAMI

Failed technical attempts: Q039 V1-V4 and early FGPROFILE launcher/static failures; workflow/infrastructure only, no scientific evidence.

Graveyard: calibration-only, precision-only, calibration+precision, native foreground-profile freedom as sufficient explanations. Synthetic cross-likelihood nuisance/core hybrids rejected as scientifically invalid designs.

7. MODEL CHANGE

Before: C-036-IMPL active with several possible isolated implementation causes.

After: C-036-IMPL remains active, but single-block explanation space is strongly narrowed.

Change: CONSTRAINED.

Reason: validated negative Q039 interventions.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Backend: class_edc "5a131c91d657dd9a7c6364cc45b038710f8d0d97"

Q032: run 33994305721, commit "4dc873a5e880d40858d831a3b421456728f0c032"

Q039 V5: run 34161368438, commit "365217d4a4ca21e0af3b296c0e6b23df7e07ba60",

workflow "q039-implementation-block-intervention-v5.yml"

FGPROFILE: run 34184346582, commit "449fe096e49a793446c0c859cf8caf0fcdaf4c8f",

workflow "q039-native-foreground-profile-v1.yml"

Results: R-Q039-EDE-IMPLEMENTATION-BLOCK-INTERVENTION-001; R-Q039-EDE-NATIVE-

FOREGROUND-PROFILE-001. FINAL_RESULT_GATE PASS.

9. REPRODUCIBILITY

Checkout recorded commits obtain frozen Q032 endpoints/support run recorded → →
workflows with frozen parameters reproduce geometry/LOO outputs apply 0.10 → → ≤
sufficiency criterion.

10. CONCLUSION

Q039 closes INCONCLUSIVE but strongly constraining. The CamSpec-HiLLiPoP discrepancy

survives all scientifically clean tested single-block interventions. Its remaining cause lies, if identifiable, in deeper coupled implementation structure. This is methodological evidence only.

11. NEXT-Q HANDOFF

Next Q: Q040

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0.42

Why: determine whether a finite, scientifically defensible coupled structural intervention can explain C-036-IMPL without constructing a meaningless synthetic likelihood.

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: No tested clean single implementation block is sufficient; coupled deeper structure remains unresolved.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Calibration, precision, calibration+precision and native foreground-profile freedom killed as sufficient explanations.

MODEL CHANGE SUMMARY: C-036-IMPL retained but substantially constrained.

REPRODUCIBILITY / GITHUB: Runs 34161368438 and 34184346582; commits and workflows above.

NEXT-Q HANDOFF: Q040 — test whether a scientifically valid coupled structural bridge can be defined.

BUBBLEVERSE Q JOURNALS

0.42

Q040

Q041

BUBBLEVERSE — Q-JOURNAL

1. IDENTITY

Q-ID: Q040

Date/time: 2026-09-08 22:47 CEST

Status: INCONCLUSIVE

Question: Can a finite, preregistered and scientifically defensible coupled structural intervention involving documented CamSpec-versus-HiLLiPoP differences in foreground-model form, data-vector construction/weighting and covariance/likelihood construction materially reduce C-036-IMPL on the Q032 matched-support architecture without constructing a scientifically meaningless synthetic likelihood?

2. STARTING STATE

Model entering Q: MOD-EDE-N3, n_scf=3 — ACTIVE / CONSTRAINED / NOT ESTABLISHED

NEW PHYSICS.

Inherited: Q036–Q038 established reproducible, distributed CamSpec–HiLLiPoP geometry discrepancy; Q039 rejected calibration, precision coupling and native foreground profiling as sufficient explanations.

Constraints: Frozen Q032 TT support, 7D geometry, threshold 0.10; CamSpec/HiLLiPoP remain native separate likelihoods; no synthetic likelihood; A_planck explicit.

3. HYPOTHESES

Tested:

H1: Valid common-CMB latent representation exists.

H2: Single-Gaussian nuisance marginalization can support the test.

H3: Defensive non-Gaussian RQMC can produce validated endpoints.

Rejected: H2 — CamSpec boundary/non-PD Hessian and HiLLiPoP multistart instability.

Surviving: H1 mathematically supported; H3 numerically unresolved. C-036-DEEP-STRUCTURAL remains active/unresolved.

4. METHOD & EVIDENCE

Methods: Native nuisance marginalization; Laplace/Schur validation; defensive importance sampling + Owen-scrambled Sobol RQMC, 4 replicates, m=9–16.

Sources: K-044 CamSpec; K-049 HiLLiPoP; K-053/K-054 official implementations; K-055 boundary statistics; K-056 mixed Laplace; K-057 defensive importance sampling; K-058

RQMC.

Programs/workflows: "q040_defensive_rqmc_v4.py"; ".github/workflows/q040-defensive-rqmc-v4.yml"; GitHub run 34258155387.

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Tests: Frozen $\Delta\chi^2$ convergence 0.05; Bank A/B 0.05; downstream science endpoints $\leq$ allowed only after validation.

5. KEY RESULTS

Single-Gaussian compression: COMPRESSION_VALIDATION_FAIL.

RQMC reached hard cap $m=16$ / 393,216 nodes per replicate/implementation.

CamSpec: transition 16.5899; Bank A/B 28.4438.

HiLLiPoP: transition 61.1260; Bank A/B 63.1116.

Required: 0.05. $\leq$

Final result: "DEFENSIVE_MARGINALIZATION_NUMERICAL_VALIDATION_FAIL".

Science endpoints, final geometry and 9/9 LOO were correctly blocked.

Limitation: No conclusion on whether a successfully validated structural bridge would reduce C-036-IMPL.

Contradictions: None; Q036–Q039 remain compatible.

6. NEGATIVE RESULTS / MIKAMI

Failed attempts: CMB-space V1–V3 technical failures; RQMC V1 serialization, V2 GitHub artifact transport, V3 deployment failure. No physical evidence.

Graveyard additions: Single-Gaussian Laplace/Schur route killed for Q040 science; finite

Q040 RQMC campaign killed as numerically validated bridge at frozen hard cap.

7. MODEL CHANGE

Before: C-036-IMPL active; deeper structural origin unresolved.

After: Same physical/model status, but deeper bridge is mathematically defined and tested numerical routes constrained.

Change: CONSTRAINED / REFINED.

Reason: Both Gaussian compression and finite defensive RQMC failed preregistered validation.

8. GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Commit: 855a28c58246dbdd52d01cae1f40e0611103d96b

Workflow: ".github/workflows/q040-defensive-rqmc-v4.yml"

Program: "q040_defensive_rqmc_v4.py"

Result: R-Q040-EDE-DEFENSIVE-RQMC-CMB-MARGINAL-004

Artifacts: "q040-rqmc-final-v4"; "q040-rqmc-integration-final-v4"; "q040-rqmc-state-m16-v4".

Provenance: Frozen Q032/Q037 architecture + native CamSpec/HiLLiPoP likelihoods; preregistered finite campaign completed through hard cap.

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9. REPRODUCIBILITY

Checkout recorded commits; obtain Q032/Q037 parent artifacts and native likelihoods; run

Q040-RQMC-V4 with frozen parameters; reproduce m9–m16 states; apply unchanged 0.05

validation gates and 0.10 science criterion.

10. CONCLUSION

A scientifically defensible common native-nuisance-marginalized CMB representation exists mathematically, but Q040 did not obtain a numerically validated implementation capable of testing whether it materially reduces C-036-IMPL. The discrepancy therefore remains active and unexplained; no pipeline superiority, EDE confirmation/falsification or new physics is established.

11. NEXT-Q HANDOFF

Why: The dominant source of RQMC instability remains unidentified.

Next Q: Q041

Question: Is Q040's finite defensive-RQMC nonconvergence dominated by a small identifiable subset of nuisance dimensions, boundaries, proposal strata or likelihood-specific directions, or is it broadly distributed?

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: INCONCLUSIVE — valid bridge definition exists, but finite numerical validation failed.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Gaussian compression and Q040 finite-RQMC science route rejected under tested conditions.

MODEL CHANGE SUMMARY: Model unchanged; methodological space narrowed.

REPRODUCIBILITY / GITHUB: Morfindien/Bubbleverse, commit 855a28c..., run

34258155387, Q040-RQMC-V4.

NEXT-Q HANDOFF: Q041 — localize the source of RQMC instability.

LATER ROUTING NOTE (2026-09-24): The Q040 handoff above is preserved as historical research provenance. The Q041 execution that was ultimately preregistered and run addressed downstream scientific-consequence portability rather than the proposed RQMC-instability localization. Q040 is not reopened by this routing change.

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Q041

BUBBLEVERSE — Q-JOURNAL

IDENTITY

Q-ID: Q041

Date/time: 2026-09-24 CEST

Status: INCONCLUSIVE / COMPUTATIONAL CAMPAIGN COMPLETE / ORIGINAL SCIENTIFIC QUESTION UNRESOLVED

Question: Does the validated CamSpec–HiLLiPoP fitted-geometry difference in $n=3$ EDE produce a materially different downstream cosmological inference when the Planck likelihood arms are combined with matched external information?

STARTING STATE

Model entering Q: MOD-EDE-N3, $n_{\text{scf}}=3$ — ACTIVE / CONSTRAINED / NOT ESTABLISHED

NEW PHYSICS.

Inherited: Q035 showed that the historical stable/non-stable basin contrast is classifier dependent; Q036 established a material continuous CamSpec–HiLLiPoP geometry difference; Q037 showed that the difference survives exact-common-TT support; Q038 localized the compact multivariate structure to ω_b , ω_{cdm} , f_{EDE} , and $\theta_{\text{i,scf}}$; Q039 rejected calibration, precision coupling, calibration+precision, and native foreground-profile freedom as sufficient explanations; Q040 defined a legitimate common-CMB-space marginalization target but failed numerical validation before science endpoints.

Hard constraints: the Planck arms remain separate; both arms use the same frozen $n=3$ EDE backend and matched external data; cross-arm absolute likelihood/ χ^2 subtraction is forbidden as physical evidence; failed or unconverged chains are not scientific evidence;

Q040-CMBSPACE-* and Q040-RQMC-* scientific products are forbidden as Q041 inputs.

HYPOTHESES / DECISION CLASSES

V19 preregistered three possible scientific classes after all required gates:

MATERIAL_DOWNSTREAM_DIFFERENCE, SCIENTIFICALLY_EQUIVALENT_CONSTRAINTS,
or CONSTRAINED_MIXED.

A technical or convergence gate failure instead yields NO_SCIENTIFIC_RESULT. That state is a control outcome and is not a fourth physical class.

METHOD & EVIDENCE

Authoritative program: Q041-PLANCKPORT-V19; result R-Q041-EDE-DOWNSTREAM-PORTABILITY-019; workflow [.github/workflows/q041-planck-portability-v19.yml](#).

V19 arms: camspec and hillipop. Data combinations: P, P_A6, P_L6, P_D2, P_A6_L6_D2, P_L6_D2, P_A6_D2, P_A6_L6. Two independent chains per arm/combination produced 32 logical chains.

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External inputs: ACT DR6 primary through DR6-ACT-lite v1.0.1 commit

880eacb40d66722eb1c32d7b5621e91662b4d808; ACT DR6 lensing through act_dr6_lenslike v1.2.1 commit b386ddbb5821c1216c709f051c9289292f174d30; DESI DR2 BAO through bao_data v2.6 commit b7b8a36e9bccb063081f811f323cada21ab5fbdd and Cobaya definition commit b76b6fed2a6c8c5594c6f92d5058bef10079746a.

Frozen parents and software: Q032 result R-Q032-EDE-PLANCK-TT3PAIR-BRIDGE-002, GitHub run 33994305721, execution commit

4dc873a5e880d40858d831a3b421456728f0c032; class_edc commit

5a131c91d657dd9a7c6364cc45b038710f8d0d97; HiLLiPoP commit

a09dde3e7ce11df99f74685feb1f1764cafb251; Cobaya 3.5.6.

Overlap policy: without A6, use the full authoritative Q032 exact-common-TT support. With A6, further restrict Planck primary TT to $\ell \leq 599$ while ACT primary TT/TE/EE begins at $\ell = 600$; cross-likelihood covariance is assumed zero only after that disjoint primary-multipole construction.

Sampler: Cobaya MCMC; Rminus1_stop = 0.02; Rminus1_cl_stop = 0.2; hard science R-hat gate ≤ 1.05 ; max_samples = 20,000 accepted samples per chain; maximum eight successful compute segments per chain; 300-minute segment soft stop; 330-minute GitHub job timeout.

Execution architecture: one fixed linear eight-stage workflow, 32 matrix jobs per segment, max-parallel = 6. Each segment transports only the exact immediate-predecessor same-run artifact. Cross-version chain-state reuse is forbidden. HiLLiPoP resume is locked to the serialized planck_2020_hillipop.TT component in chain.updated.yaml; Cobaya --allow-changes is not used.

KEY RESULTS

GitHub Actions run 34979609004 completed successfully on attempt 4 at execution commit a962ba70077422f8b69642dea4e8272e1d00e5ff.

Final artifact: q041-final-v19, artifact ID 10754769330, SHA-256

2284518ecbaaf1ff0d0093977827d5162ec29549ba2b81880f266c250311ca2b.

The final JSON reports scientific_classification = NO_SCIENTIFIC_RESULT; outcome_type = CONTROLLED_NO_SCIENTIFIC_RESULT; no_science_reason =

MAX_SAMPLES_WITHOUT_CONVERGENCE; actual_computed_result = false;

technical_failure = false; final_outcome_valid = true; status = PASS.

Thirty logical chains are listed as MAX_SAMPLES_WITHOUT_CONVERGENCE. The two remaining chains — CamSpec / P_A6 / c1 and CamSpec / P_A6_L6_D2 / c1 — reached segment 8 still PARTIAL after all eight compute segments. The combined record therefore contains 0/32 chains with documented COMPLETE status.

Final science gates: CHAIN_COMPLETENESS = BLOCKED; ALL_RHAT_LE_1P05 = BLOCKED;

BOTH_ARMS_ALL_COMBINATIONS = BLOCKED; Q040_FIREWALL = PASS;

SEGMENTED_RESUME_COMPLETENESS = CONTROLLED_STOP.

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No authoritative full-matrix standardized shifts, Bhattacharyya coefficient, or leave-one-out portability classification may be interpreted scientifically because the prerequisite convergence/completeness gates were not satisfied.

CONTRACT AUDIT

The executed V19 preregistration is not identical to the broader Q041 scientific design that preceded it. The broader design required matched CamSpec and HiLLiPoP arms, both Λ CDM and $n=3$ EDE, ACT DR6 primary, ACT DR6 lensing, DESI DR2, one common frozen supernova likelihood, a FULL combination plus four leave-one-out combinations, within-arm Λ CDM-versus-EDE model-preference information, and the original materiality/overlap decision rules.

V19 instead executes MOD-EDE-N3 only, omits a separate Λ CDM matrix and the supernova likelihood, uses three leave-one-out combinations, and applies its own six-coordinate standardized-shift and Gaussian Bhattacharyya classification.

This is a material scope/provenance difference. It does not retroactively invalidate V19: V19 had its own preregistration before the final outcome. It does mean that V19 cannot be described as the complete execution of the broader original downstream contract.

NEGATIVE RESULTS / MIKAMI

Killed interpretation: Q041 non-convergence => CamSpec and HiLLiPoP physically differ.

Killed interpretation: Q041 non-convergence => CamSpec and HiLLiPoP are scientifically equivalent.

Killed interpretation: a green GitHub Actions conclusion => a green cosmological result.

Retained prohibition: failed technical runs, timeouts, runner failures, dependency failures, and resume bugs carry no physical evidentiary weight.

Retained prohibition: Q040 invalidated science endpoints may not be resurrected as Q041 evidence.

MODEL CHANGE

Physical model change: NONE.

C-036-IMPL remains active as a methodological likelihood-geometry discrepancy. Q041 does not determine its downstream physical significance.

New computational knowledge: under the frozen V19 parameterization, sampler settings, and compute/sample budget, a complete converged posterior matrix was not obtained.

New provenance knowledge: the executed V19 test is narrower than the broader original

Q041 scientific contract.

GITHUB / PROVENANCE

Repository: Morfindien/Bubbleverse

Branch: main

Execution commit: a962ba70077422f8b69642dea4e8272e1d00e5ff

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GitHub Actions run: 34979609004; attempt 4; status/conclusion: completed / success.

Workflow: .github/workflows/q041-planck-portability-v19.yml

Program: q041_planck_portability_v19.py

Preregistration: q041_planck_portability_preregister_v19.json

Source lock: q041_planck_portability_source_lock_v19.json

Validator: q041_planck_portability_tests_v19.py

Final artifact: q041-final-v19; ID 10754769330; SHA-256

2284518ecbaaf1ff0d0093977827d5162ec29549ba2b81880f266c250311ca2b.

q041_final_v19.json SHA-256:

0062874ed3529004f46cfb542a708d5b8e8dde55d1b4fd1c2ac7c394e6c8be79.

q041_final_tests_v19.json SHA-256:

95fe1e63c78033de8559f609f0d8a1f4d48ee1b4123f3bb8f845b3ac6e5797ab.

GitHub status success validates the workflow-defined terminal control state. It does not convert blocked scientific gates into passed scientific gates.

REPRODUCIBILITY

Reconstruct V19 from the recorded execution commit; verify the preregistration and source lock; reproduce the same external source versions and Q032 parent; preserve the eight-stage same-run resume contract; compare the final artifact and validator output; require the same science gates before any posterior portability interpretation.

Any successor campaign must receive a new version/result identity. Raising max_samples or changing the sampler after observing V19 without a new preregistered configuration would retroactively change the test rather than reproduce it.

CONCLUSION

Q041-V19 is complete as a computational campaign and valid as a controlled no-science

result. It did not produce a converged downstream cosmological portability classification.

The original downstream physical question therefore remains unresolved. V19 establishes neither material downstream difference nor scientific equivalence nor dataset-conditional mixed behaviour.

Because the executed V19 contract is also narrower than the broader original design, a future test must explicitly state which scientific contract it is executing rather than silently merging the two.

NEXT-Q HANDOFF

Next Q: Q042 — HIGH

Question: Can the original downstream-portability test be executed with a new preregistered, convergence-robust inference strategy that preserves both native Planck arms, Λ CDM and $n=3$ EDE, ACT DR6 primary, ACT DR6 lensing, DESI DR2, one common frozen supernova likelihood, FULL plus four leave-one-out tests, and the original

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materiality/model-preference rules, without reusing Q040 invalidated science endpoints or moving the criteria after results are seen?

Recommended start motor: BUBBLEVERSE — AUTONOMOUS EXECUTION-MECHANISM / **NUMERICAL / HPC / MOTOR-BUILDER ENGINE.**

Constraint: diagnose existing V19 chains/progress/checkpoints before authorizing another large compute campaign. Do not simply “run longer.”

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: CONTROLLED_NO_SCIENTIFIC_RESULT — the V19 campaign did not reach the convergence/completeness gates required for scientific classification; the broader original downstream question remains unresolved.

Q-JOURNAL: This entry.

MIKAMI GRAVEYARD UPDATE: Non-convergence cannot be used as evidence for either physical agreement or disagreement; green workflow status is not a scientific result.

MODEL CHANGE SUMMARY: No physical model change; computational and provenance state refined.

REPRODUCIBILITY / GITHUB: Q041-PLANCKPORT-V19; R-Q041-EDE-DOWNSTREAM-

PORTABILITY-019; run 34979609004; execution commit

a962ba70077422f8b69642dea4e8272e1d00e5ff; final artifact digest above.

NEXT-Q HANDOFF: Q042 — design and validate the correct convergence-robust execution mechanism for the unresolved downstream scientific contract.

Q042

BUBBLEVERSE - Q-JOURNAL

IDENTITY

Q-ID: Q042 Case-ID: NOT DOCUMENTED Date/time: 2026-10-07 08:00 CEST closure state Status: CLOSED - INCONCLUSIVE
GENERAL FEASIBILITY / NO NEW COSMOLOGICAL INFERENCE

QUESTION

Can the original Q041 downstream-portability test be executed using a newly preregistered, convergence-robust strategy that preserves both native Planck arms, LCDM and $n=3$ EDE, ACT DR6 primary, ACT DR6 lensing, DESI DR2, one identical frozen supernova likelihood, FULL plus four leave-one-out tests and the original materiality/model-preference rules, without reusing invalidated Q040 science endpoints, cross-arm absolute-objective arithmetic or retrospective criteria changes?

STARTING STATE

Q041-V19 was complete as a controlled computational campaign but produced NO_SCIENTIFIC_RESULT: 30 chains reached the sample limit, two remained partial, and 0/32 had documented COMPLETE status. Its executed contract was narrower than the original downstream specification because it omitted the separate LCDM matrix and common supernova block and used three rather than four leave-one-out combinations. The physical downstream significance of the CamSpec-HiLLiPoP fitted-geometry difference therefore remained unresolved.

Q040 science endpoints remained forbidden inputs. Cross-arm absolute likelihood or chi-squared subtraction remained forbidden as physical evidence. A green workflow status could not substitute for convergence, completeness or scientific qualification.

FROZEN SCIENTIFIC CONTRACT

The retained contract required two native Planck arms x two models (LCDM and $n=3$ EDE) x FULL plus four leave-one-out combinations, for 20 required inference cells. External blocks remained ACT DR6 primary, ACT DR6 lensing, DESI DR2 and one identical frozen supernova likelihood. Original priors, overlap rules, materiality rules and model-preference criteria were unchanged.

Scientific specification SHA-256: 41e4d8f026933f64dbe41048f894002f479ba2b504432b33dc155f4dfe71378c.

PRODUCTION HISTORY

The original Q042 posterior branch failed before completed posteriors were produced. Initial sampler-option leakage was a configuration failure, not a measured convergence failure. All 80 optimizer records survive: 78 flag-0 records and two flag-3 failures. Their existence does not certify global minima and does not provide posterior portability evidence.

Subsequent bounded diagnostics progressively isolated a numerical reference-construction problem. V22 tested bounded branch durability/checkpoint behaviour. V23 exposed positive-node spline overshoot in the original binary. V24 recorded a nonfinite upper optical-depth trial followed by successful return and zero bisection. V25 reproduced a source-consistent finite-input helium-cutoff overflow followed by propagation of nonfinite state. V26-V28 were retained as RAW_NOT_QUALIFIED diagnostics. V29 initialized the context and 24 explicit interval-box prerequisites but did not qualify a complete history or a scientific result. V30 produced partial conditional histories and failed its frozen inflation construction.

V31 BOUNDED COMPARISON

V31 produced partial conditional histories plus a separate necessary support-cap incompatibility. Both local attempts stopped at the parent 600-second deadline.

UPPER: 3,741 retained certificates; 3,233/3,334 post-entry native nodes; last retained certificate endpoint $z = 1.501953125$. LOWER: 3,579 retained certificates; 3,210/3,334 post-entry native nodes; last retained certificate endpoint $z = 1.849609375$.

Final committed endpoints, complete final counters and peak memory after the parent kill are NOT DOCUMENTED. No full tau value was computed.

CONDITIONAL MATHEMATICAL RESULT

Frozen support-candidate cap: 32. Retained conservative candidate counts: UPPER 138; LOWER 3,002. Existing minimum upper bounds: UPPER 0.00024287858999792338; LOWER 0.00024212042562975905. Certified missing-tail source lower bounds: UPPER 1.1628411809721158; LOWER 0.08148057013890592.

Under the frozen source extension, global rails and append-only retention of the existing intervals, the missing tail cannot lower the relevant minimum threshold or remove the existing candidates. Completing that unchanged representation therefore cannot satisfy cap32.

This is a conditional failure of an interval representation and output policy. It is not evidence for 138 or 3,002 physical minima, an executed tau failure, or impossibility of every alternative method.

QUALIFICATION BOUNDARY

The construction concerns existing absolutely continuous solutions of the frozen real-table right-hand side under its stated domain assumptions. It does not establish existence or uniqueness for the original switched right-hand side, original-binary arithmetic accuracy, upstream physical/table accuracy, a justified downstream prediction/likelihood error budget, a valid complete reference, or the convergence-qualified 20-cell comparison.

CURRENT GATES

Q completion: PASS - documented inconclusive closure. Whole history: PARTIAL. Actual support/spline/tau pipeline: NOT EXECUTED. Unchanged support-cap condition: FAIL within the stated conditional scope. Original-binary arithmetic accuracy: UNRESOLVED. Upstream accuracy: UNRESOLVED.

Downstream acceptance budget: NOT DOCUMENTED. Reference truth: BLOCKED. Numerical final result: UNRESOLVED.
Production restart: NOT AUTHORIZED.

NEGATIVE RESULTS / MIKAMI

Killed as sufficient repair: simply running the unchanged V31 retained-interval representation longer. Faster or longer execution cannot remove the independently documented cap32 obstruction under the frozen append-only/global-rail assumptions.

Not killed: LCDM, $n=3$ EDE, the underlying common-latent mathematical target, or every future contract-preserving numerical strategy. Technical failures and unqualified diagnostics remain non-physical evidence.

MODEL CHANGE

Physical model change: NONE. H0 benchmarks are unchanged. MOD-EDE-N3 remains ACTIVE / CONSTRAINED. C-036-IMPL remains a methodological CamSpec-HiLLiPoP geometry tension with unresolved downstream cosmological significance.

Execution-state change: Q042 is no longer an active unchanged execution campaign. The documented tested strategies did not qualify the original contract, and Q042 closes inconclusively. A materially new qualified design would require a new-case or explicit reopening decision.

GITHUB / PROVENANCE

Relevant remote prerequisite: Q042-CONTEXT-V29, GitHub Actions run 37362125253, attempt 1, execution commit 72cf9e92fc794c122f593a77b6a555e6e97f6a2e, status success. Its scope is prerequisite-only.

V31 is local, not committed as a GitHub workflow result: RUN Q042_COMPARISON_V31_LOCAL_001; result R-Q042-COMPARISON-V31-001; CONFIG SHA-256 2f7a5d08e7566c613de93328fe4d4d6760d8ede81f6c3b789e230f64ca202dd1; CLASS_EDE source commit 5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Conclusion-critical internal source IDs: I-Q042-COMPARISON-DESIGN-001; I-Q042-COMPARISON-V31-001; I-Q042-COMPARISON-V31-VALIDATION-001; I-Q042-V31-INGESTION-001. The complete Q042 ingestion decision/audit preserves the 91-source register and inherited claim maps without renumbering legacy K-IDs.

REPOSITORY STATE

At the documented revision point, Morfindien/bubbleverse-model remained accepted through Q041 at model head 10fa83797b2f7740519d4e43b1bebf0a6e35bada, while Morfindien/Bubbleverse remained at execution head 72cf9e92fc794c122f593a77b6a555e6e97f6a2e. README and program-registry status lagged the supplied Q042 closure. No remote synchronization was performed as part of this journal closure.

CONCLUSION

The available tested strategies did not establish execution and scientific qualification of the exact original Q041 downstream-portability contract. General feasibility remains

inconclusive. Q042 therefore closes as a documented inconclusive feasibility investigation, not as an affirmative production result and not as a universal impossibility theorem.

The downstream physical question remains unresolved. No posterior portability class, model preference, physical falsification, pipeline superiority, or new physics follows from Q042.

NEXT HANDOFF

No new scientific Q-ID is allocated solely from this closure. The next concrete task is operational: TASK-Q042-M14-SYNC-001 - synchronize the Q042 closure into one consistent repository, registry and candidate-model state while preserving the existing evidential boundaries and previous accepted snapshot until the controlled update passes its gates.

REQUIRED END-OF-Q OUTPUT

FINAL ANSWER: CLOSED - INCONCLUSIVE GENERAL FEASIBILITY. The documented strategies did not qualify the original contract; materially different contract-preserving feasibility remains unresolved. MIKAMI: Run-longer is rejected as sufficient repair of unchanged V31. Technical failure is not physical failure. MODEL CHANGE: No physical change; execution feasibility and qualification limits refined. NEXT: Repository/model synchronization task; no new scientific Q allocated.

Q043 — Repository integration preparation and validation

Q-ID: Q043. Status: CLOSED. Result ID: R-Q043-REPOSITORY-INTEGRATION-001. Task lineage: TASK-Q042-M14-SYNC-001. CASE-ID: NOT DOCUMENTED.

QUESTION: Can the Q042 closure be incorporated into a consistent candidate/control package while preserving source and qualification boundaries and passing the existing gates?

ANSWER: Yes, locally at the pinned repository bases. Result: PASS_LOCAL_PREPARATION_AND_VALIDATION_ONLY. No remote synchronization, candidate admission, promotion, release or numerical production was performed.

STARTING STATE: Q042 CLOSED, general feasibility INCONCLUSIVE; reference-truth gate BLOCKED; final-result gate UNRESOLVED; production NOT AUTHORIZED. Accepted remote model v0.3/R000003 through Q041. Original Q041 physical portability question remains OPEN.

METHOD: Prepare and validate controlled diffs for Morfindien/bubbleverse-model and Morfindien/Bubbleverse. Preserve all 91 late-Q042 source objects, both claim-map trees, all prior accepted snapshots, physical model content, and historical workflows/manifests/artifacts. Proposed v0.4/R000004 stored under provenance/prepared_candidates/Q042/; status CANDIDATE_PREPARED_NOT_PROMOTED.

VALIDATION: Candidate 8/8 PASS; scientific 10/10 PASS; formal 13/13 PASS; Q041 historical 10/10 PASS; failure-injection 10/10 PASS; public healthcheck 20/20 PASS; launcher resolver 4/4 PASS; shared closure fields 14 matched; protected execution files 1,526 unchanged; clean-base diff PASS; read-only technical review CLEAN.

GITHUB BOUNDARY: Last inspected model HEAD 10fa83797b2f7740519d4e43b1bebf0a6e35bada; execution HEAD 72cf9e92fc794c122f593a77b6a555e6e97f6a2e. V29 run 37362125253 succeeded for prerequisites only; live registry remained ACTIVE for V29. Prepared package had not been applied at the inspected heads.

LIMITATIONS: Internal technical consistency is not independent physical validation. Q042 optimizer trajectories and cap proof were preserved rather than rerun. No scientific portability classification, H0 shift, EDE detection or falsification follows. Physical model NOT MATERIALLY CHANGED.

NEXT HANDOFF: TASK-Q043-M14-APPLY-001 to REPOSITORY / MODEL UPDATE ENGINE for controlled application and separately gated admission/promotion. This journal does not authorize dispatch. PRED-EDE-PORTABILITY-001 and PRED-Q039-COUPLED-001 remain OPEN.

SOURCES: EVD-Q043-JOURNAL; EVD-Q043-RESULT; I-Q043-PREPARATION-VALIDATION-001; I-Q042-V31-INGESTION-001. q043_integration_result.json SHA-256 4486e5173ba8ef3ac659c743bb578284c93446ce9e5b3d0dae71728beadbb9ff. Revision timestamp: 2026-10-09 06:32:29 UTC (historical revision time, not a new inspection).

Q044 | Closed research journal | 9 October 2026

Research question and classification

Q-ID: Q044. STATUS: CLOSED — epistemically inconclusive. Original scientific classifier: NOT_AVAILABLE.

Under the frozen Q041 matched-data and prior contract, does the CamSpec–HiLLiPoP implementation difference materially alter H_0 constraints or the $n_{\text{scf}} = 3$ EDE-versus-Lambda-CDM conclusion once numerical prediction accuracy and posterior convergence are qualified?

ANSWER: No qualified downstream cosmological classification is available. The scoped component analysis yields a conditional mathematical result, not a validated H_0 or model-preference inference.

Methods, evidence and quantitative result

A support-specific adjoint real-arithmetic enclosure was obtained under the inherited hydrogen source rails and fixed knots. UPPER: 138 supports (3195–3332), native-plus hull [0.70676532823050475, 0.7067869701940579]. LOWER: 3002 supports (331–3332), hull [0.0016297839913017342, 0.0018091545475532051]. The received checks included 64 rational fixtures, 18 native-C diagnostics and 17 V29 recovery matches. The current publication edit did not rerun those diagnostics.

Limitations: conditional source rails, correlated source inputs, incomplete physical histories, unqualified native-binary rounding, no validated prediction/likelihood error budget and no converged original 20-cell matrix. No cosmological parameter benchmark changes.

Scientific state, negative results and provenance

The representation/storage problem is partially cleared. The historical plus-versus-minus optical-depth functional difference remains physically unqualified. No physical hypothesis is falsified or newly confirmed. Q041 V19 non-convergence and Q042 incomplete feasibility remain numerical limitations rather than physical verdicts.

Accepted remote model: v0.5 / R000005 through Q043, following technical installation run 37927147545. Q044 local evidence has not been remotely admitted. No repository write, workflow dispatch or production restart was authorized by the supplied Motor 14 revision.

Provenance: R-Q044-SPLINE-ADJOINT-001; R-Q044-SPLINE-INGESTION-002; I-Q044-SPLINE-CHECKS-001; I-Q044-CONTRACT-001; I-Q044-M14-REPOSITORY-001. Repository pins: model dd7575e7146382957d206e87274ca6246c2587b7; execution 72cf9e92fc794c122f593a77b6a555e6e97f6a2e; native 5a131c91d657dd9a7c6364cc45b038710f8d0d97.

Next-question handoff

Q045 — PROPOSED, canonicalization required. For frozen Q041 Lambda-CDM and $n_{\text{scf}} = 3$ EDE configurations, test whether replacing only the native optical-depth integration functional with a convergent Thomson-integral reference at the same requested τ_{reio} changes CMB TT/TE/EE and native likelihoods enough to threaten the original H_0 /EDE decision margins.

Starting motor: Bubbleverse Autonomous Mathematics Engine. This question is not yet answered. The full matched native-arm posterior portability question remains open as PRED-EDE-PORTABILITY-001. Additional open predictions: PRED-Q039-COUPLED-001 and PRED-H0-INDEPENDENT-001.